Learning a Codebase

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Contents

1 Repository **4**
1.1 Glossary - Summary . . . 4
1.2 Repository Summary Table . . . 5
1.3 Sample scatter touched against commit - . . . 6
1.4 Total commits by developer - Developer Commits . . . 7
1.5 Histogram components touched by developer - For components with total . . . 9
1.6 Time series components touched - For packages touched for commit . . . 18
1.7 Time series components touched - For packages touched for each period . . . 20
1.8 Time series components touched - For classes touched for commit . . . 22
1.9 Time series components touched - For classes touched for each period . . . 24
1.10 Time series components touched - For methods touched for commit . . . 26
1.11 Time series components touched - For methods touched for each period . . . 28
1.12 Time series components touched - For packages touched for commit . . . 30
1.13 Time series components touched - For packages touched for each period . . . 32
1.14 Time series components touched - For classes touched for commit . . . 34
1.15 Time series components touched - For classes touched for each period . . . 36
1.16 Time series components touched - For methods touched for commit . . . 38
1.17 Time series components touched - For methods touched for each period . . . 40
1.18 Time series components touched - For packages touched for first ten commits . . . 42
1.19 Time series components touched - For classes touched for first ten commits . . . 44
1.20 Time series components touched - For methods touched for first ten commits . . . 46
1.21 Anova generate - For packages touched for commit . . . 48
1.22 Anova generate - For classes touched for commit . . . 48
1.23 Anova generate - For methods touched for commit . . . 48

2 Repository: 14 **50**
2.1 Time series components touched - 14 For packages touched for commit . . . 50
2.2 Time series components touched - 14 For packages touched for each period . . . 52
2.3 Time series components touched - 14 For classes touched for commit . . . 54
2.4 Time series components touched - 14 For classes touched for each period . . . 56
2.5 Time series components touched - 14 For methods touched for commit . . . 58
2.6 Time series components touched - 14 For methods touched for each period . . . 60
2.7 Time series components touched - 14 For packages touched for commit . . . 62
2.8 Time series components touched - 14 For packages touched for each period . . . 64
2.9 Time series components touched - 14 For classes touched for commit . . . 66
2.10 Time series components touched - 14 For classes touched for each period . . . 68
2.11 Time series components touched - 14 For methods touched for commit . . . 70
2.12 Time series components touched - 14 For methods touched for each period . . . 72
2.13 Time series sustained components touched by developer - 14 For developer commits and components touched. . . . 74
2.14 Time series sustained components touched by developer - 14 For developer day and components touched. . . . 77

3 Repository: 21 **79**
3.1 Time series components touched - 21 For packages touched for commit . . . 79
3.2 Time series components touched - 21 For packages touched for each period . . . 81
3.3 Time series components touched - 21 For classes touched for commit . . . 83
3.4 Time series components touched - 21 For classes touched for each period . . . 85
3.5 Time series components touched - 21 For methods touched for commit . . . 87
3.6 Time series components touched - 21 For methods touched for each period . . . 89
3.7 Time series components touched - 21 For packages touched for commit . . . 91
3.8 Time series components touched - 21 For packages touched for each period . . . 93

3.9 Time series components touched - 21 For classes touched for commit 95
3.10 Time series components touched - 21 For classes touched for each period 97
3.11 Time series components touched - 21 For methods touched for commit 99
3.12 Time series components touched - 21 For methods touched for each period 101
3.13 Time series sustained components touched by developer - 21 For developer commits and components touched. 103
3.14 Time series sustained components touched by developer - 21 For developer day and components touched. 106

1 Repository

1.1 Glossary - Summary

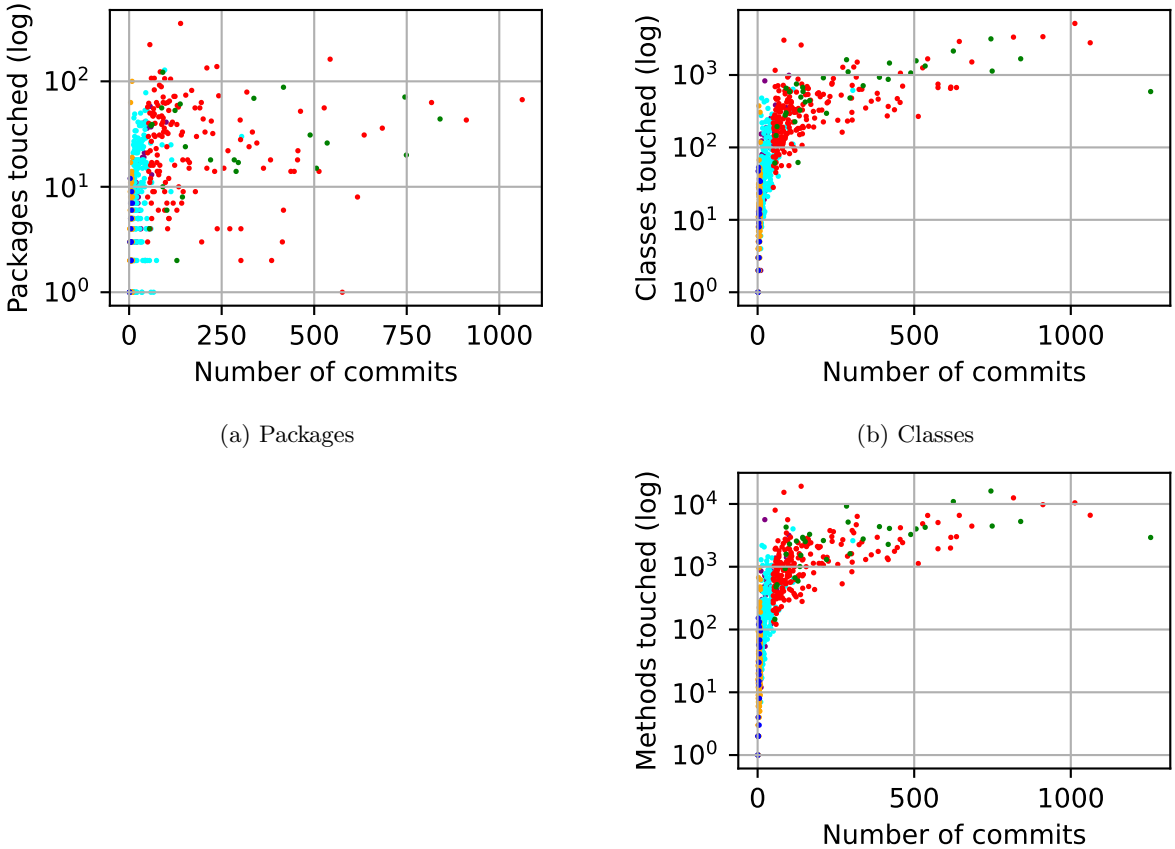
- The founder developer starts in the first six months of a project.
- Late joiner developers begin after six months.
- Transient developers have ten (10) or fewer commits.
- Moderate developers have more than ten (10) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

1.2 Repository Summary Table

Table 1: Summary of fifteen open-source repositories identified from GitHub that have at least 1000 pull requests and at least three sustained late joiner developers. Sustained later joiner developers joining the project after six months and contributed at least 50 commits over a period of 250 days or more. Please note that five developers each worked on two repositories.

ID	Repo Name	Transient Founder	Moderate Founder	Sustained Founder	Transient Joiner	Moderate Joiner	Sustained Joiner	Commit	Start	End
3	activiti	4	7	6	47	20	5	8662	2010-Jun-18	On going
5	airbyte-platform	3	2	1	45	32	6	6784	2020-Aug-04	2023-Sep-27
7	ambari	0	3	0	34	32	15	4247	2011-Sep-22	On going
10	automq	2	0	0	40	10	3	1874	2011-Aug-10	2017-Apr-21
14	buck	8	5	6	119	85	21	14876	2013-Mar-21	2022-May-05
15	camel	0	4	1	257	63	13	36528	2007-Mar-19	On going
18	checkstyle	0	0	1	57	42	9	17338	2001-Jun-22	On going
20	cxf	6	6	3	27	17	3	7893	2008-Apr-29	On going
21	intellij-community	0	3	7	113	93	84	101247	2004-Nov-11	2018-Apr-18
26	guava	0	1	0	47	8	3	9685	2009-Sep-15	On going
28	jenkins	1	1	1	144	39	4	30252	2006-Nov-05	On going
33	openmrs-core	0	1	0	123	26	3	15757	2006-May-03	On going
36	presto	0	1	3	162	71	18	30605	2012-Aug-09	On going
37	quarkus	1	3	5	126	25	8	81612	2018-Jun-14	On going
39	selenium	3	3	0	55	17	8	23354	2004-Nov-03	On going
	Total	28	40	34	1396	580	203	390714		

1.3 Sample scatter touched against commit -



(a) Packages

(b) Classes

(c) Methods

Figure 1: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at 203 from six (6) categories: Transient Founder (Blue, 28), Moderate Founder (Purple, 40), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

1.4 Total commits by developer - Developer Commits

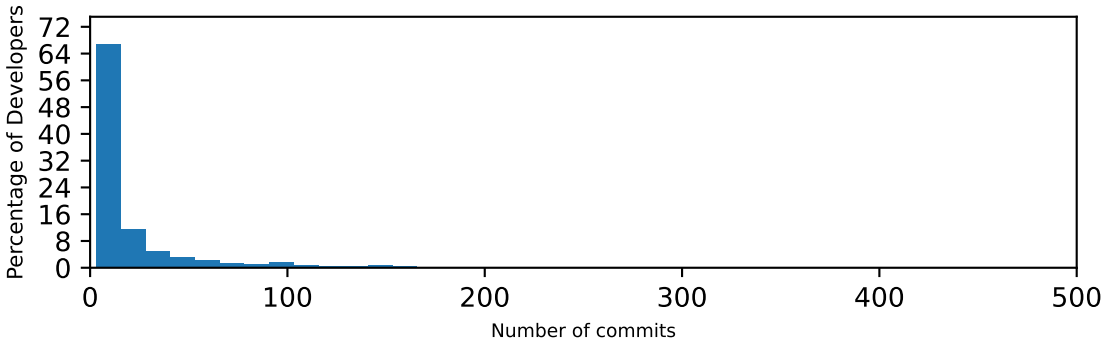
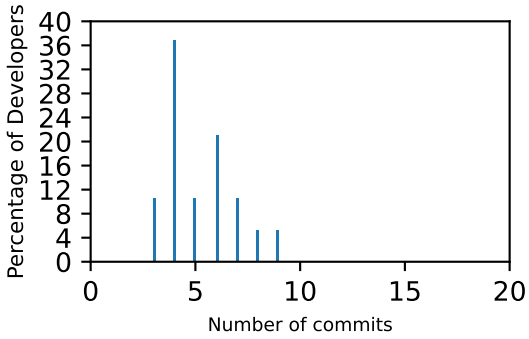
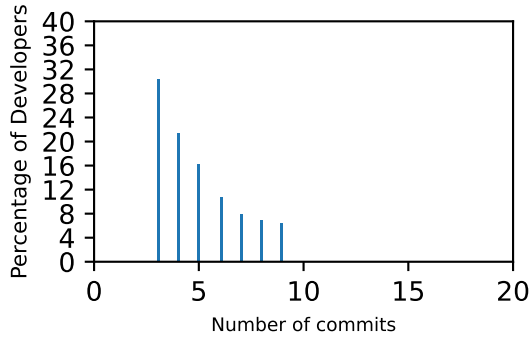


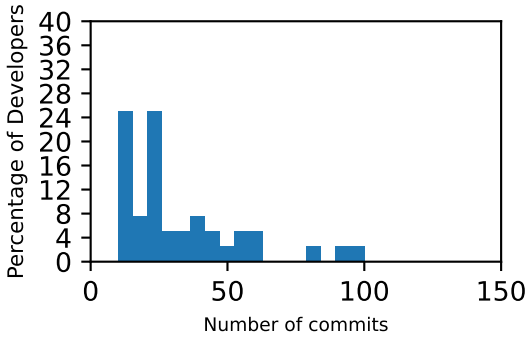
Figure 2: Histogram of the number from all developers (n=1906) (y-axis) against the total number of commits (x-axis) in 15 repositories sampled from GitHub.



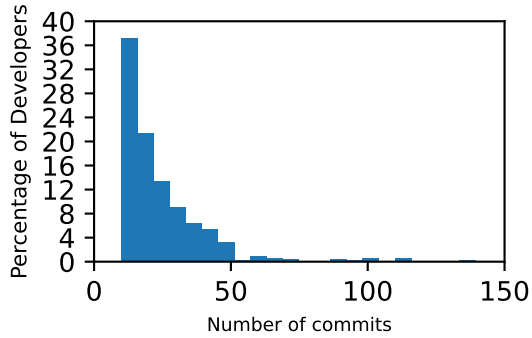
(a) Transient founder (n=19).



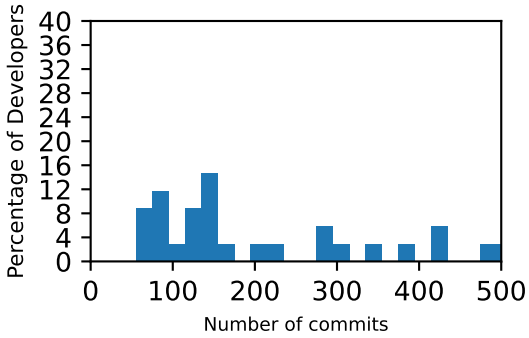
(b) Transient later joiner (n=1030).



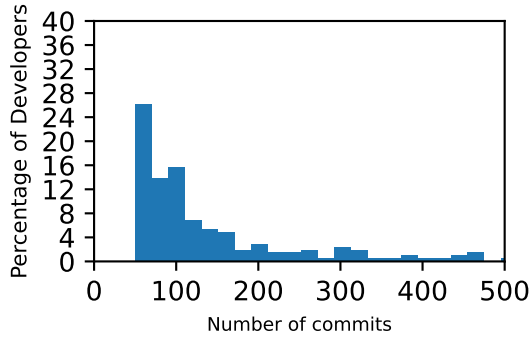
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).



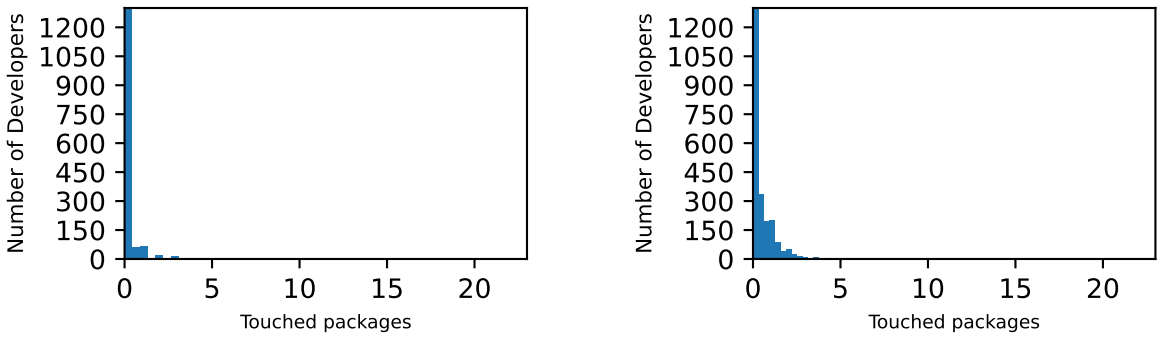
(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

Figure 3: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 15 repositories sampled from GitHub.

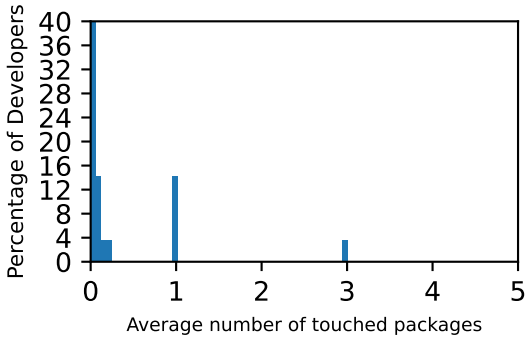
1.5 Histogram components touched by developer - For components with total



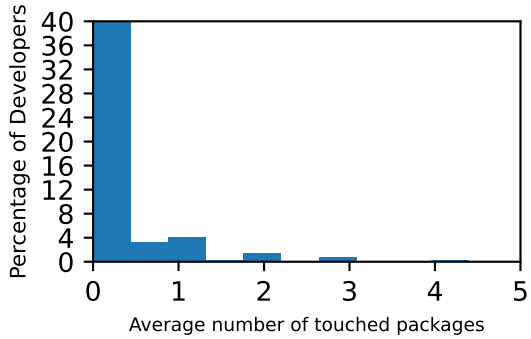
(a) By Day.

(b) By Commit.

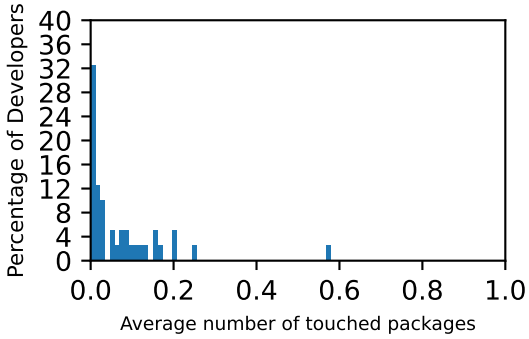
Figure 4: Histogram of the number of developers (y-axis) against the average packages touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



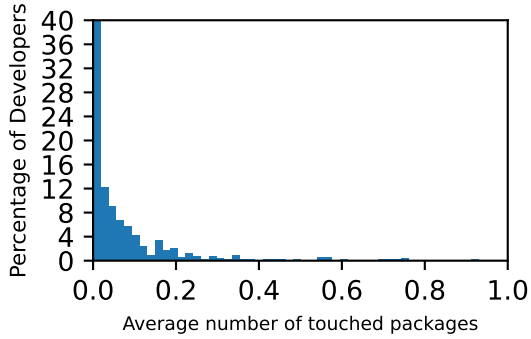
(a) Transient founder (n=28).



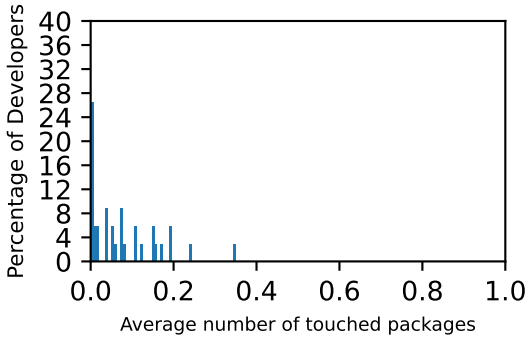
(b) Transient later joiner (n=1396).



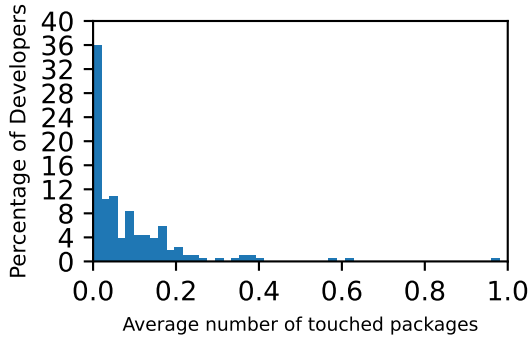
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

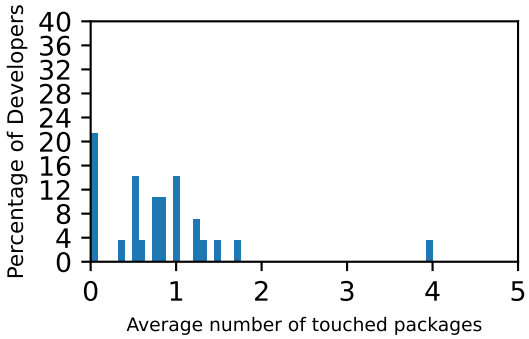


(e) Sustained founder (n=34).

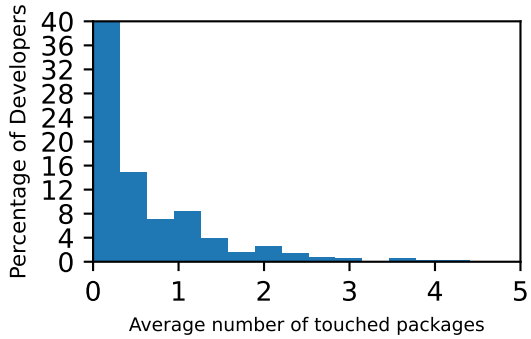


(f) Sustained later joiner (n=203).

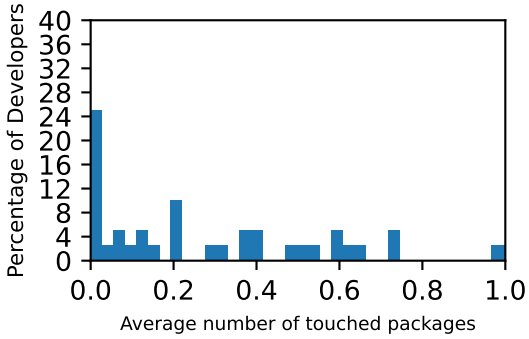
Figure 5: Histogram of the number of developers (y-axis) against the average packages touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



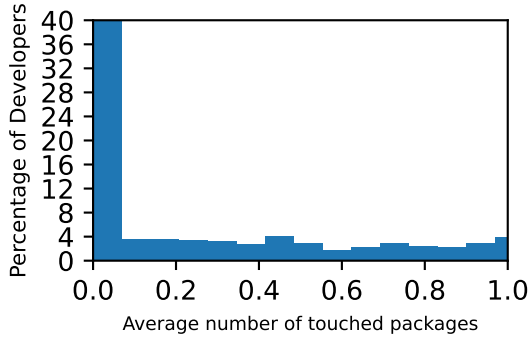
(a) Transient founder (n=28).



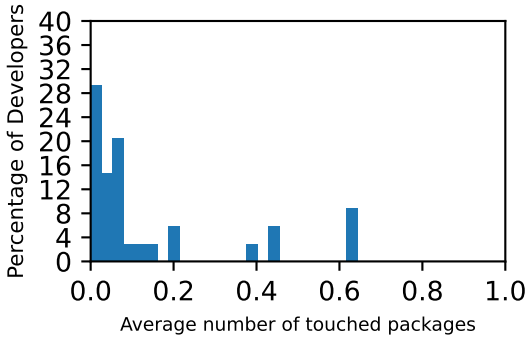
(b) Transient later joiner (n=1396).



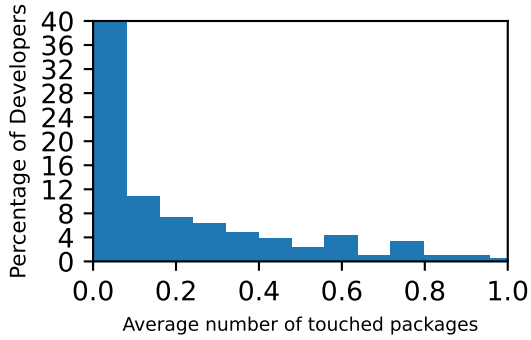
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

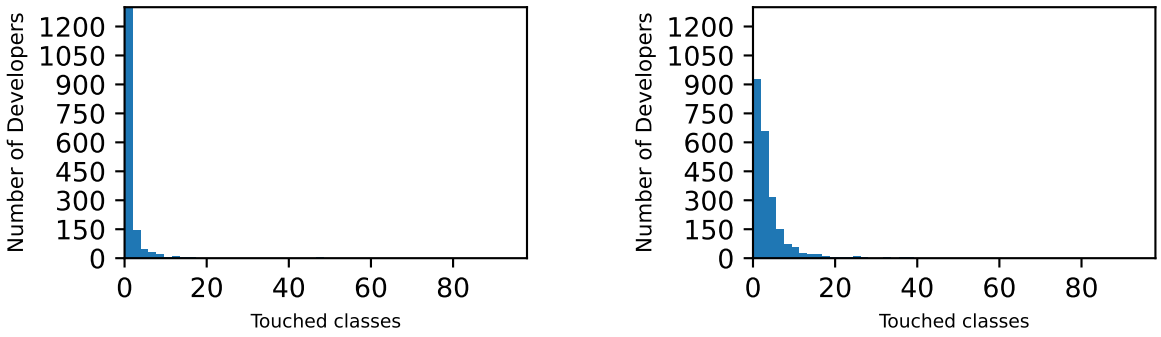


(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

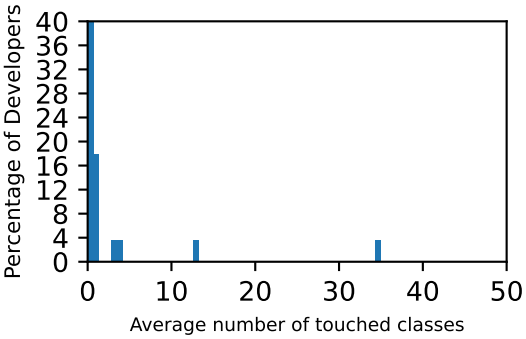
Figure 6: Histogram of the number of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



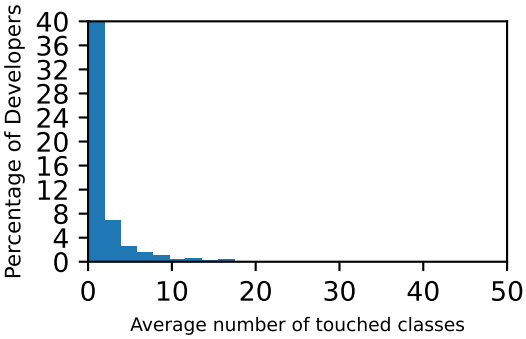
(a) By Day.

(b) By Commit.

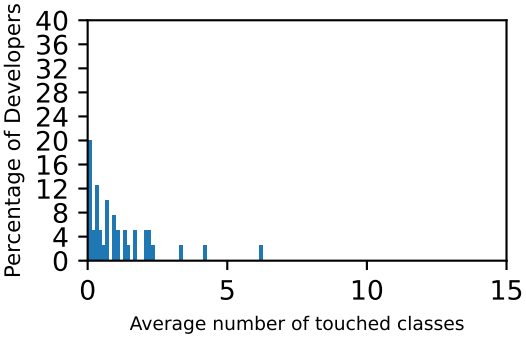
Figure 7: Histogram of the number of developers (y-axis) against the average classes touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



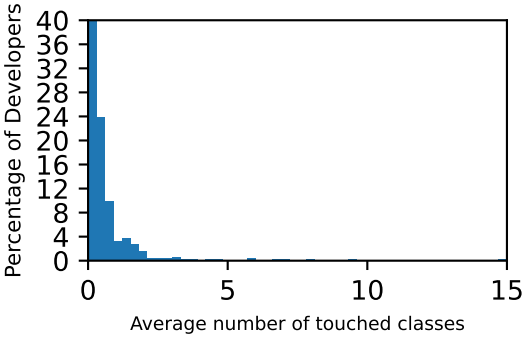
(a) Transient founder (n=28).



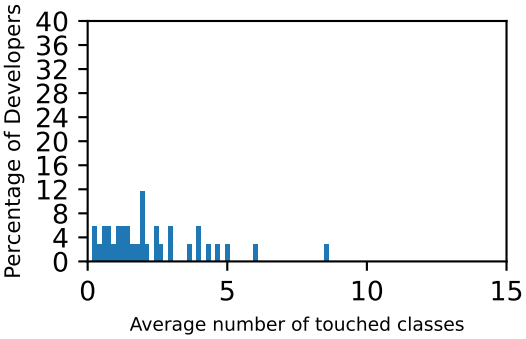
(b) Transient later joiner (n=1396).



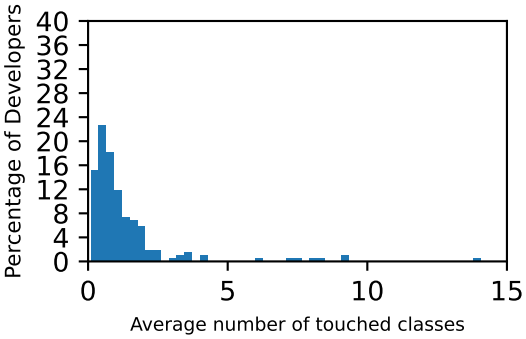
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

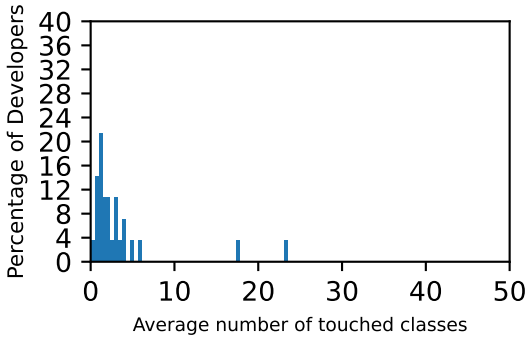


(e) Sustained founder (n=34).

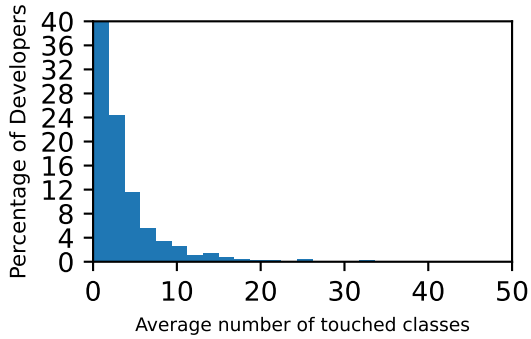


(f) Sustained later joiner (n=203).

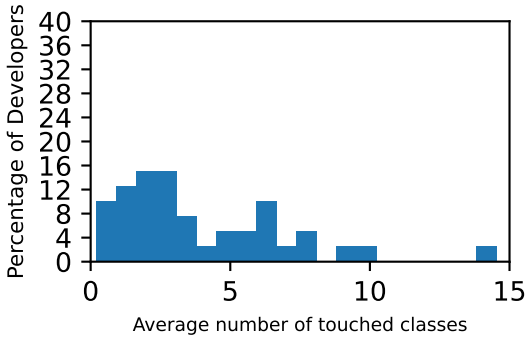
Figure 8: Histogram of the number of developers (y-axis) against the average classes touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



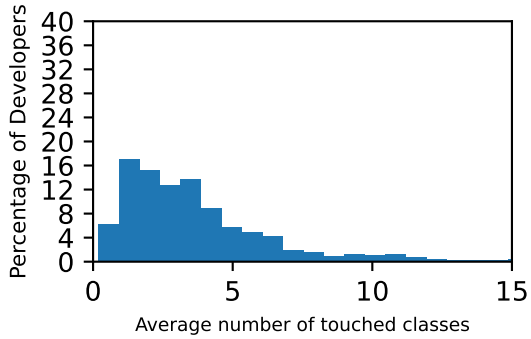
(a) Transient founder (n=28).



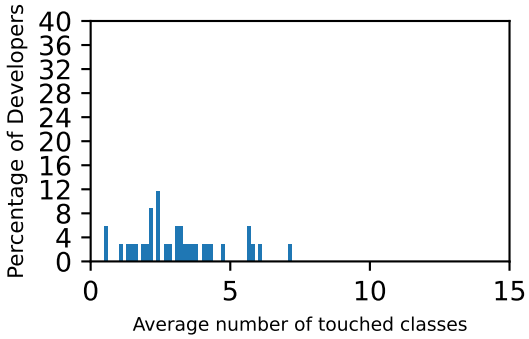
(b) Transient later joiner (n=1396).



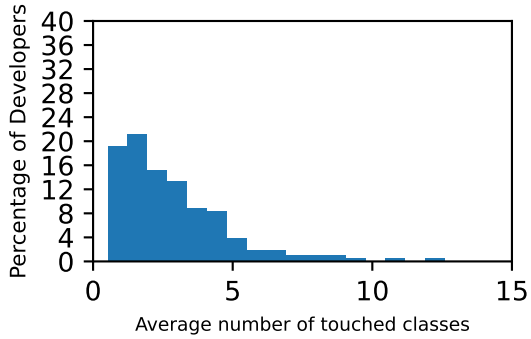
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

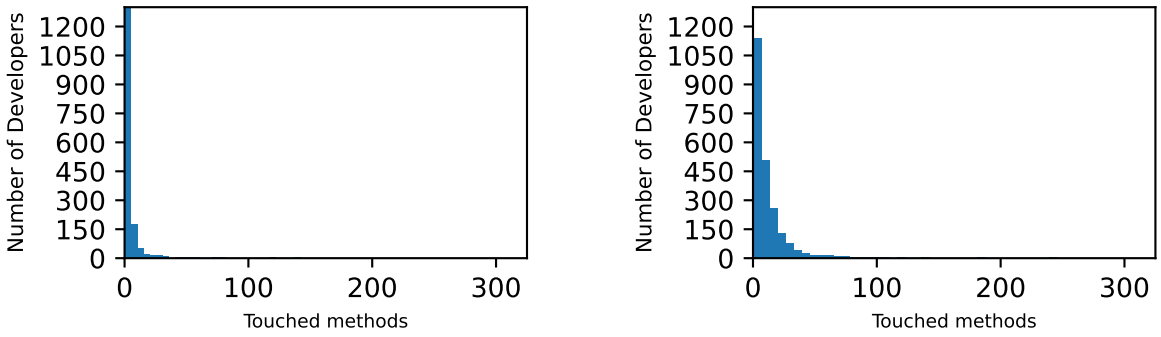


(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

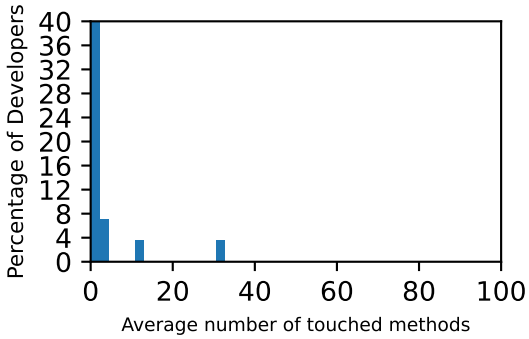
Figure 9: Histogram of the number of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



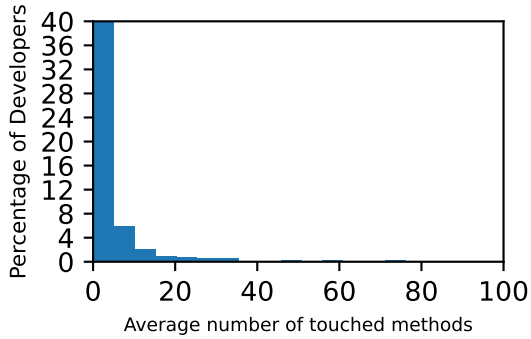
(a) By Day.

(b) By Commit.

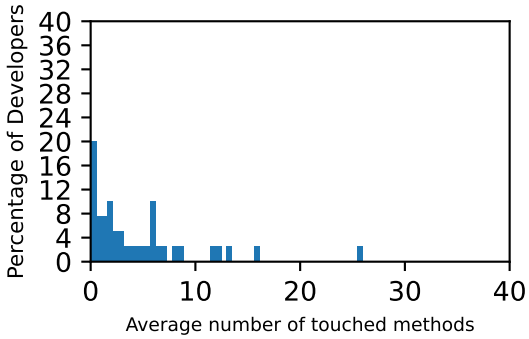
Figure 10: Histogram of the number of developers (y-axis) against the average methods touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



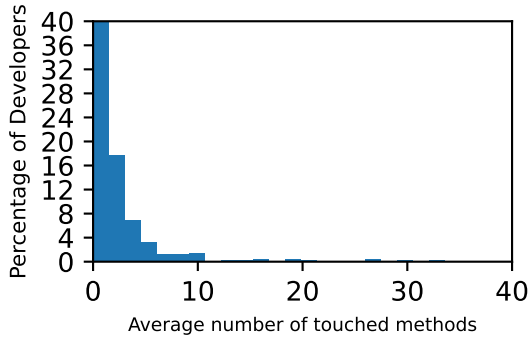
(a) Transient founder (n=28).



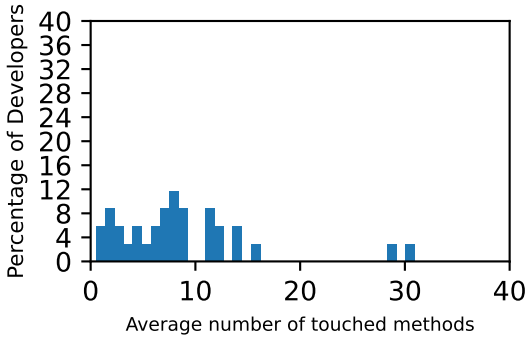
(b) Transient later joiner (n=1396).



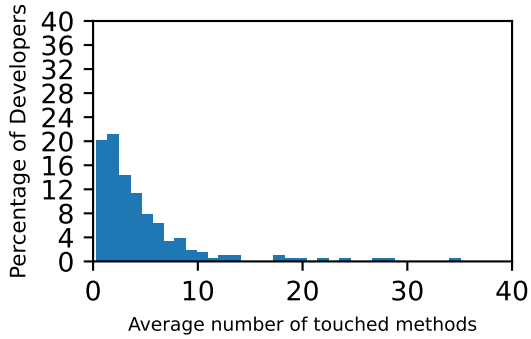
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

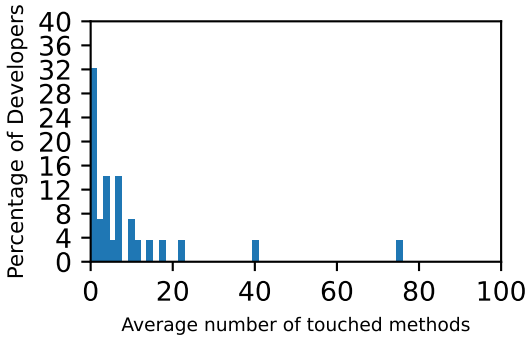


(e) Sustained founder (n=34).

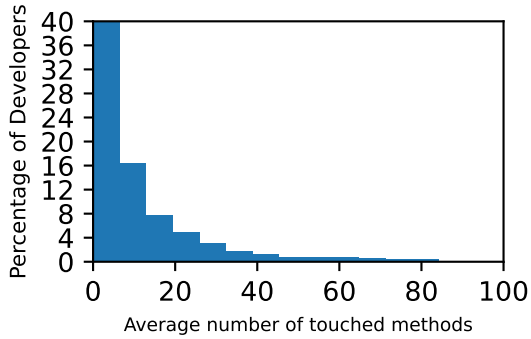


(f) Sustained later joiner (n=203).

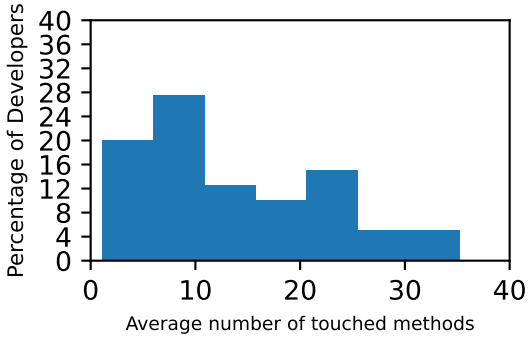
Figure 11: Histogram of the number of developers (y-axis) against the average methods touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



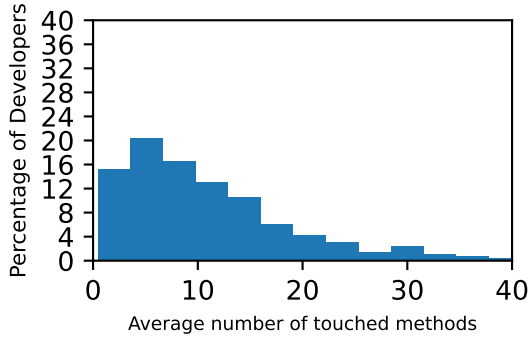
(a) Transient founder (n=28).



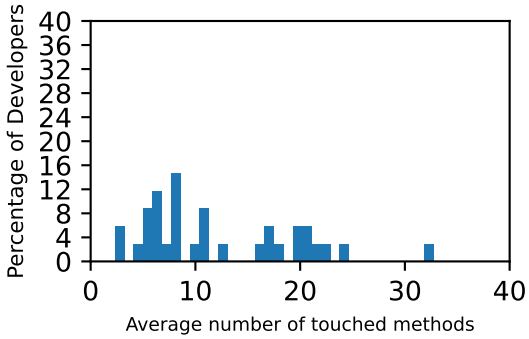
(b) Transient later joiner (n=1396).



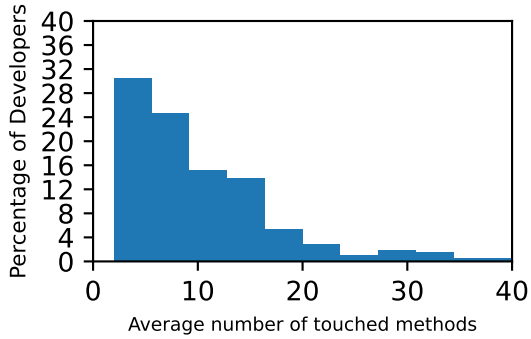
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).



(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

Figure 12: Histogram of the number of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.

1.6 Time series components touched - For packages touched for commit

A time series of packages touched on average.

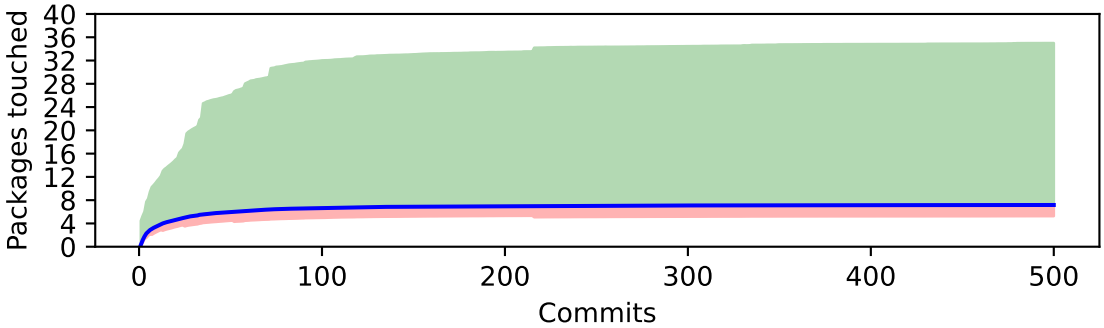
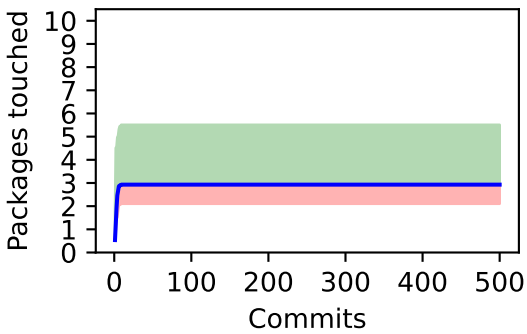
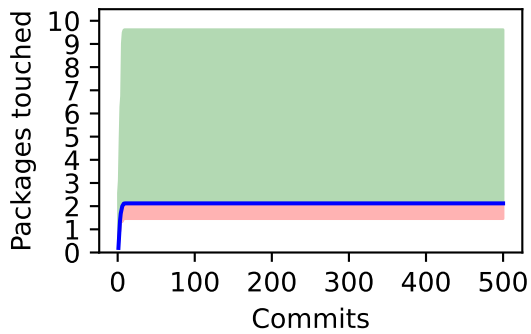


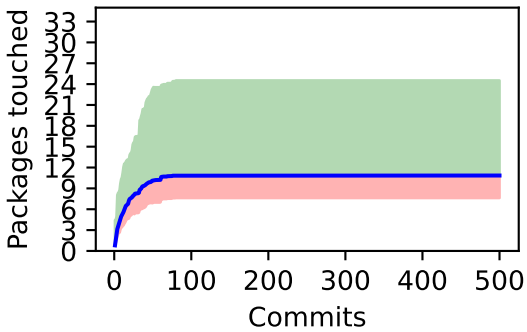
Figure 13: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



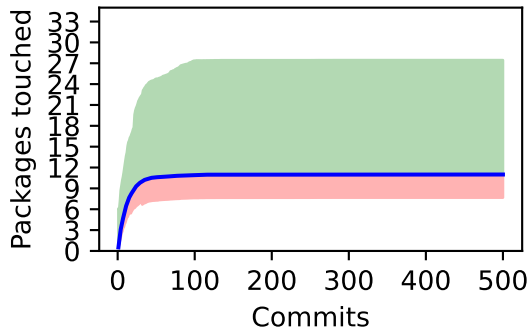
(a) Transient founder developers (n=28)



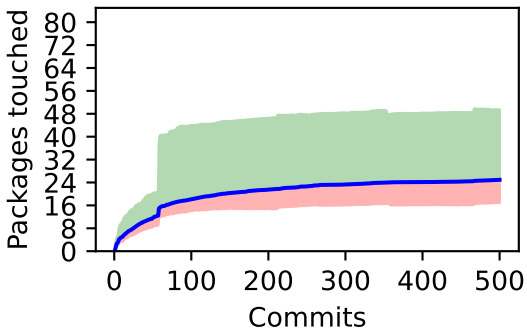
(b) Transient later joiner developers (n=1396)



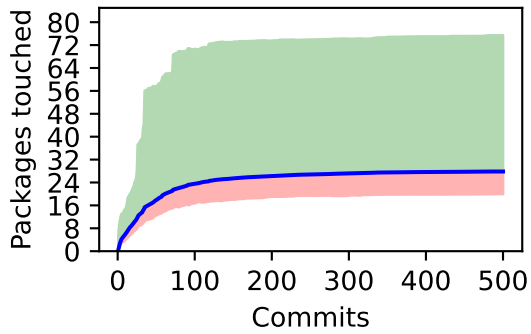
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 14: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.7 Time series components touched - For packages touched for each period

A time series of packages touched on average.

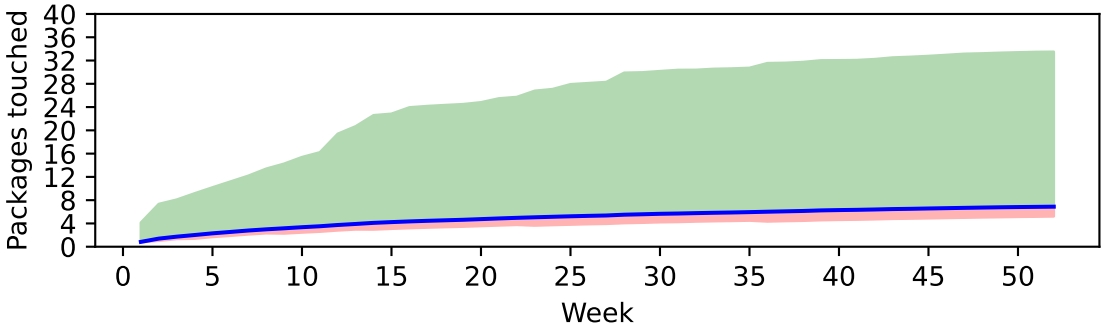
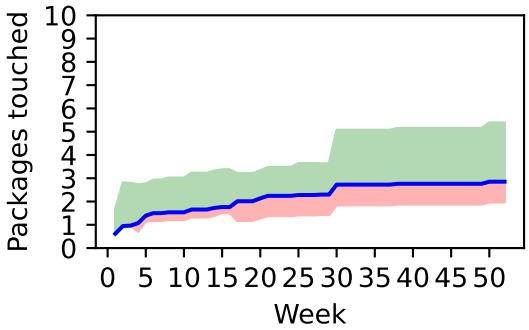
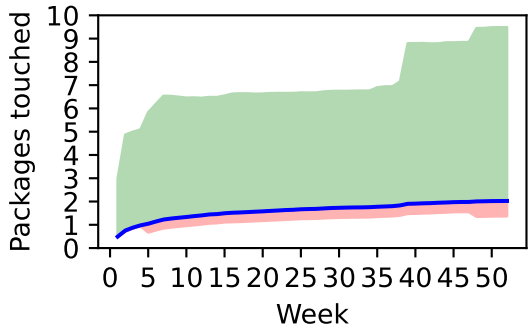


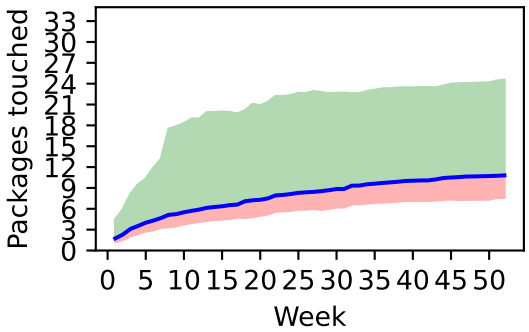
Figure 15: A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for all developers (n=2278).



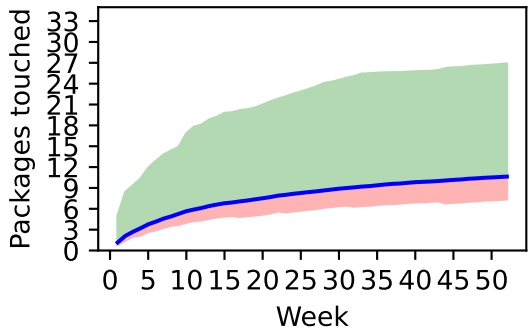
(a) Transient founder developers (n=28)



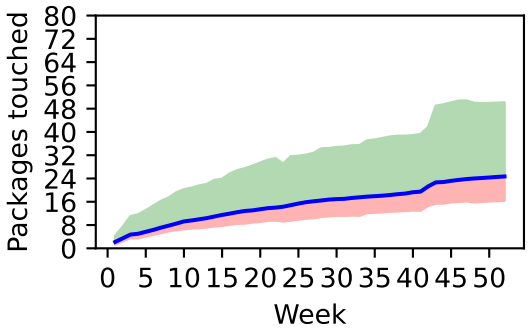
(b) Transient later joiner developers (n=1396)



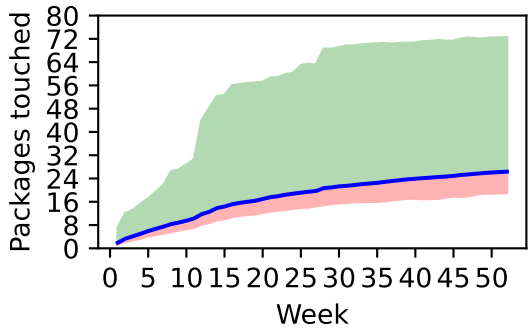
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 16: A time series of the average (mean) total packages touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

1.8 Time series components touched - For classes touched for commit

A time series of classes touched on average.

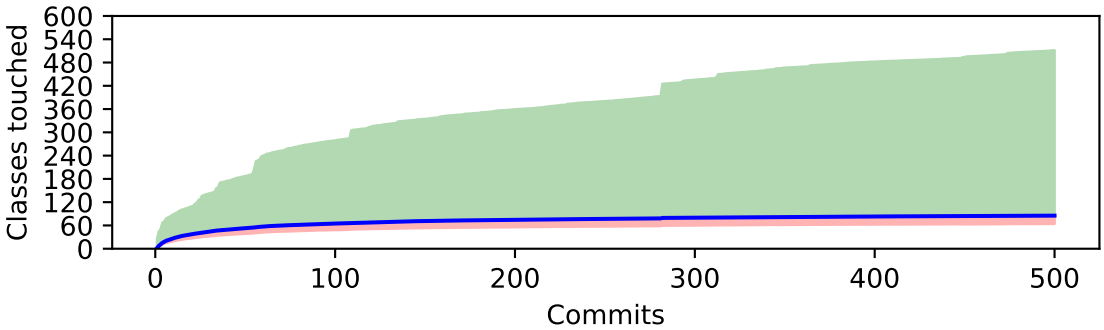
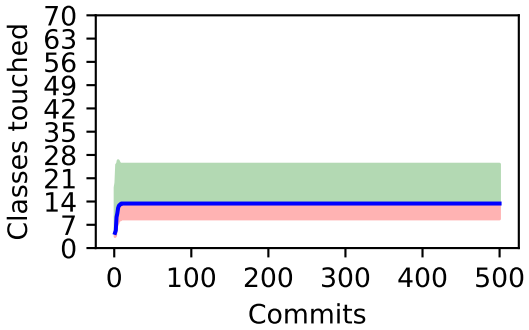
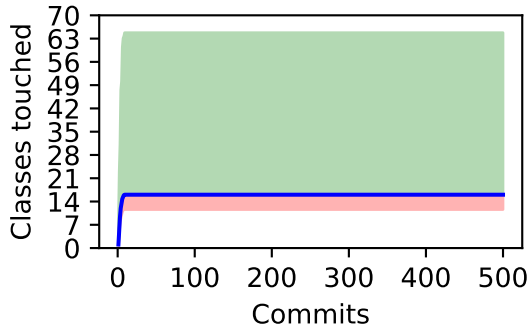


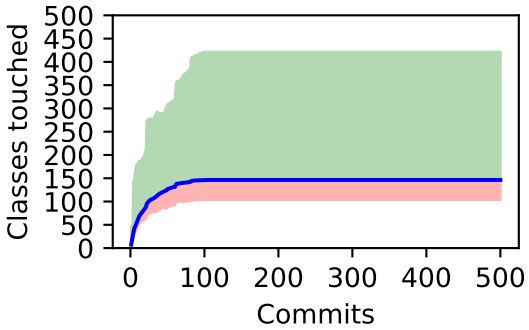
Figure 17: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



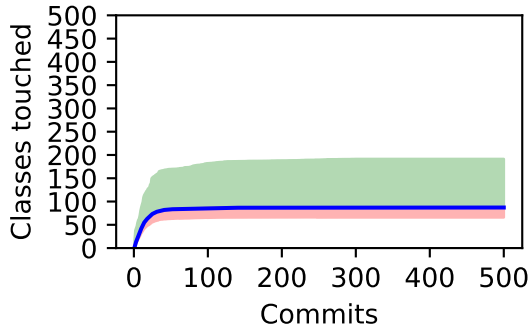
(a) Transient founder developers (n=28)



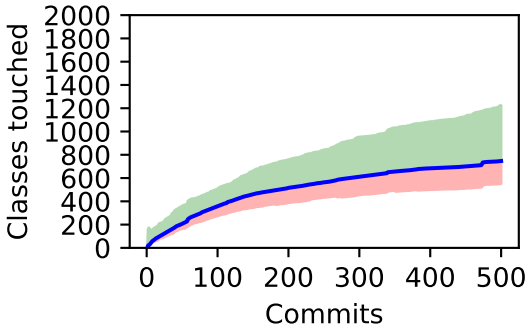
(b) Transient later joiner developers (n=1396)



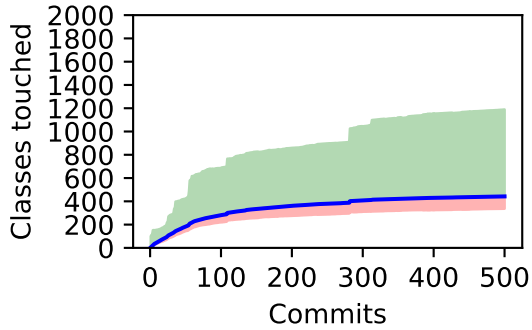
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 18: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.9 Time series components touched - For classes touched for each period

A time series of classes touched on average.

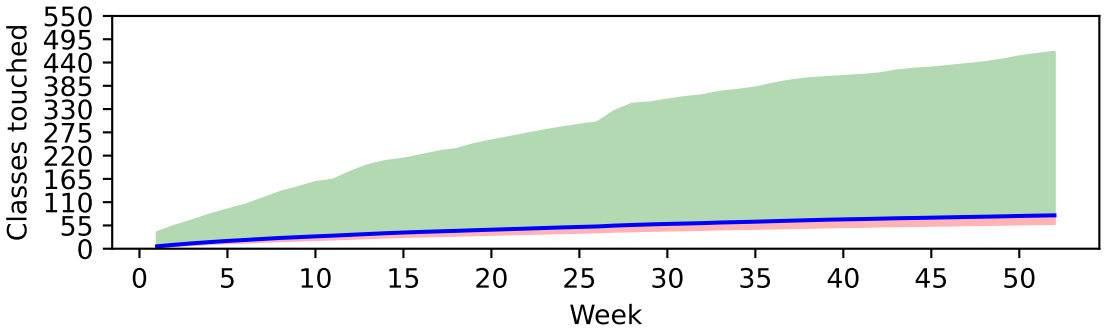
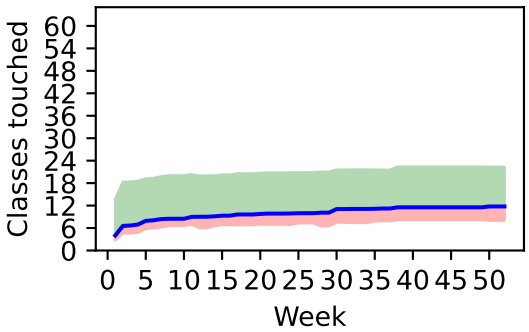
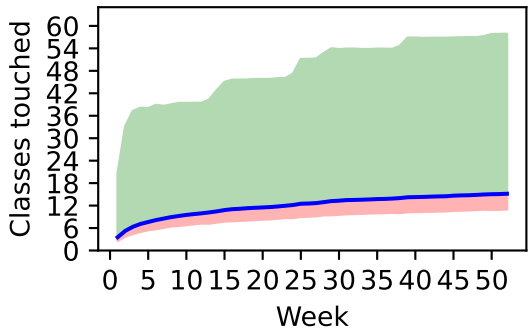


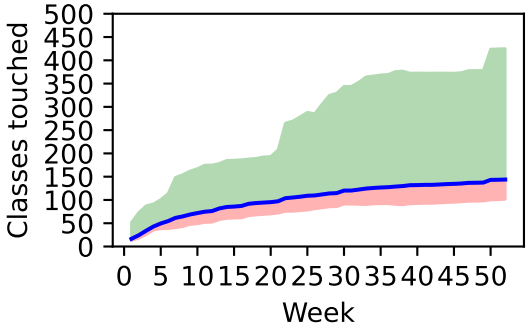
Figure 19: A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for all developers (n=2278).



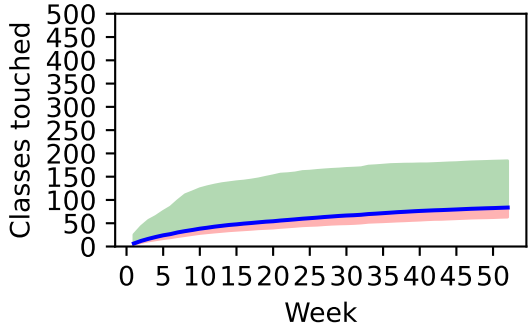
(a) Transient founder developers (n=28)



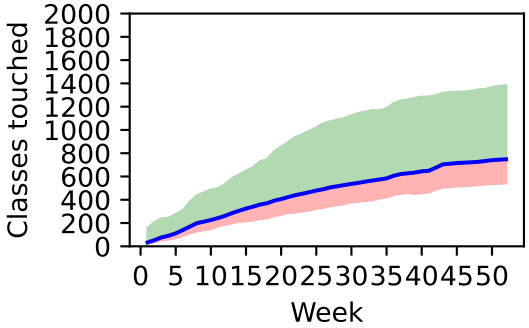
(b) Transient later joiner developers (n=1396)



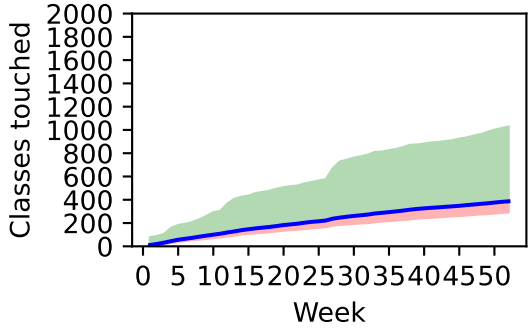
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 20: A time series of the average (mean) total classes touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

1.10 Time series components touched - For methods touched for commit

A time series of methods touched on average.

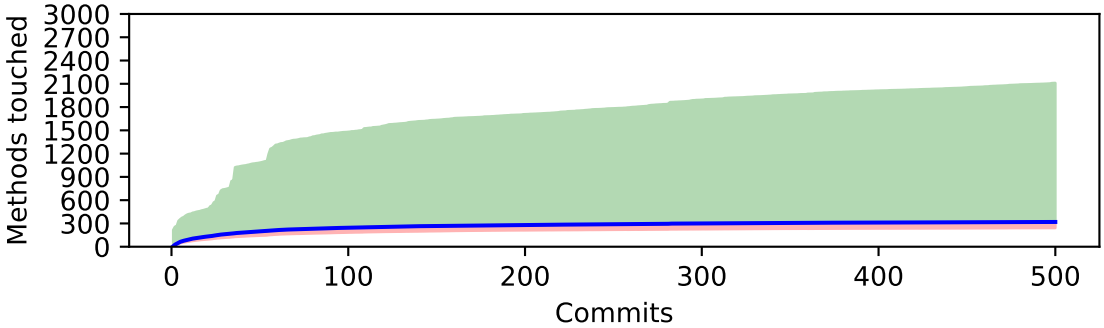
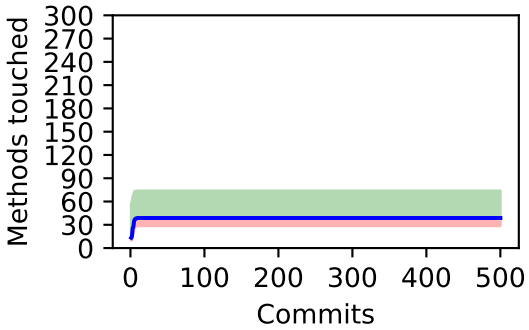
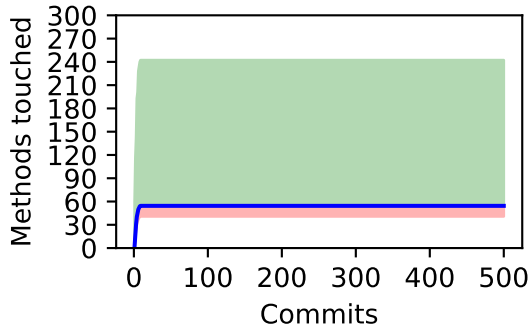


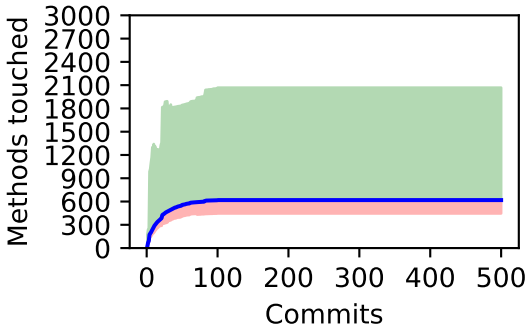
Figure 21: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



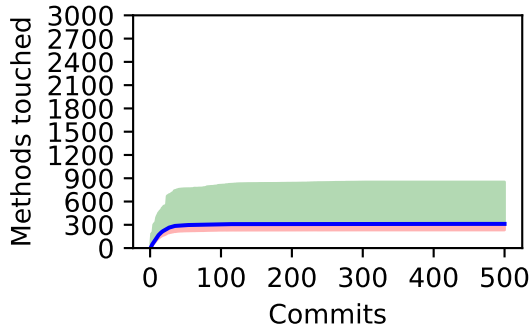
(a) Transient founder developers (n=28)



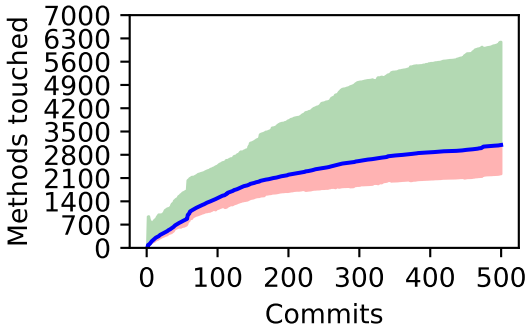
(b) Transient later joiner developers (n=1396)



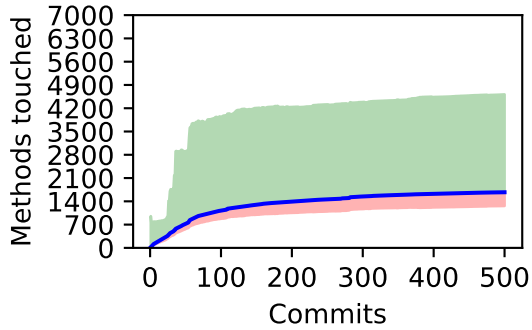
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 22: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.11 Time series components touched - For methods touched for each period

A time series of methods touched on average.

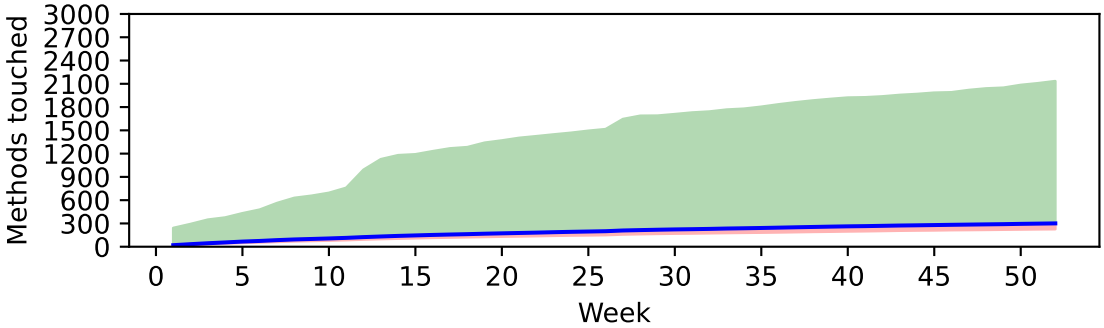
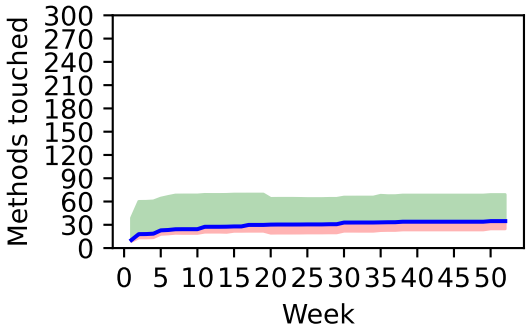
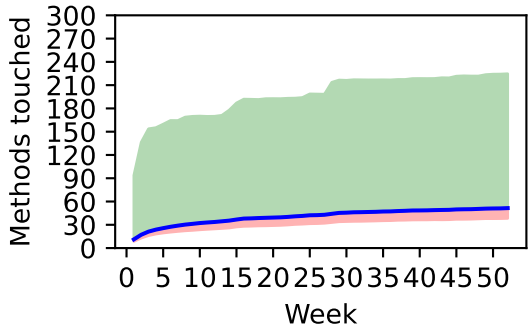


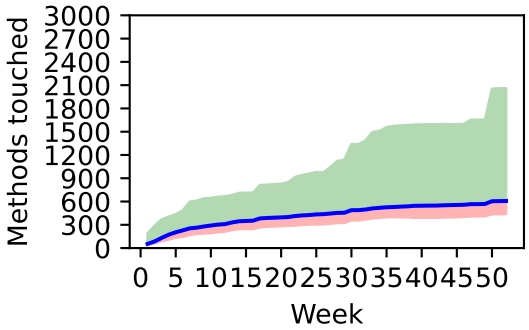
Figure 23: A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for all developers (n=2278).



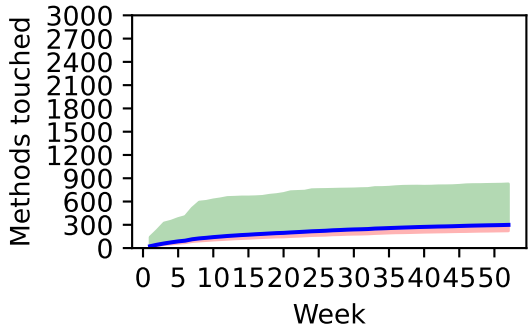
(a) Transient founder developers (n=28)



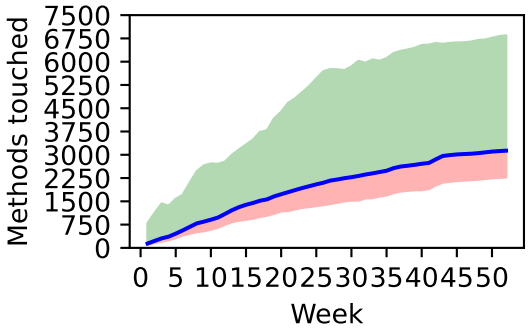
(b) Transient later joiner developers (n=1396)



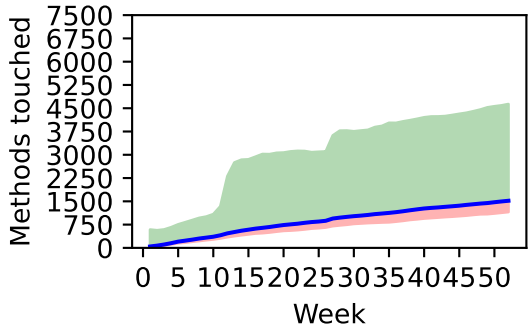
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 24: A time series of the average (mean) total methods touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

1.12 Time series components touched - For packages touched for commit

A time series of packages touched on average.

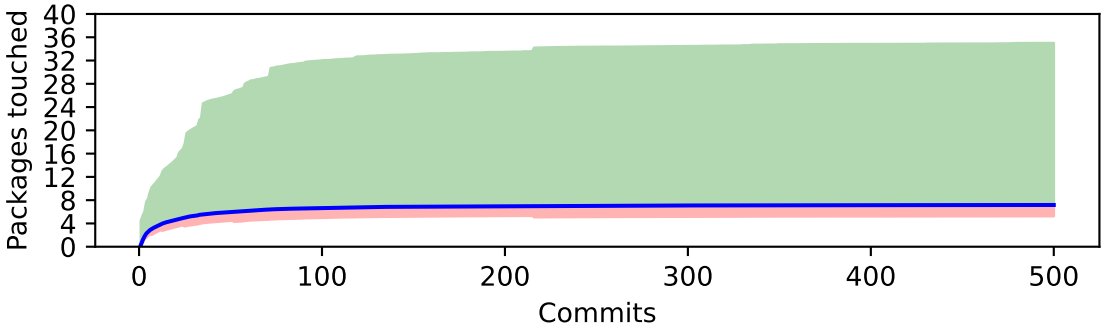
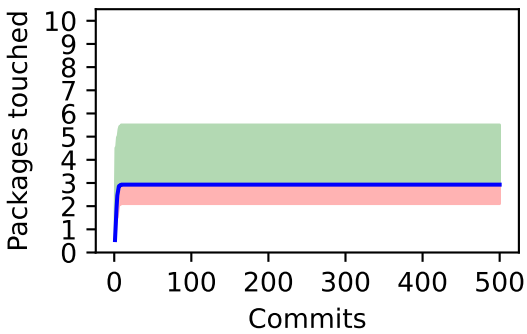
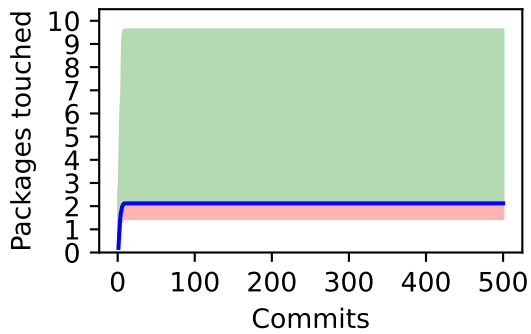


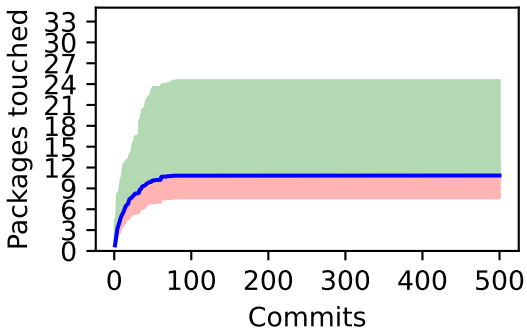
Figure 25: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



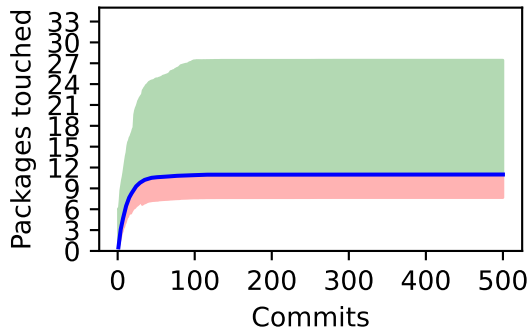
(a) Transient founder developers (n=28)



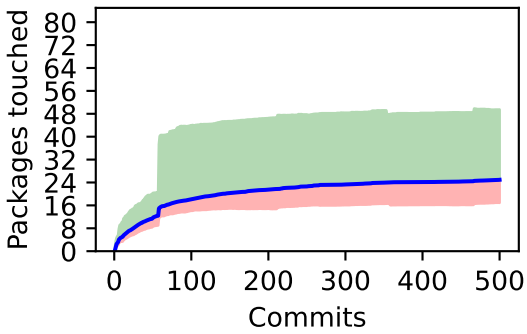
(b) Transient later joiner developers (n=1396)



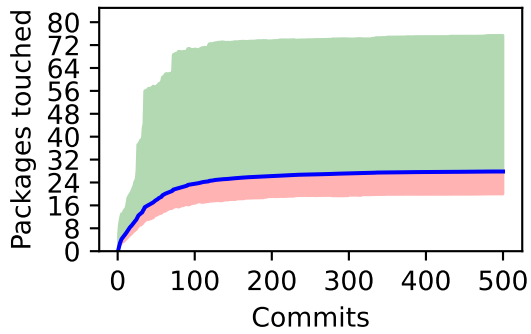
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 26: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.13 Time series components touched - For packages touched for each period

A time series of packages touched on average.

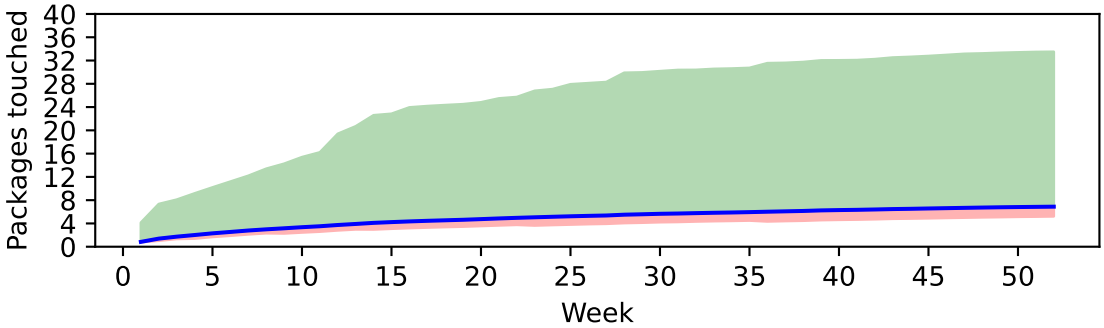
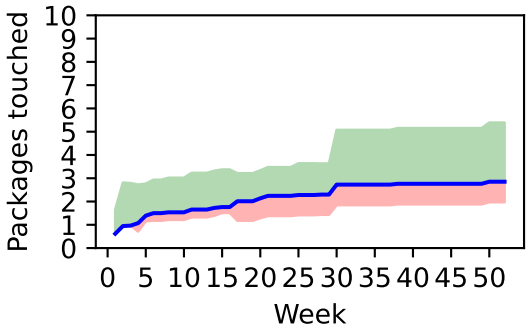
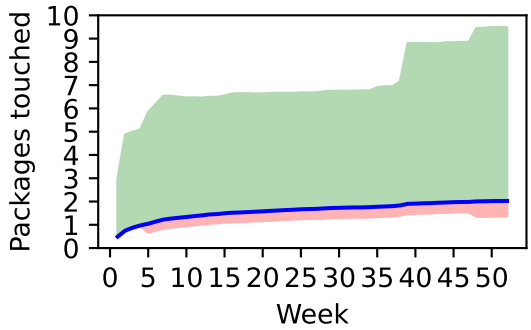


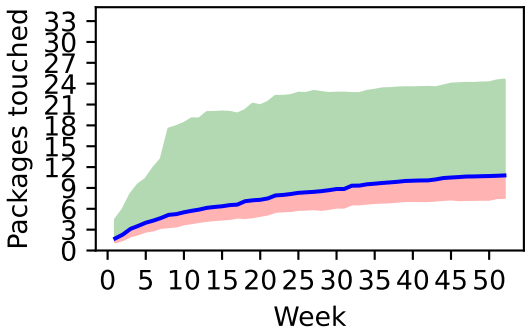
Figure 27: A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for all developers (n=2278).



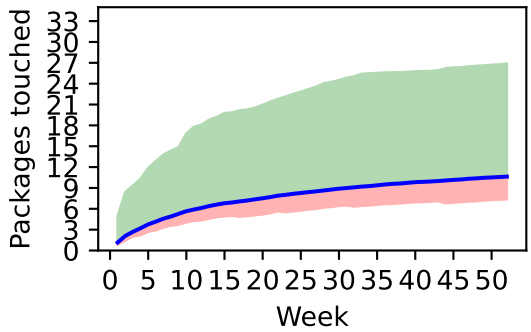
(a) Transient founder developers (n=28)



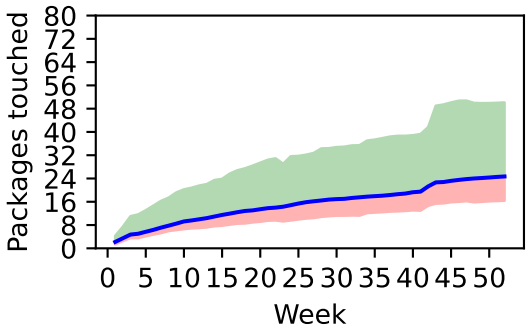
(b) Transient later joiner developers (n=1396)



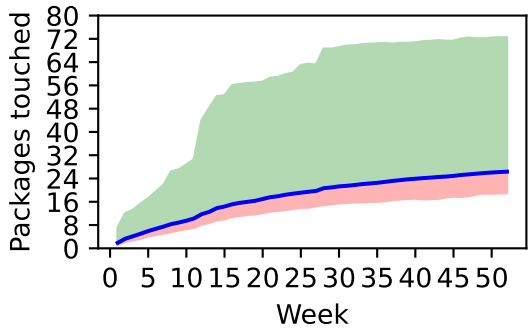
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 28: A time series of the average (mean) total packages touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

1.14 Time series components touched - For classes touched for commit

A time series of classes touched on average.

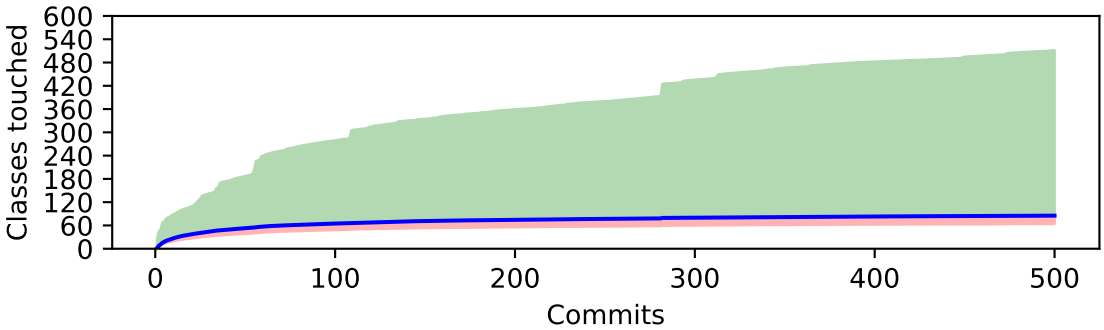
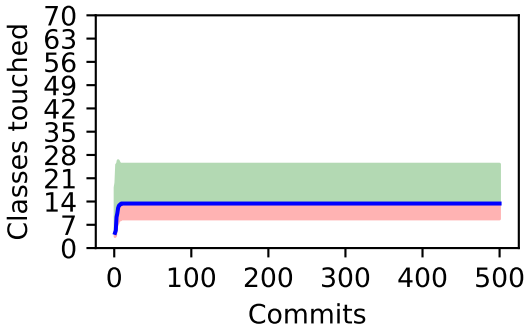
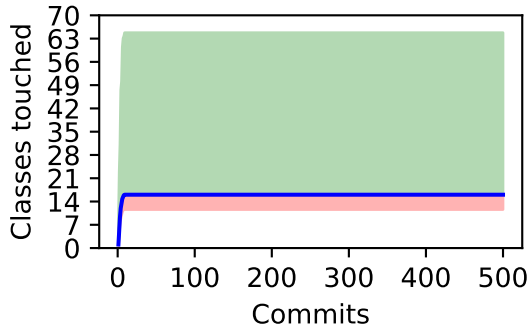


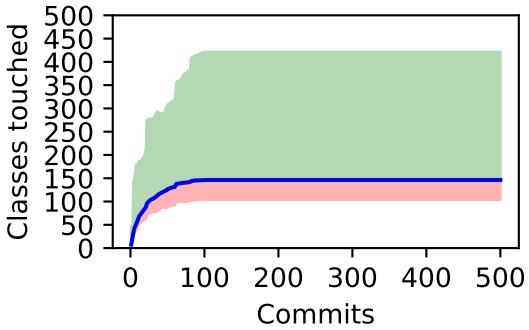
Figure 29: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



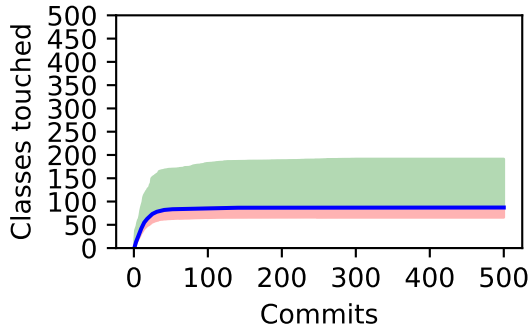
(a) Transient founder developers (n=28)



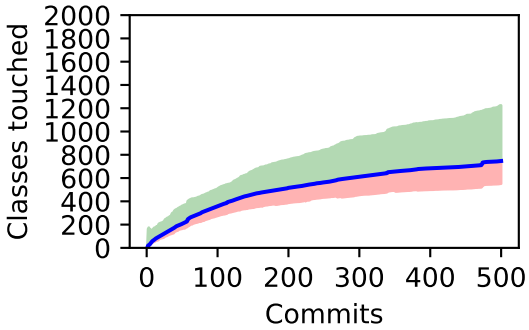
(b) Transient later joiner developers (n=1396)



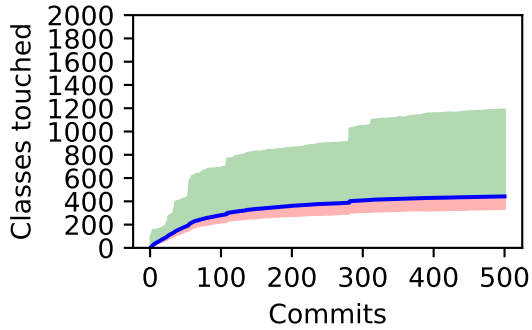
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 30: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.15 Time series components touched - For classes touched for each period

A time series of classes touched on average.

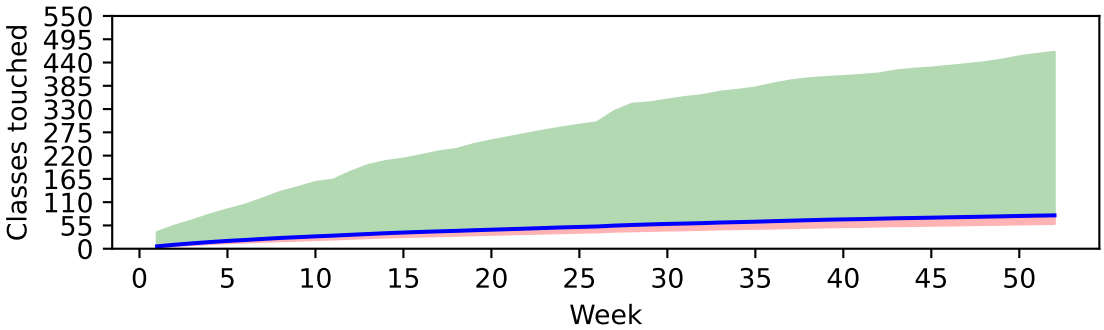
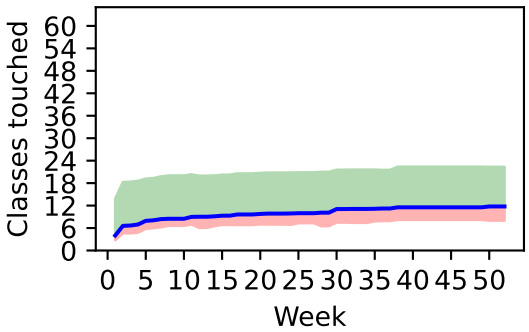
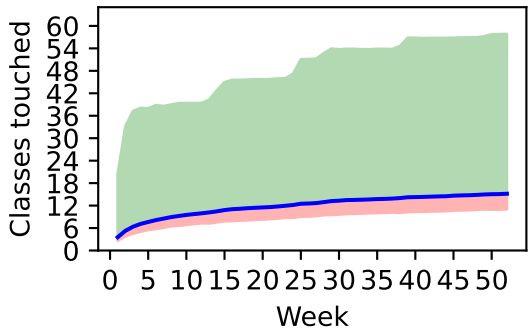


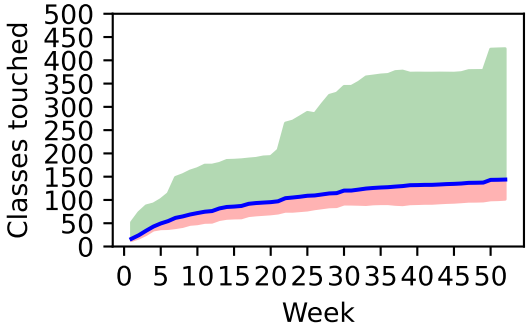
Figure 31: A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for all developers (n=2278).



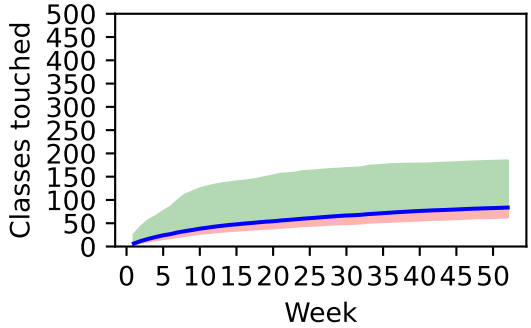
(a) Transient founder developers (n=28)



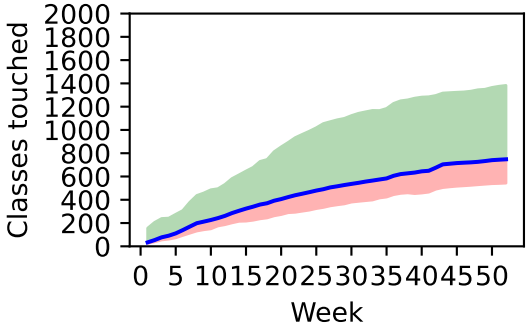
(b) Transient later joiner developers (n=1396)



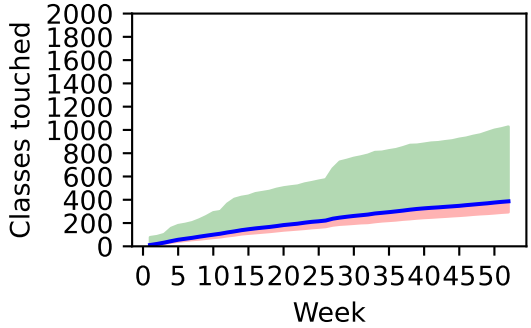
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 32: A time series of the average (mean) total classes touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

1.16 Time series components touched - For methods touched for commit

A time series of methods touched on average.

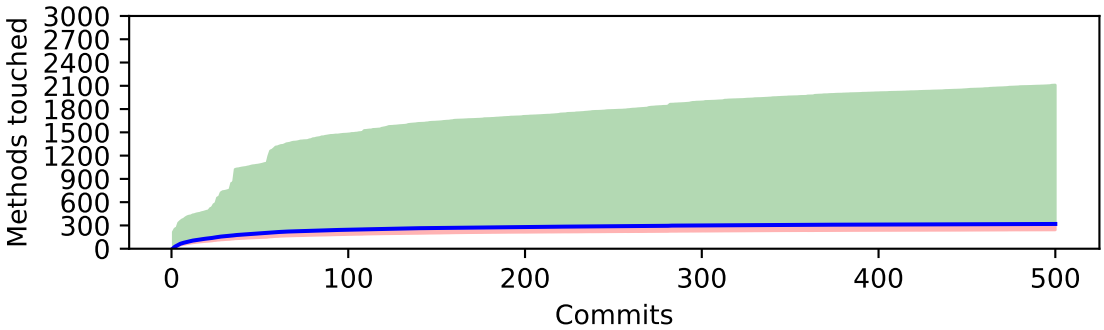
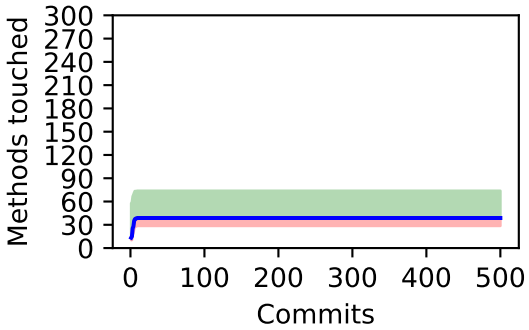
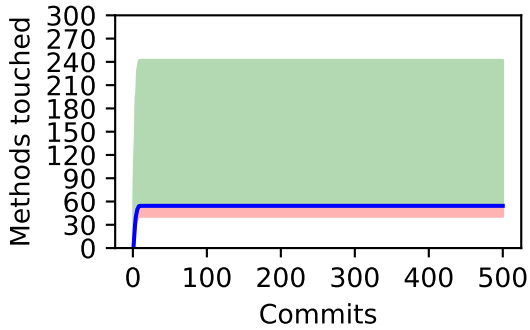


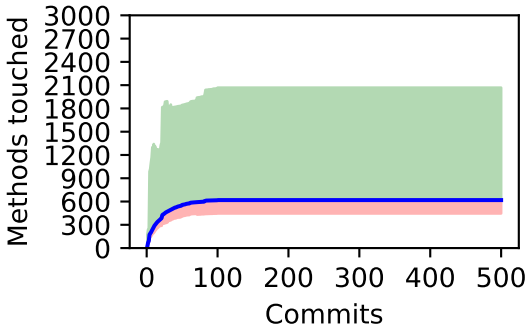
Figure 33: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



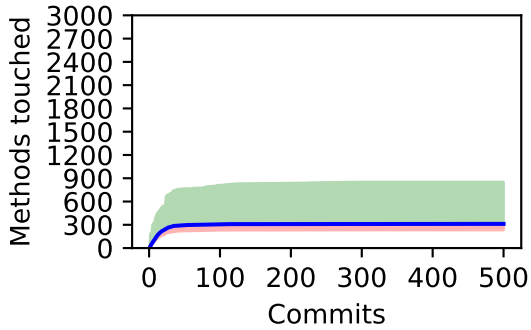
(a) Transient founder developers (n=28)



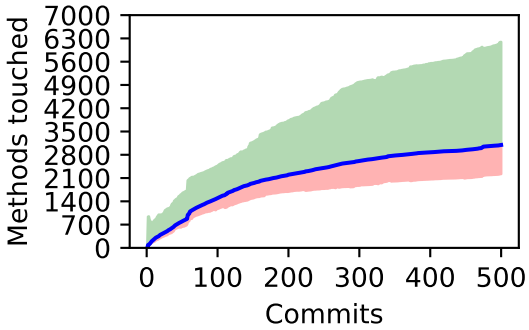
(b) Transient later joiner developers (n=1396)



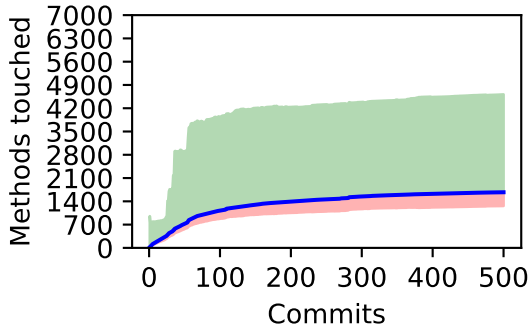
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 34: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.17 Time series components touched - For methods touched for each period

A time series of methods touched on average.

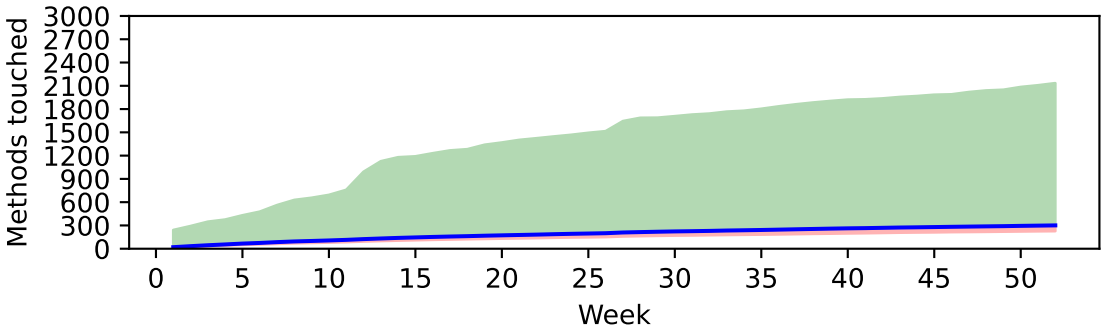
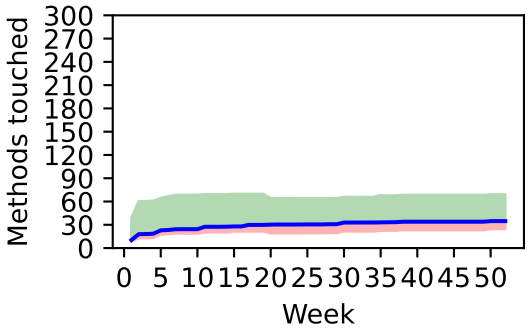
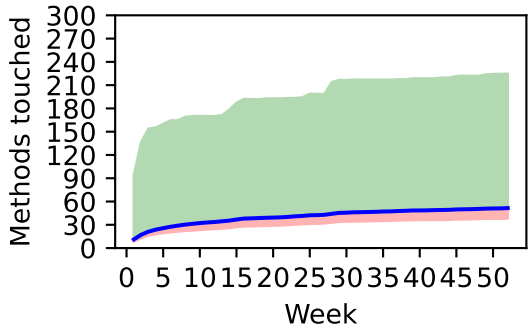


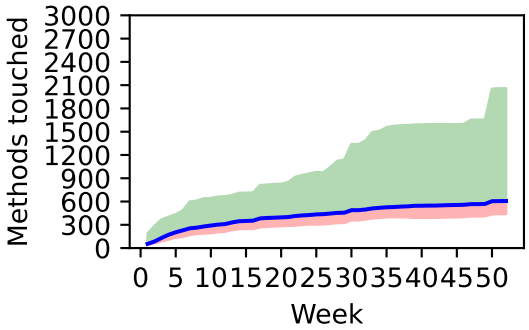
Figure 35: A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for all developers (n=2278).



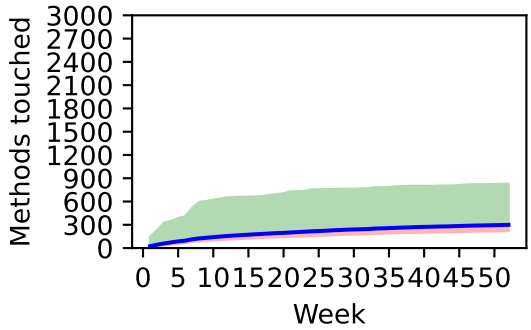
(a) Transient founder developers (n=28)



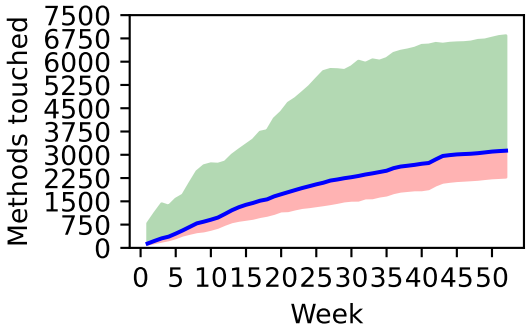
(b) Transient later joiner developers (n=1396)



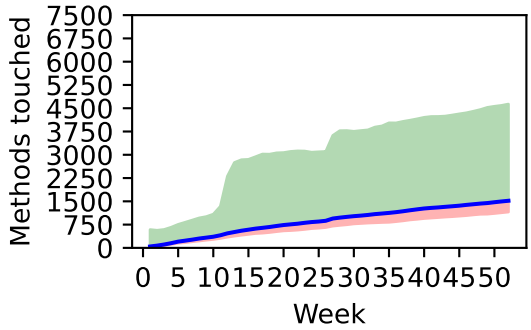
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 36: A time series of the average (mean) total methods touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

1.18 Time series components touched - For packages touched for first ten commits

A time series of packages touched on average.

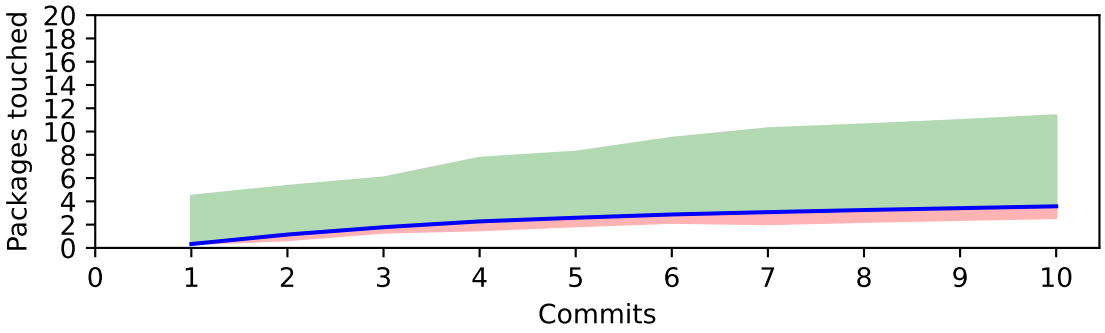
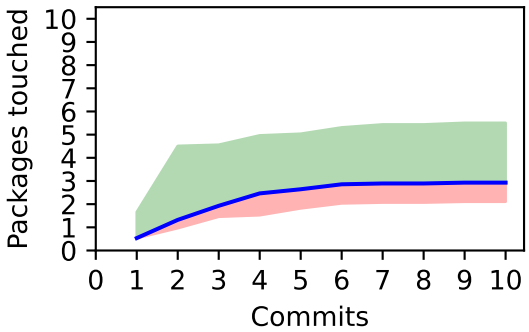
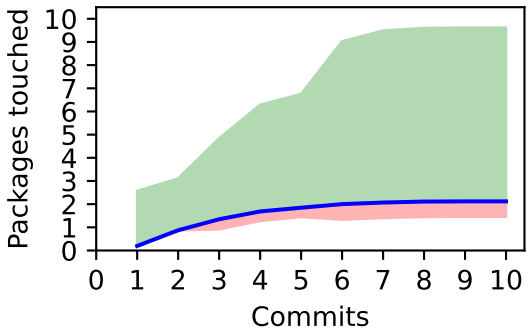


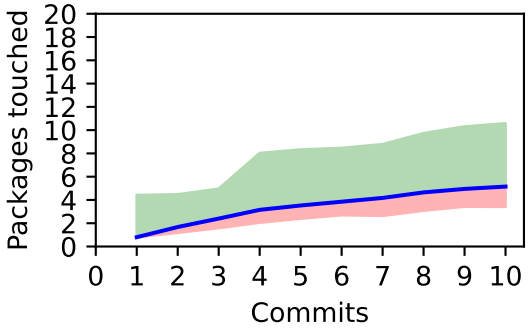
Figure 37: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



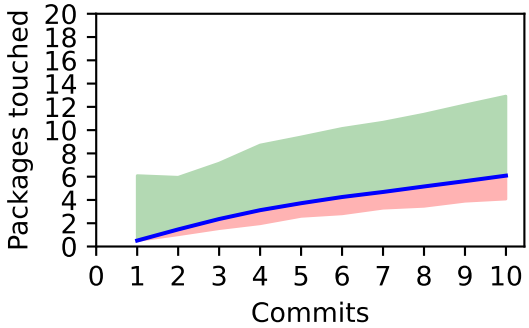
(a) Transient founder developers (n=28)



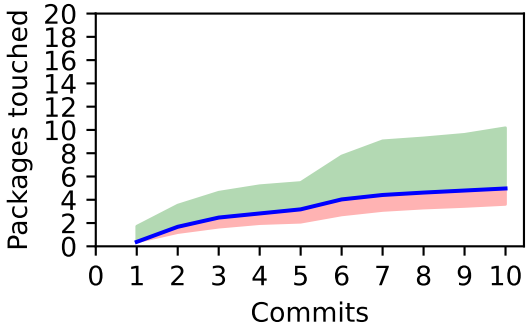
(b) Transient later joiner developers (n=1396)



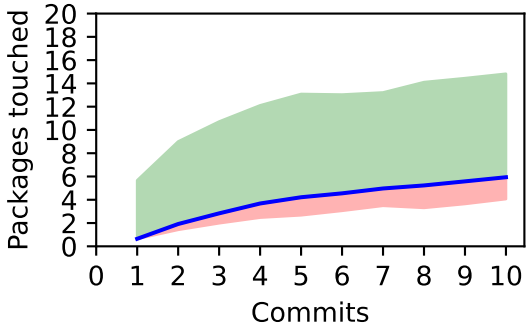
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 38: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.19 Time series components touched - For classes touched for first ten commits

A time series of classes touched on average.

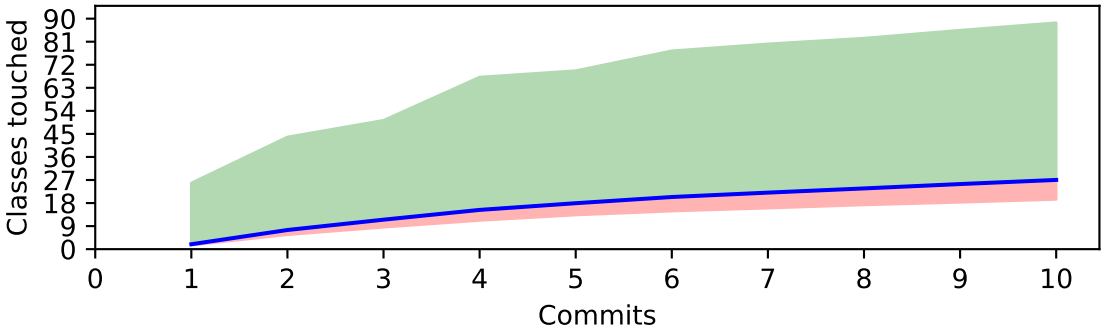
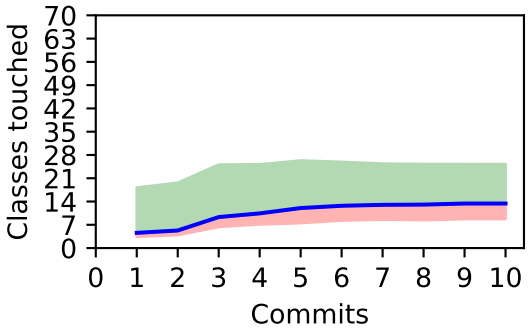
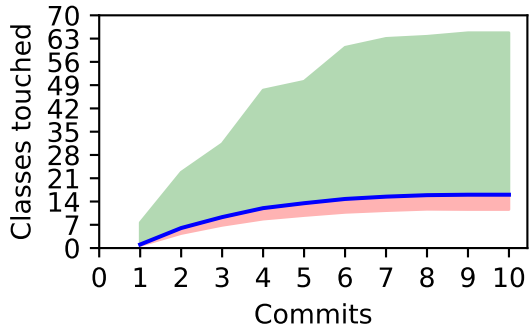


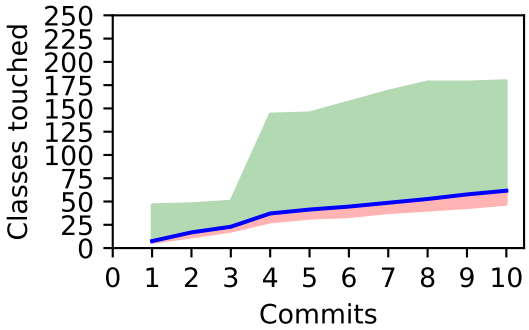
Figure 39: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



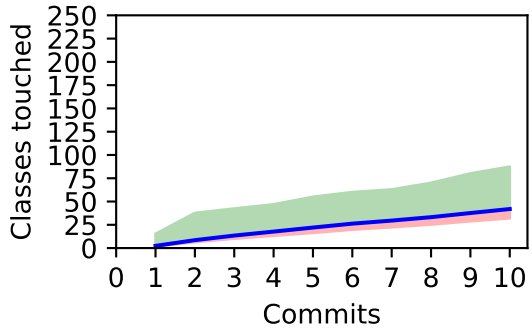
(a) Transient founder developers (n=28)



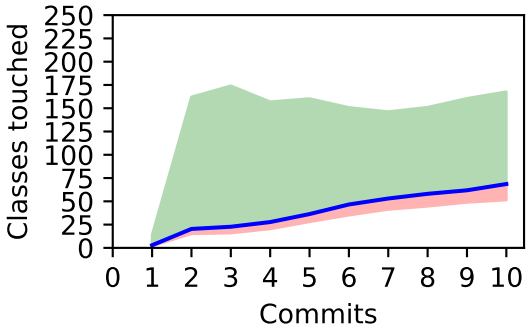
(b) Transient later joiner developers (n=1396)



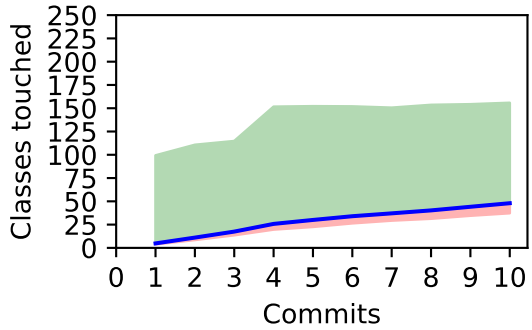
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 40: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.20 Time series components touched - For methods touched for first ten commits

A time series of methods touched on average.

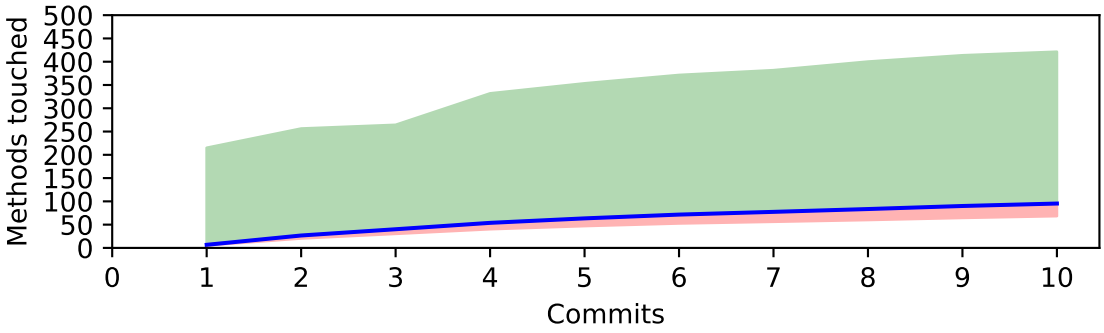
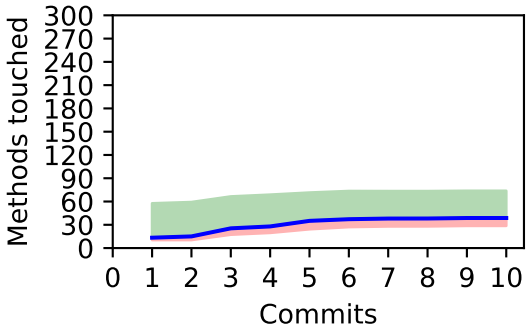
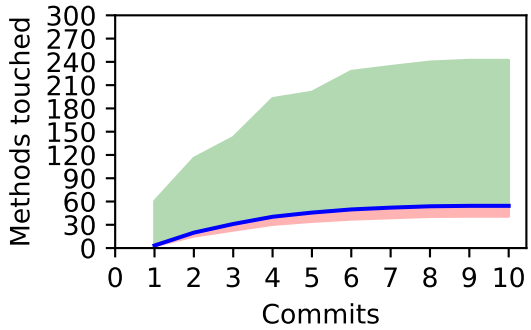


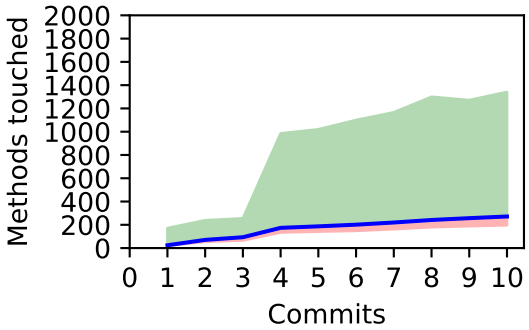
Figure 41: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for all developers (n=2278).



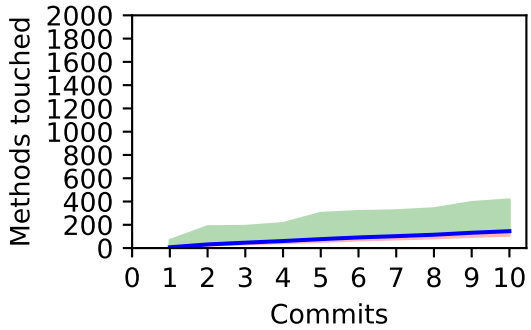
(a) Transient founder developers (n=28)



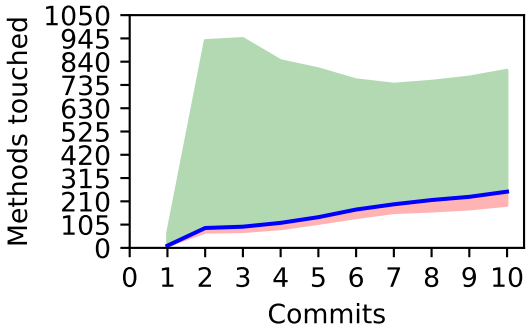
(b) Transient later joiner developers (n=1396)



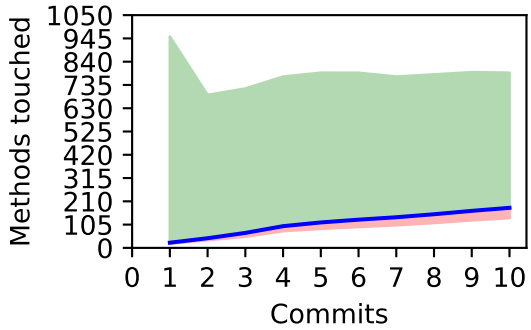
(c) Moderate founder developers (n=40)



(d) Moderate later joiner developers (n=580)



(e) Sustained founder developers (n=34)



(f) Sustained later joiner developers (n=203)

Figure 42: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

1.21 Anova generate - For packages touched for commit

An Anova table for packages touched on average.

Table 2: Two-Way ANOVA for packages touched by Developer Category

	Sum Sq	df	F	p-Value
Intercept	4708.9000	1.0000	19.8064	0.0000
C(Role)	0.7849	1.0000	0.0033	0.9542
C(Tenure)	7859.0036	2.0000	16.5281	0.0000
C(Role):C(Tenure)	214.4136	2.0000	0.4509	0.6371
Residual	540873.2417	2275.0000	NaN	NaN

1.22 Anova generate - For classes touched for commit

An Anova table for classes touched on average.

Table 3: Two-Way ANOVA for classes touched by Developer Category

	Sum Sq	df	F	p-Value
Intercept	856440.2250	1.0000	25.6264	0.0000
C(Role)	131122.2717	1.0000	3.9234	0.0477
C(Tenure)	10015617.6837	2.0000	149.8438	0.0000
C(Role):C(Tenure)	1539169.1454	2.0000	23.0275	0.0000
Residual	76030919.0919	2275.0000	NaN	NaN

1.23 Anova generate - For methods touched for commit

An Anova table for methods touched on average.

Table 4: Two-Way ANOVA for methods touched by Developer Category

	Sum Sq	df	F	p-Value
Intercept	15301690.0000	1.0000	26.3553	0.0000
C(Role)	3522420.2893	1.0000	6.0669	0.0138
C(Tenure)	172665904.0147	2.0000	148.6979	0.0000
C(Role):C(Tenure)	33151234.0089	2.0000	28.5495	0.0000
Residual	1320848642.9145	2275.0000	NaN	NaN

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{1}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).

- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{2}$$

2 Repository: 14

2.1 Time series components touched - 14 For packages touched for commit

A time series of packages touched on average.

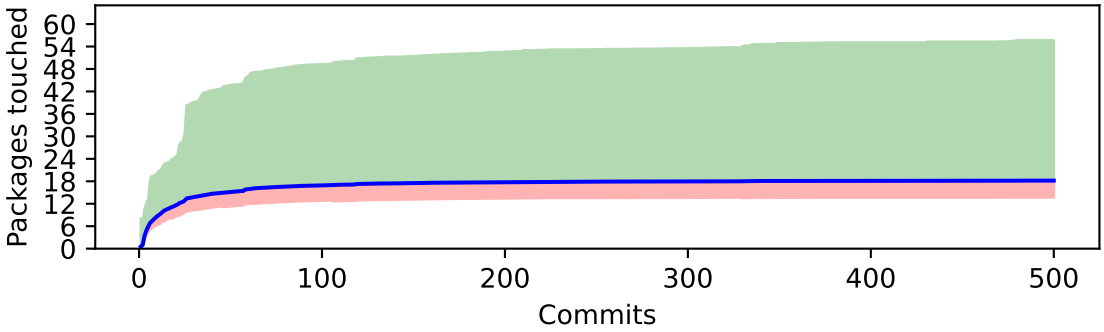
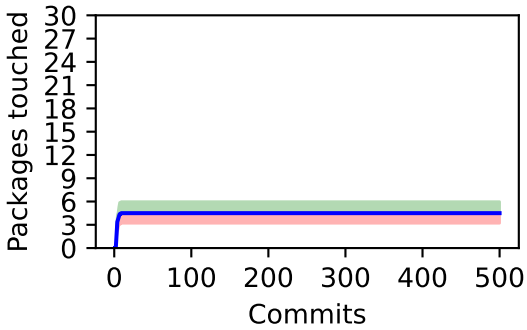
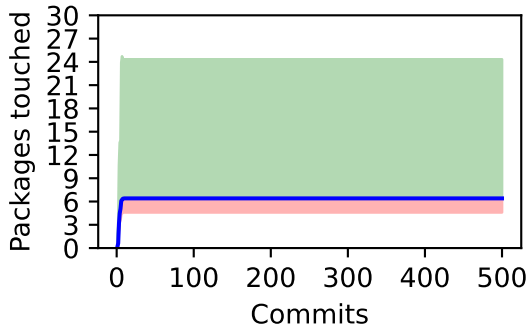


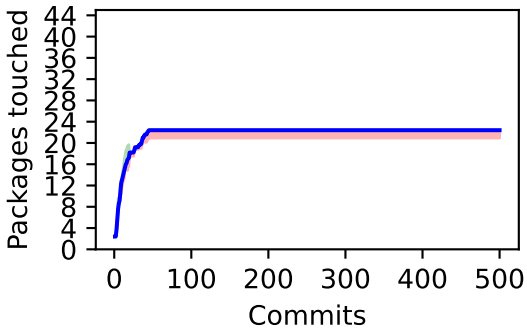
Figure 43: Repository 14 A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for all developers (n=244).



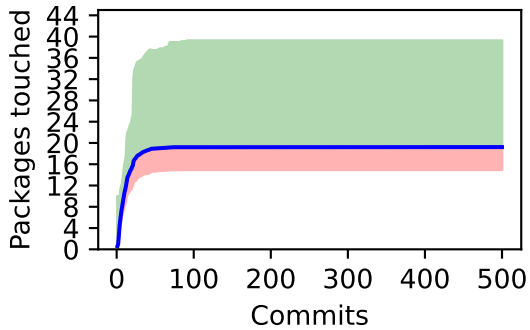
(a) Transient founder developers (n=8)



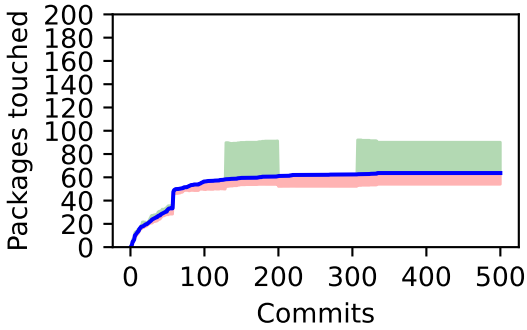
(b) Transient later joiner developers (n=119)



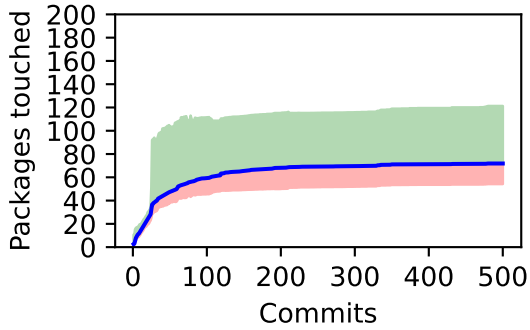
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 44: Repository 14 A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

2.2 Time series components touched - 14 For packages touched for each period

A time series of packages touched on average.

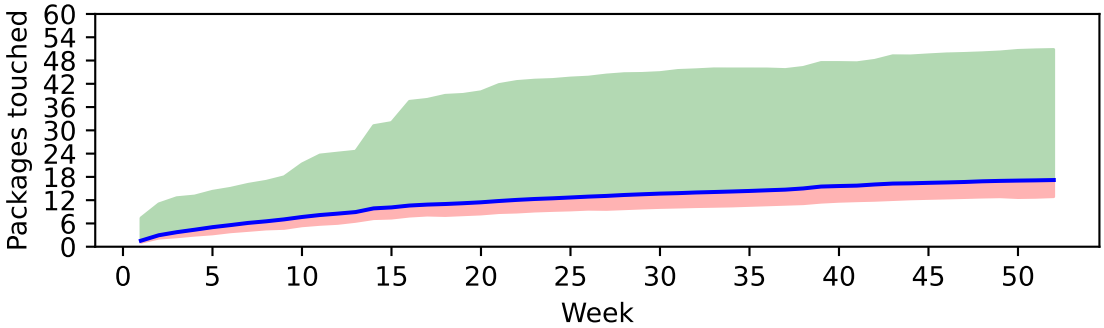
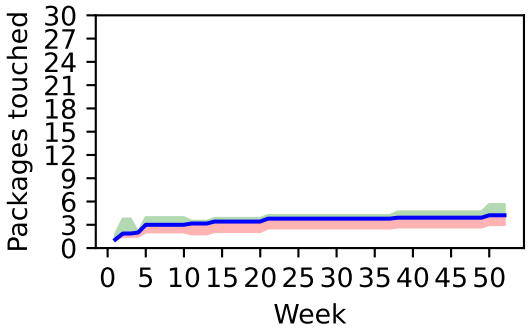
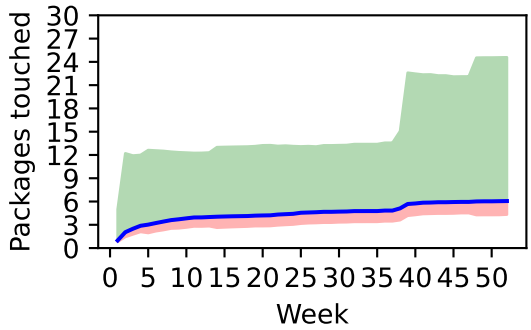


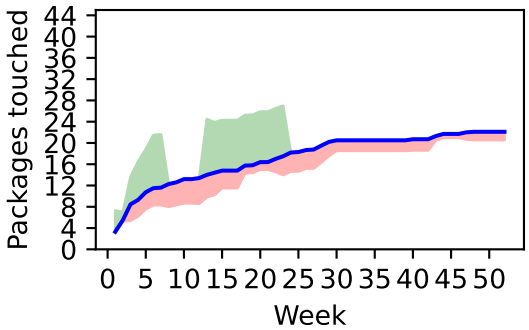
Figure 45: Repository 14 A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for all developers (n=244).



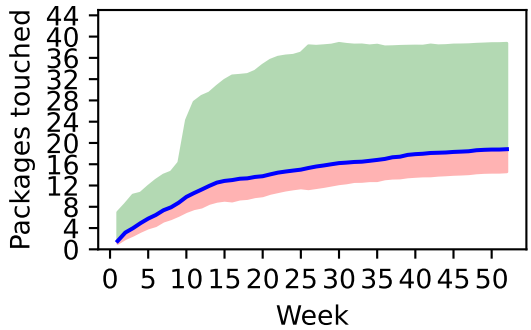
(a) Transient founder developers (n=8)



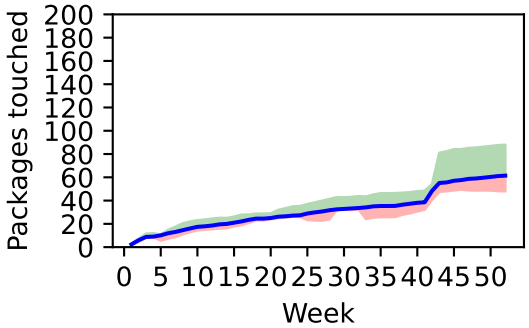
(b) Transient later joiner developers (n=119)



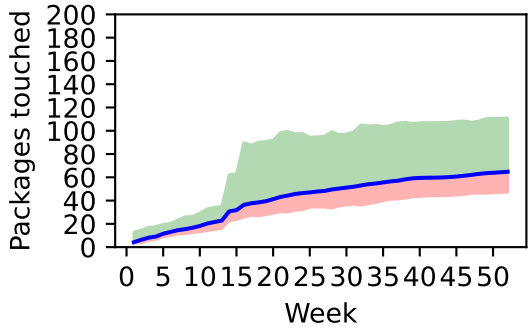
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 46: Repository 14 A time series of the average (mean) total packages touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

2.3 Time series components touched - 14 For classes touched for commit

A time series of classes touched on average.

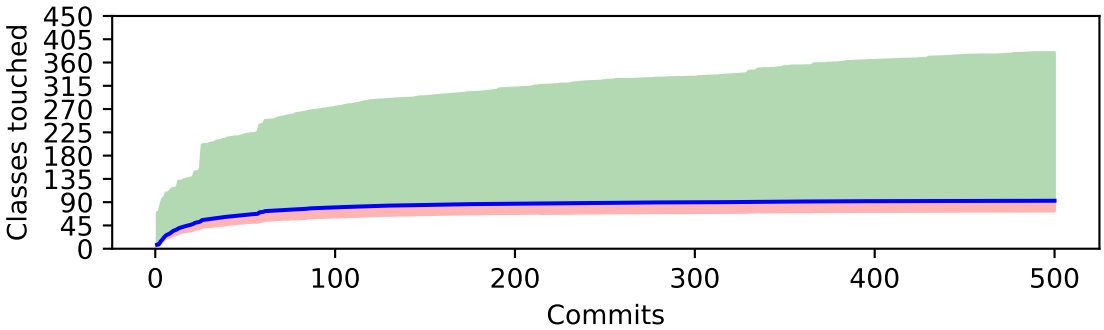
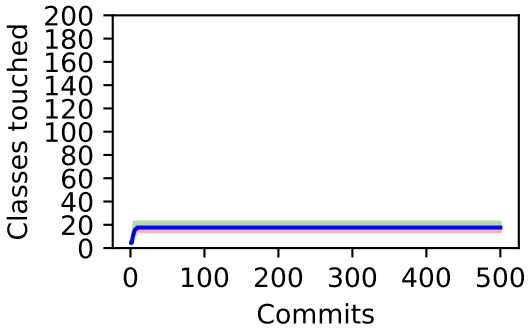
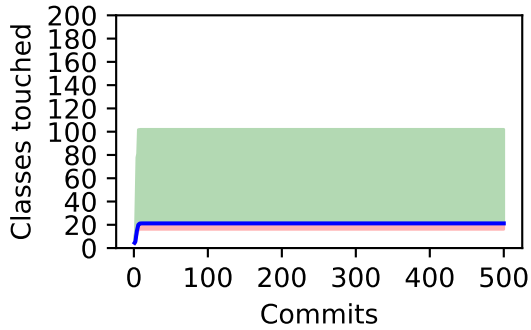


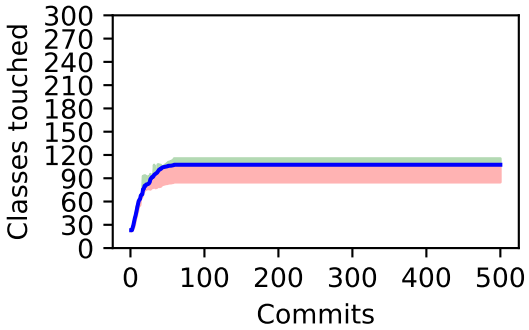
Figure 47: Repository 14 A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for all developers (n=244).



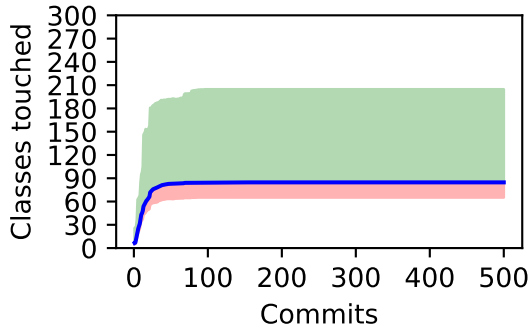
(a) Transient founder developers (n=8)



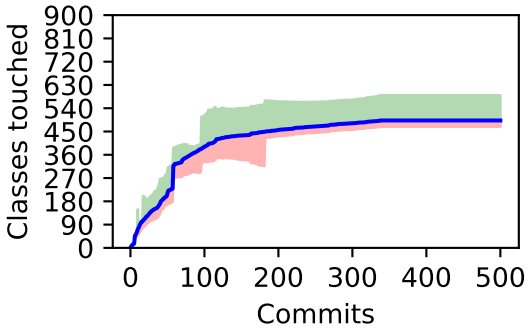
(b) Transient later joiner developers (n=119)



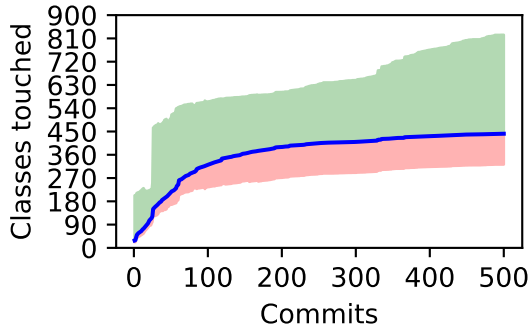
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 48: Repository 14 A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

2.4 Time series components touched - 14 For classes touched for each period

A time series of classes touched on average.

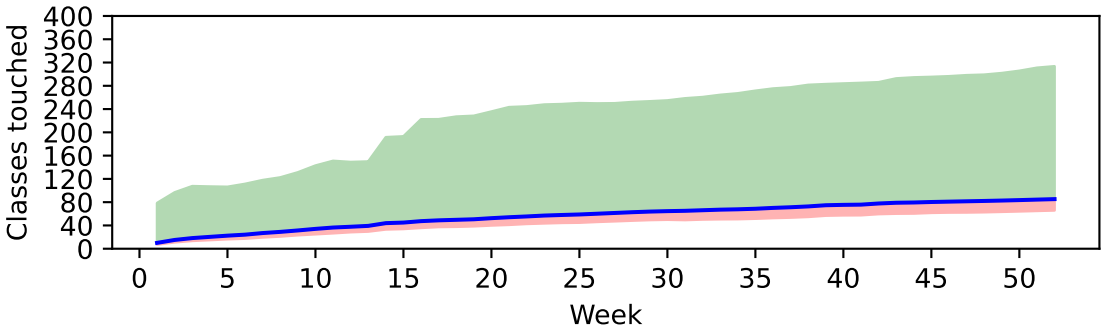
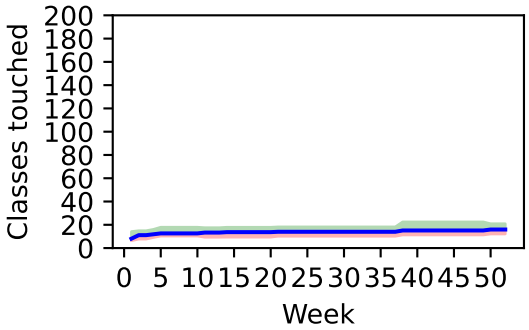
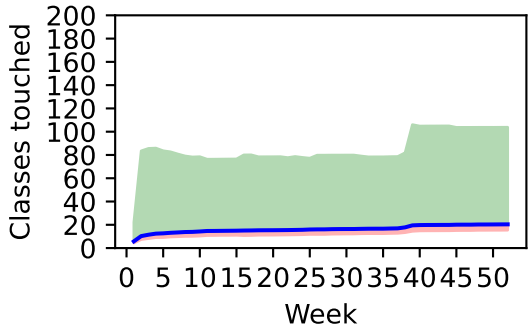


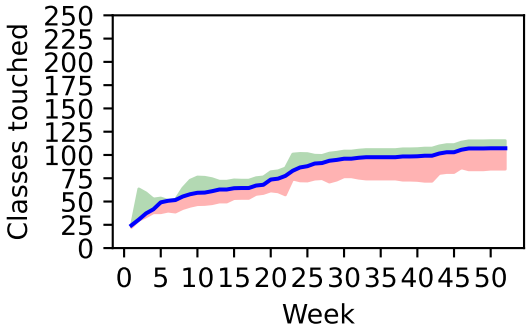
Figure 49: Repository 14 A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for all developers (n=244).



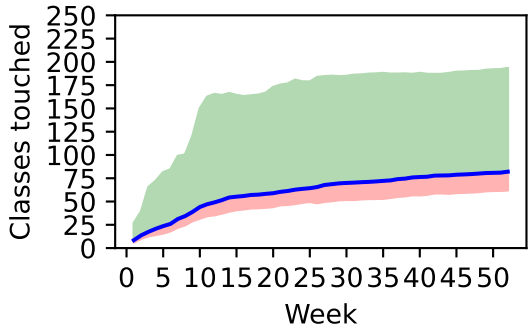
(a) Transient founder developers (n=8)



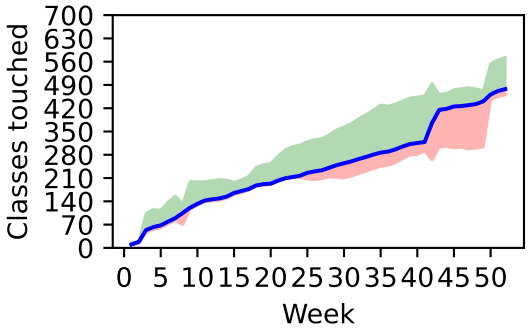
(b) Transient later joiner developers (n=119)



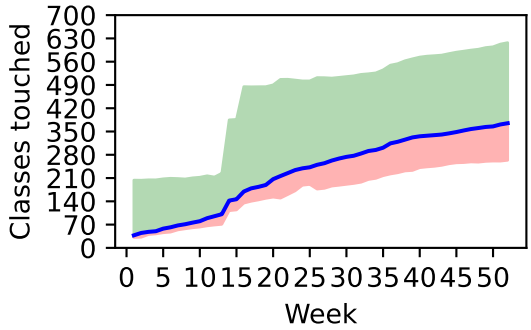
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 50: Repository 14 A time series of the average (mean) total classes touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

2.5 Time series components touched - 14 For methods touched for commit

A time series of methods touched on average.

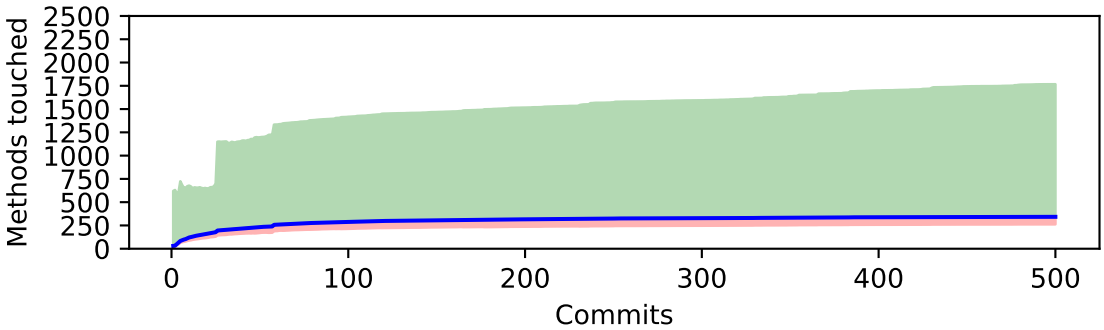
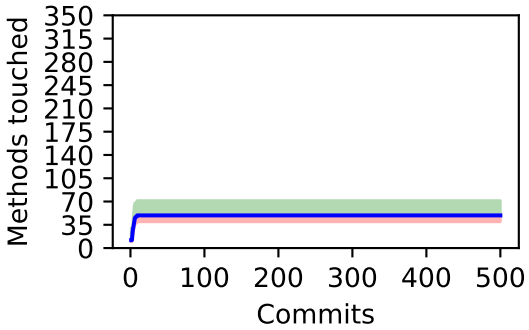
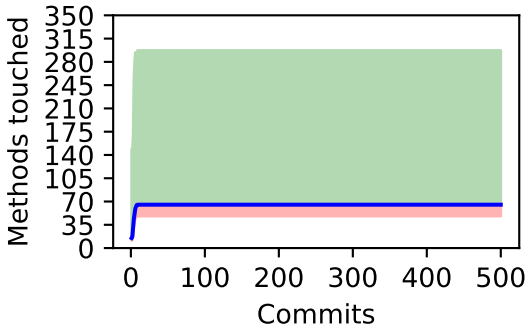


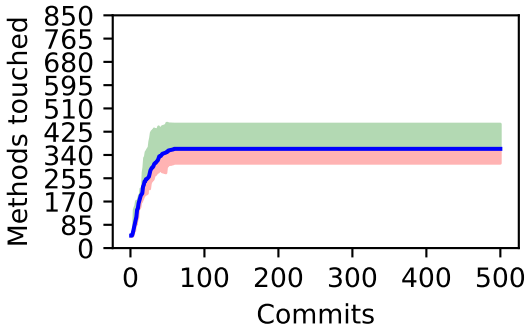
Figure 51: Repository 14 A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for all developers (n=244).



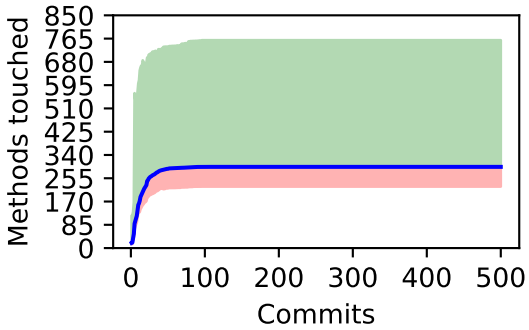
(a) Transient founder developers (n=8)



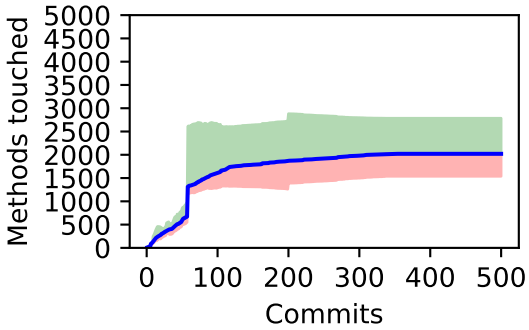
(b) Transient later joiner developers (n=119)



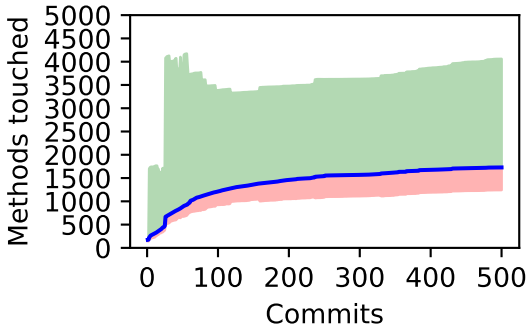
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 52: Repository 14 A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

2.6 Time series components touched - 14 For methods touched for each period

A time series of methods touched on average.

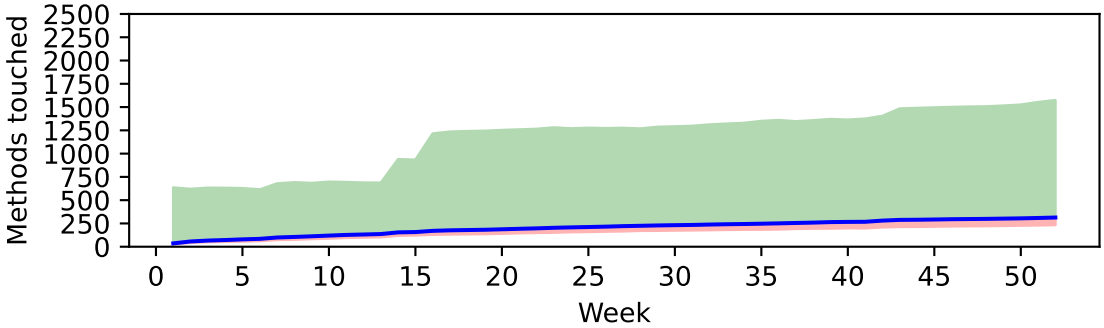
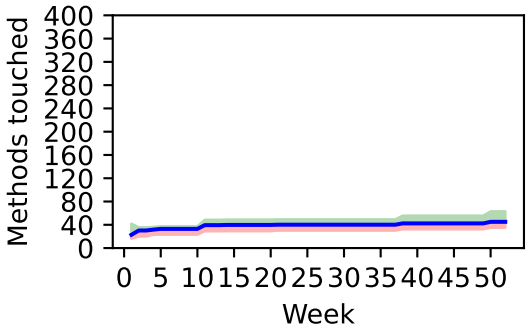
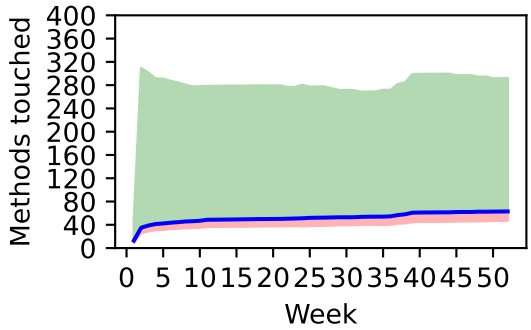


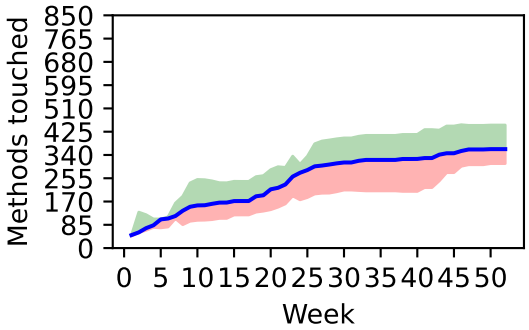
Figure 53: Repository 14 A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for all developers (n=244).



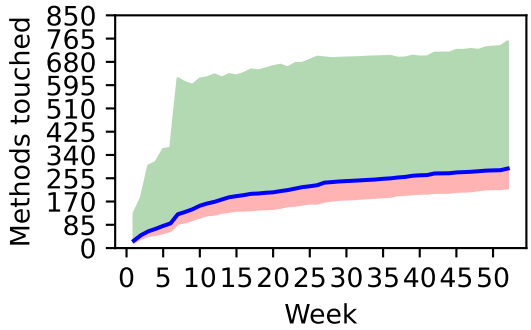
(a) Transient founder developers (n=8)



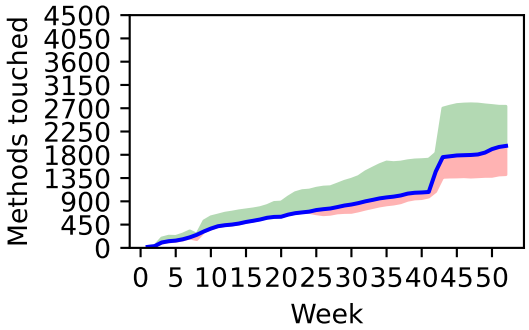
(b) Transient later joiner developers (n=119)



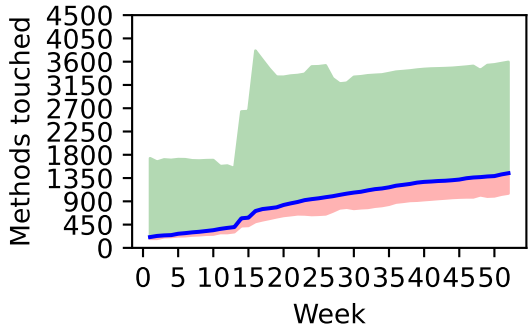
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 54: Repository 14 A time series of the average (mean) total methods touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

2.7 Time series components touched - 14 For packages touched for commit

A time series of packages touched on average.

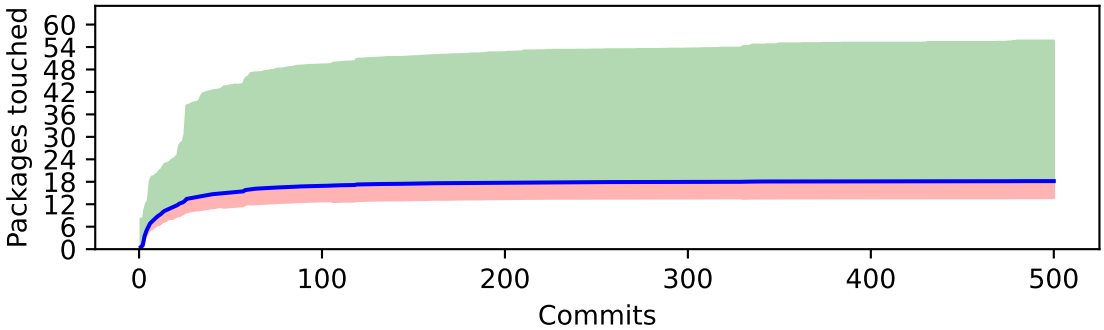
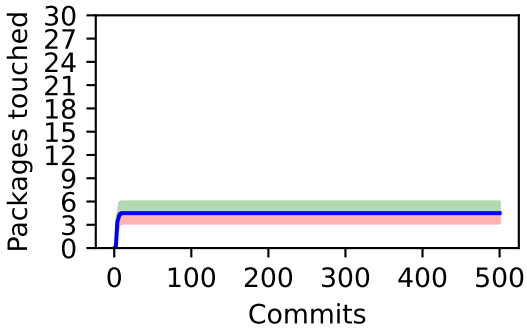
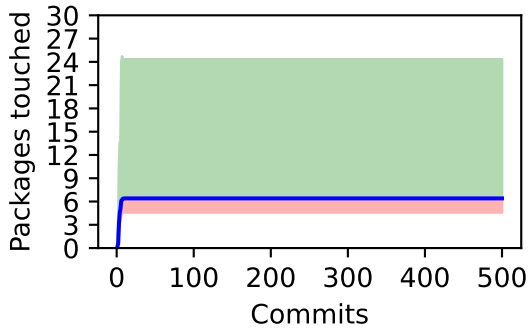


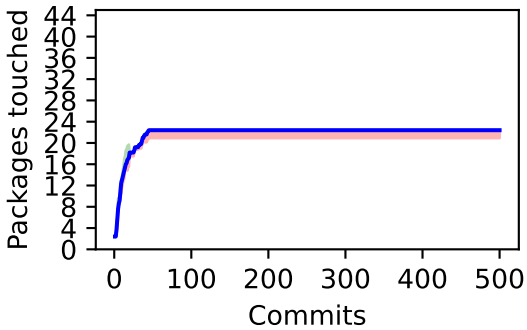
Figure 55: Repository 14 A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for all developers (n=244).



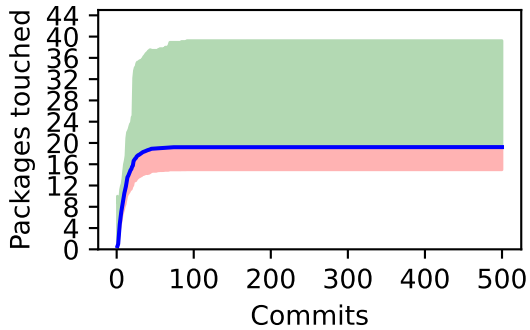
(a) Transient founder developers (n=8)



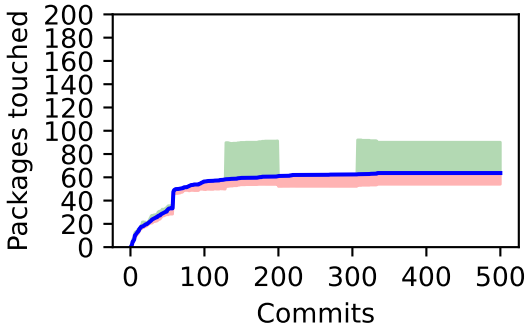
(b) Transient later joiner developers (n=119)



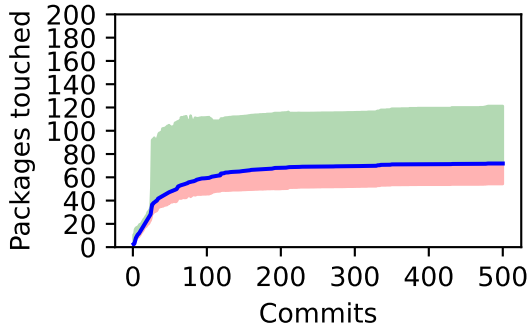
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 56: Repository 14 A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

2.8 Time series components touched - 14 For packages touched for each period

A time series of packages touched on average.

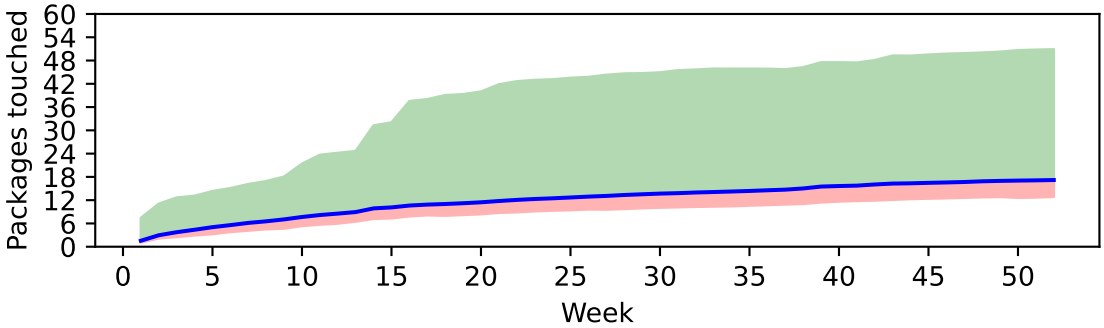
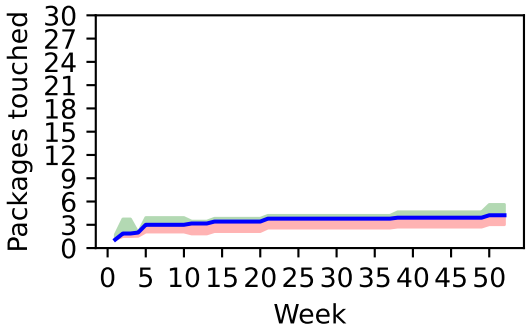
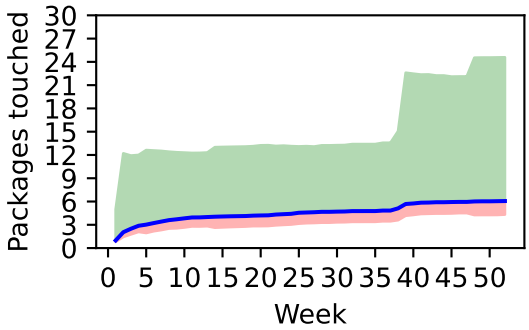


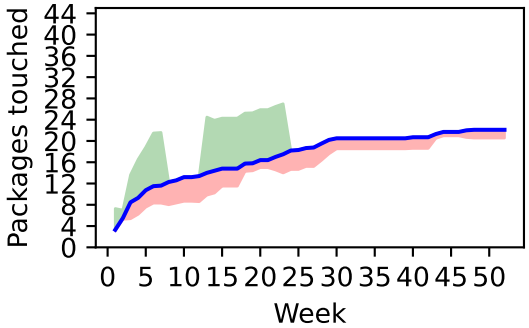
Figure 57: Repository 14 A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for all developers (n=244).



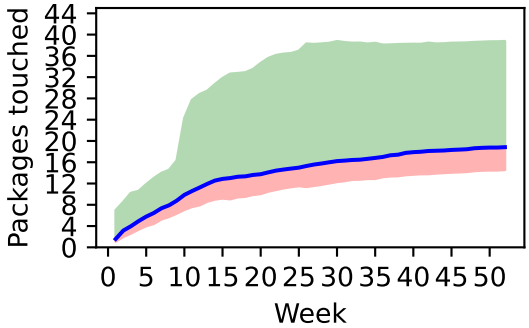
(a) Transient founder developers (n=8)



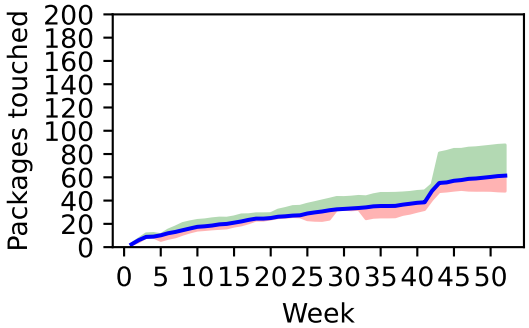
(b) Transient later joiner developers (n=119)



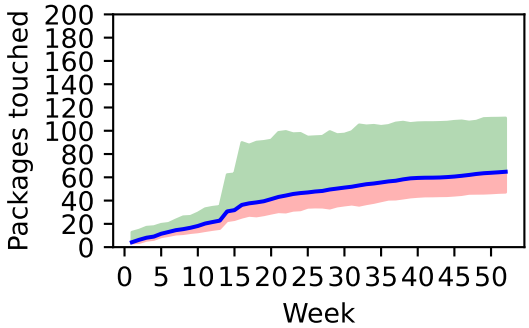
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 58: Repository 14 A time series of the average (mean) total packages touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

2.9 Time series components touched - 14 For classes touched for commit

A time series of classes touched on average.

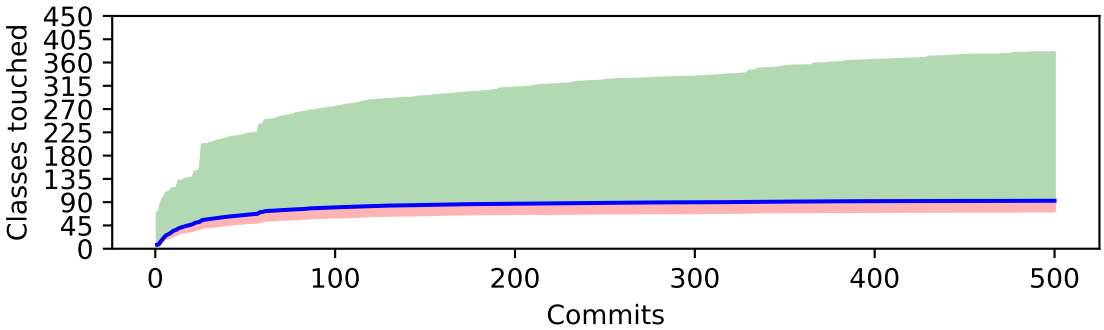
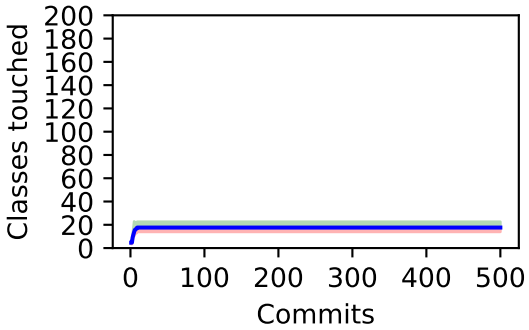
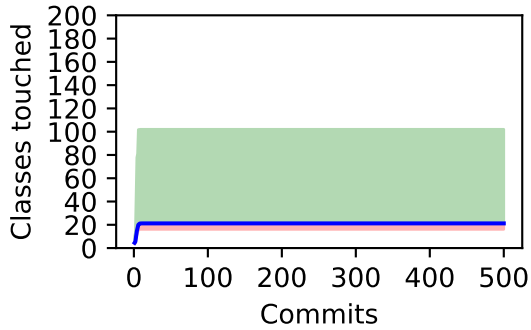


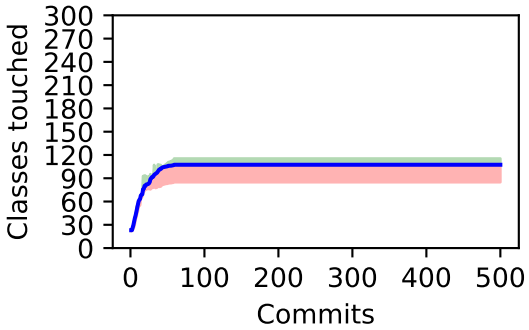
Figure 59: Repository 14 A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for all developers (n=244).



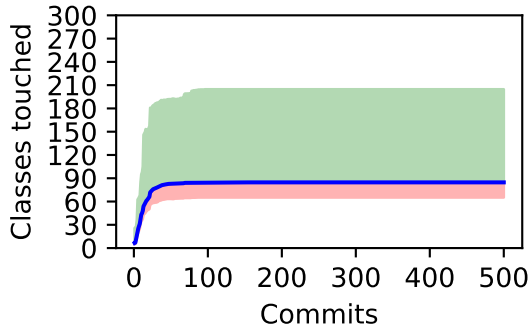
(a) Transient founder developers (n=8)



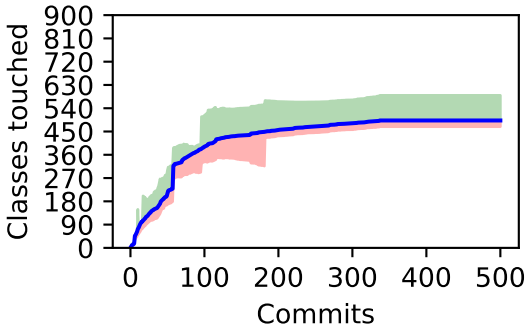
(b) Transient later joiner developers (n=119)



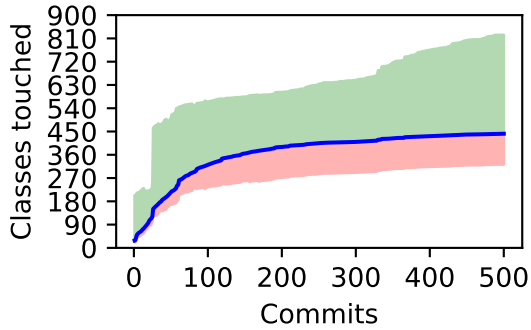
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 60: Repository 14 A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

2.10 Time series components touched - 14 For classes touched for each period

A time series of classes touched on average.

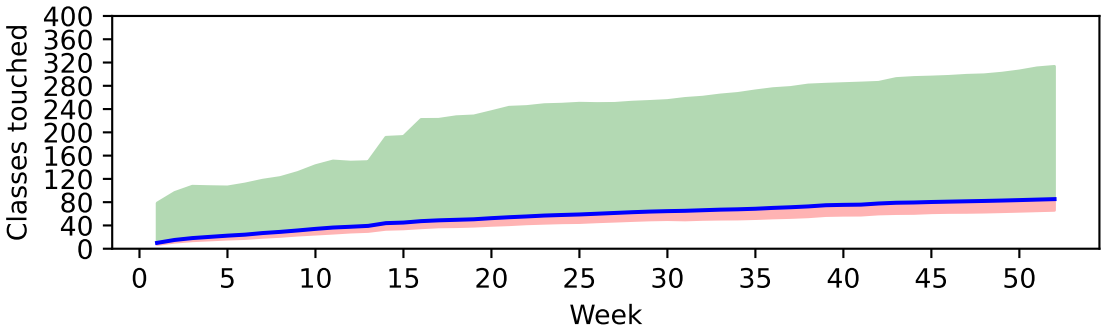
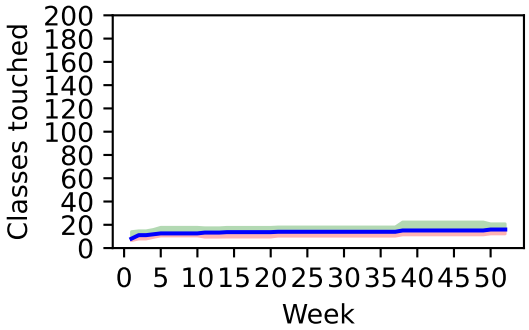
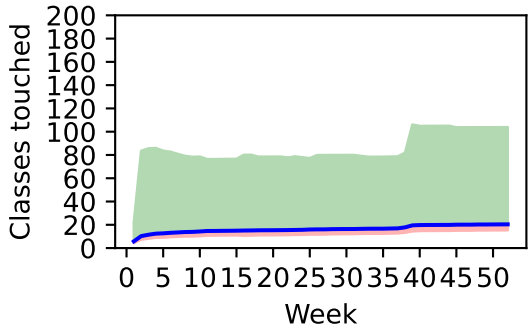


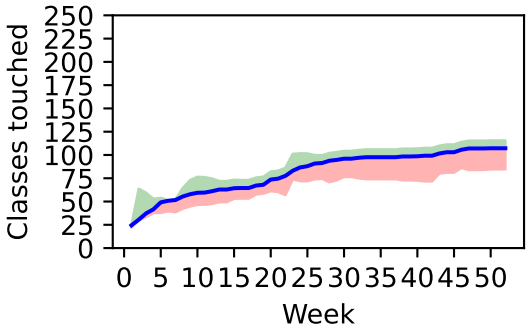
Figure 61: Repository 14 A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for all developers (n=244).



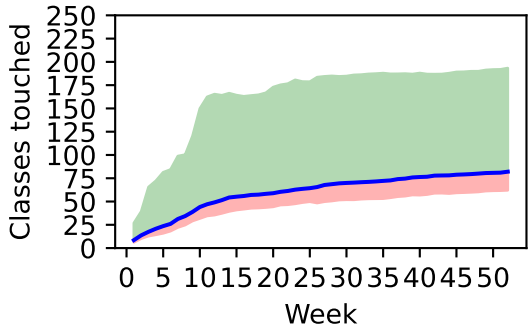
(a) Transient founder developers (n=8)



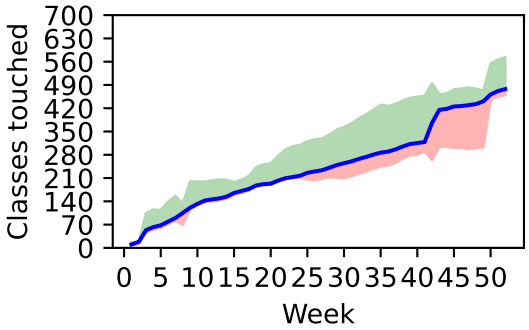
(b) Transient later joiner developers (n=119)



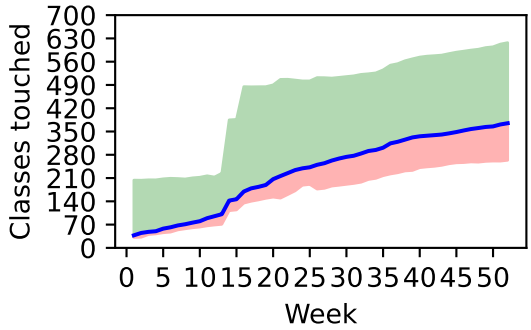
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 62: Repository 14 A time series of the average (mean) total classes touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

2.11 Time series components touched - 14 For methods touched for commit

A time series of methods touched on average.

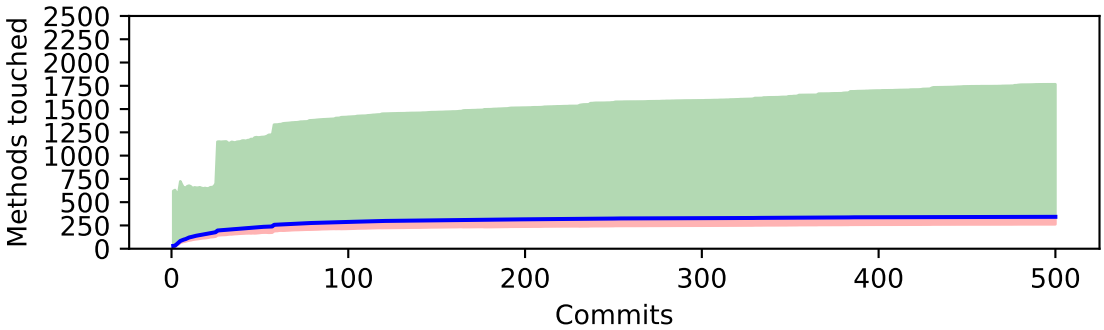
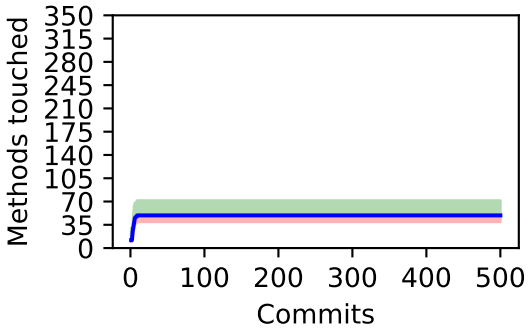
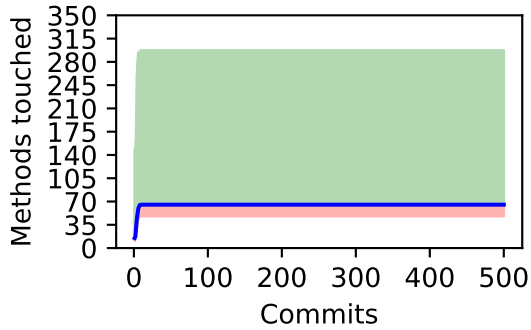


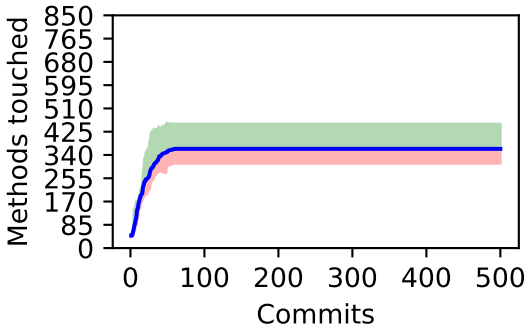
Figure 63: Repository 14 A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for all developers (n=244).



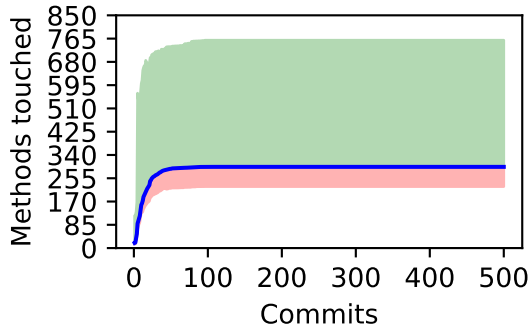
(a) Transient founder developers (n=8)



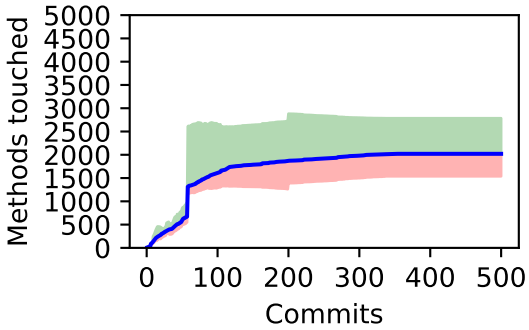
(b) Transient later joiner developers (n=119)



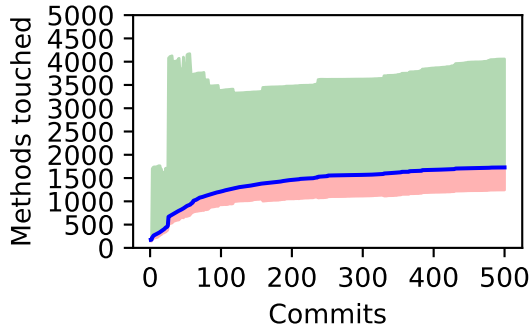
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 64: Repository 14 A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

2.12 Time series components touched - 14 For methods touched for each period

A time series of methods touched on average.

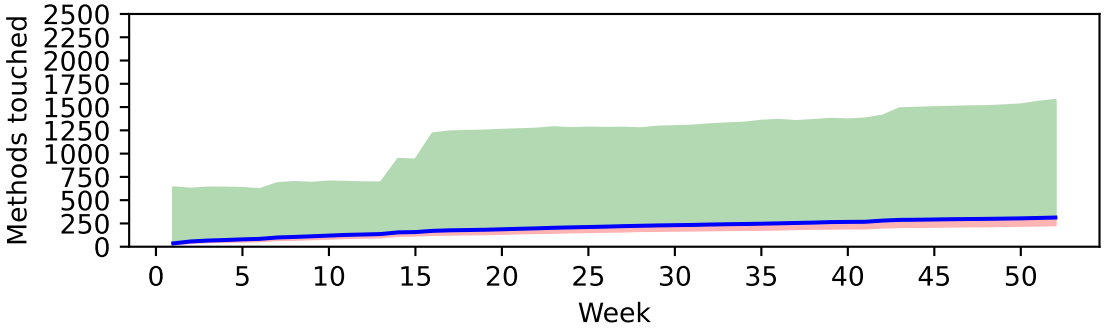
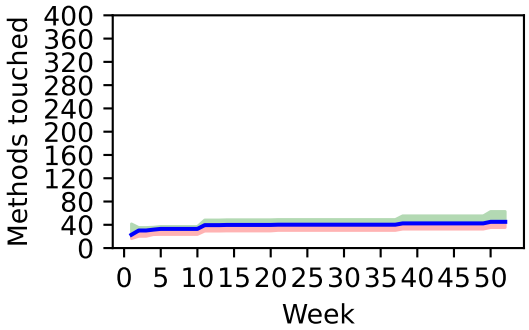
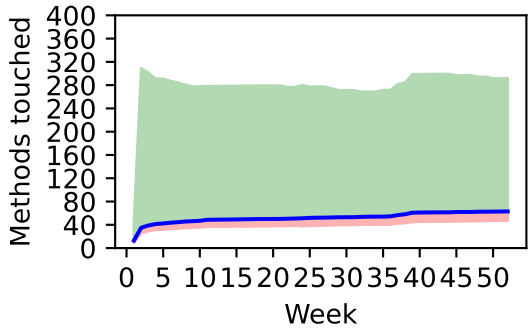


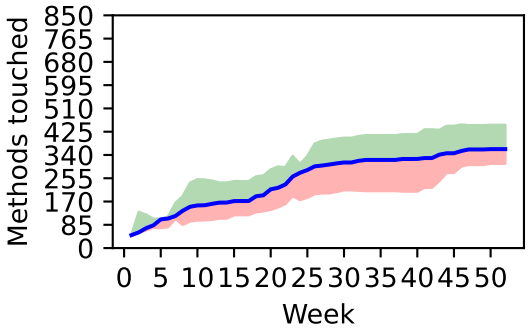
Figure 65: Repository 14 A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for all developers (n=244).



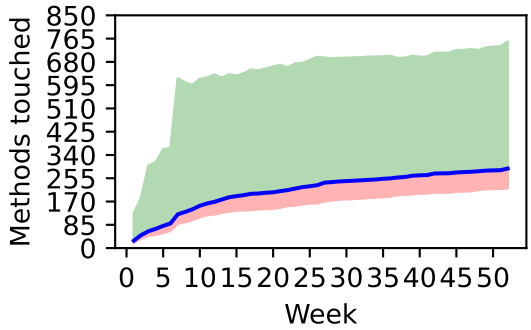
(a) Transient founder developers (n=8)



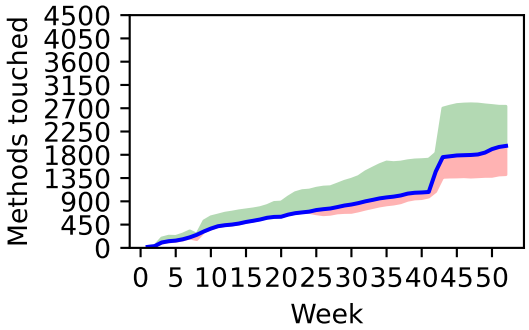
(b) Transient later joiner developers (n=119)



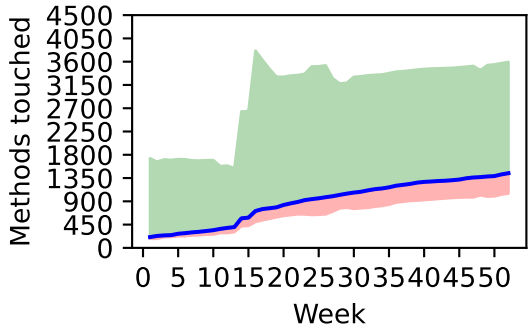
(c) Moderate founder developers (n=5)



(d) Moderate later joiner developers (n=85)



(e) Sustained founder developers (n=6)



(f) Sustained later joiner developers (n=21)

Figure 66: Repository 14 A time series of the average (mean) total methods touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

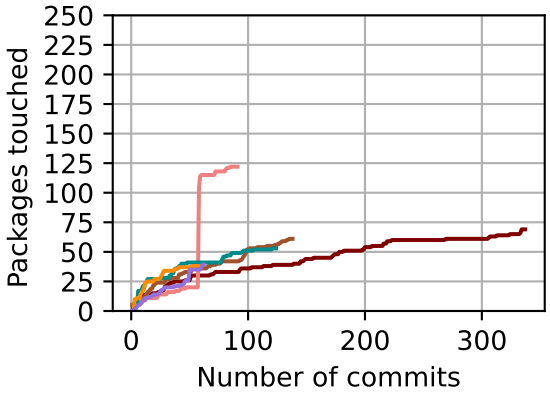
2.13 Time series sustained components touched by developer - 14 For developer commits and components touched.

Table 5: Table of first 6 sustained founder developers in the Scatter Developer graphs.

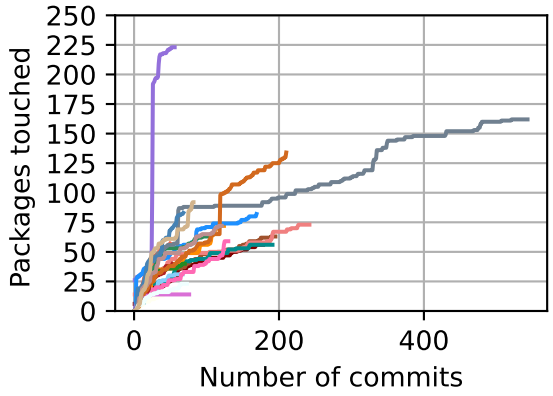
Developer ID	Colour
2930	Maroon
2932	Sienna
2934	LightCoral
2935	DarkCyan
2946	DarkOrange
2959	MediumPurple

Table 6: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

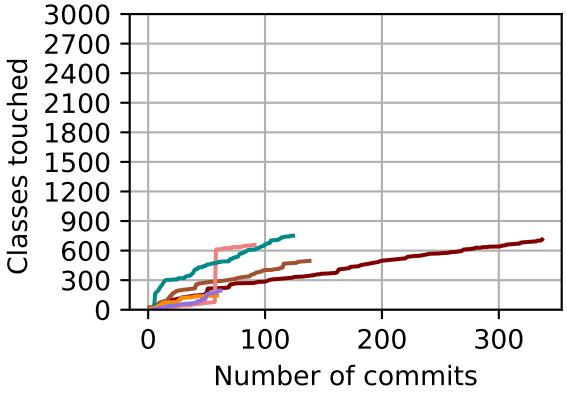
Developer ID	**Colour**
1774	Maroon
2986	Sienna
3000	LightCoral
3020	DarkCyan
3037	DarkOrange
3050	MediumPurple
3071	LightSkyBlue
3072	HotPink
3141	ForestGreen
3143	FireBrick
3163	DodgerBlue
3172	Tomato
3178	SeaGreen
3216	SlateGray
3217	Ivory
3272	Orchid
3339	Chocolate
3354	SteelBlue
3456	RosyBrown
3473	Azure
3485	Tan



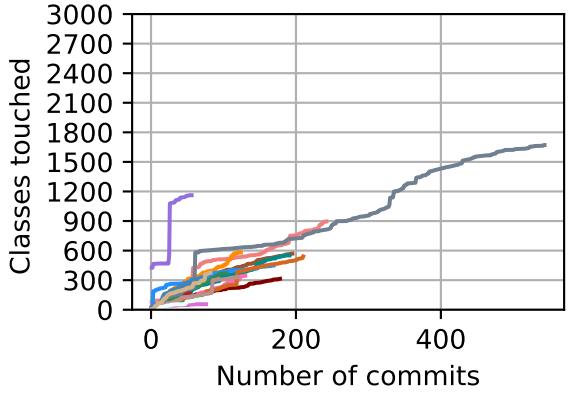
(a) Packages for sustained founder



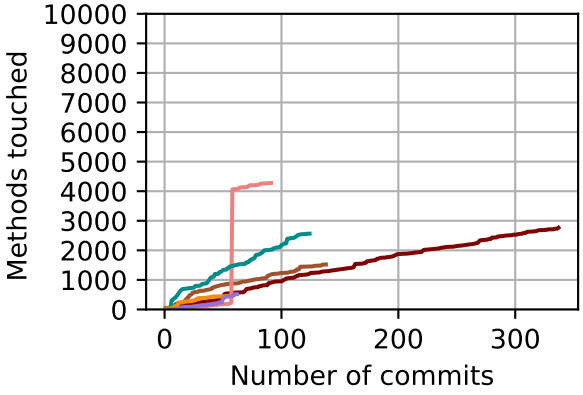
(b) Packages for sustained later joiner



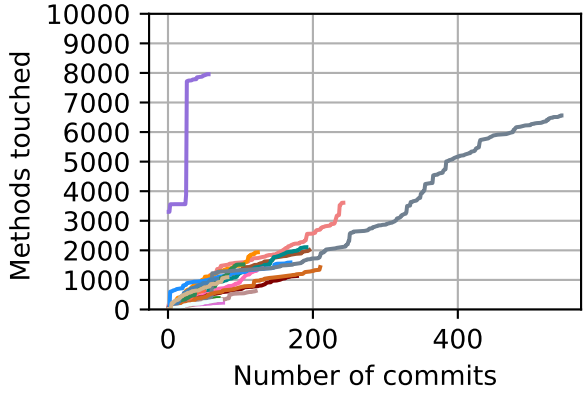
(c) Classes for sustained founder



(d) Classes for sustained later joiner



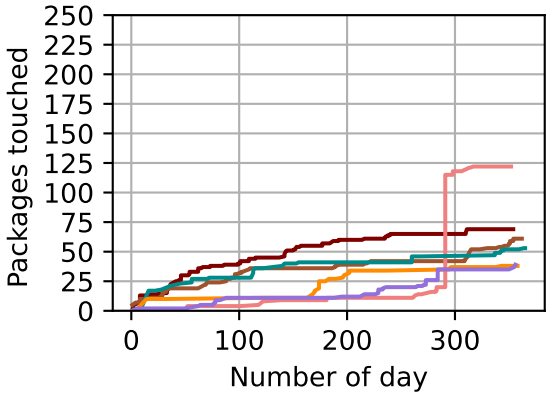
(e) Methods for sustained founder



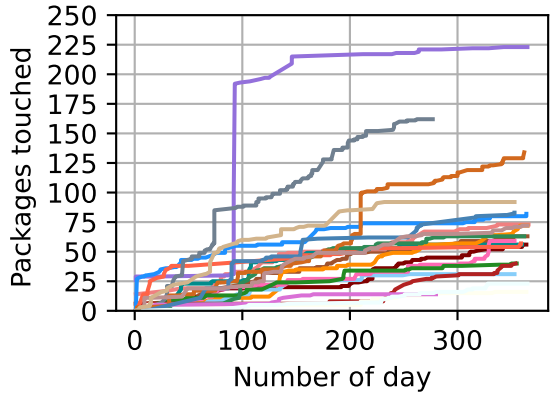
(f) Methods for sustained later joiner

Figure 67: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 5. Sustained later joiner developers with colours are shown in Table 6.

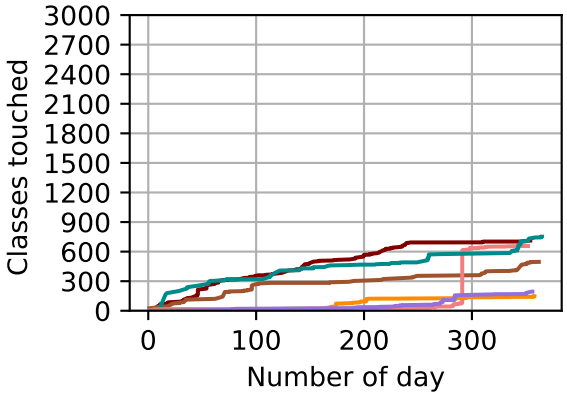
2.14 Time series sustained components touched by developer - 14 For developer day and components touched.



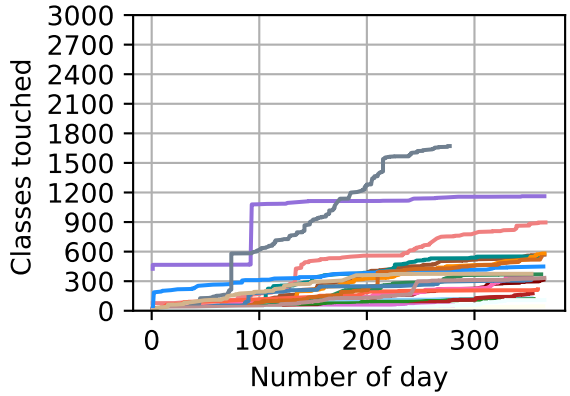
(a) Packages for sustained founder



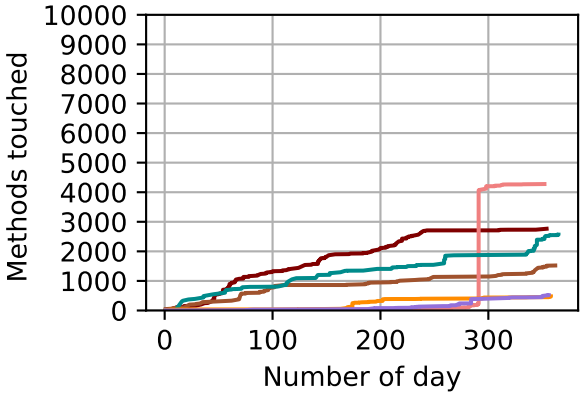
(b) Packages for sustained later joiner



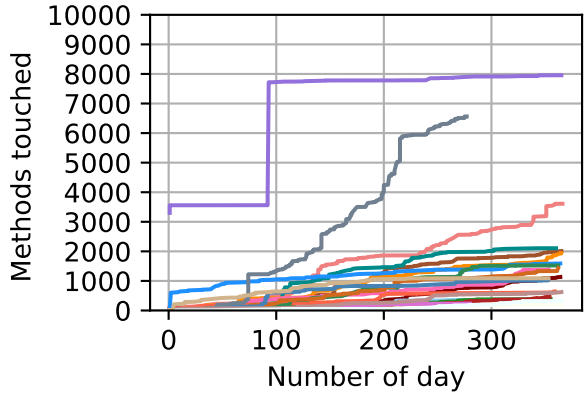
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 68: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 5. Sustained later joiner developers with colours are shown in Table 6.

3 Repository: 21

3.1 Time series components touched - 21 For packages touched for commit

A time series of packages touched on average.

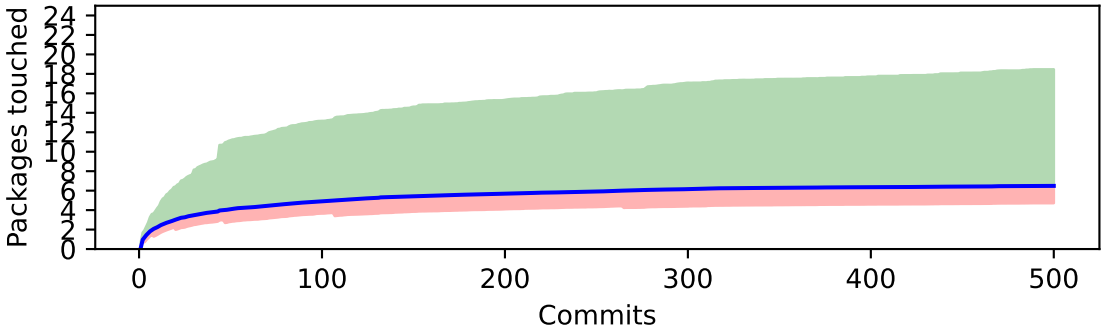
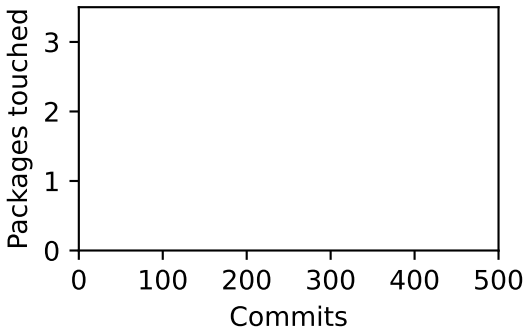
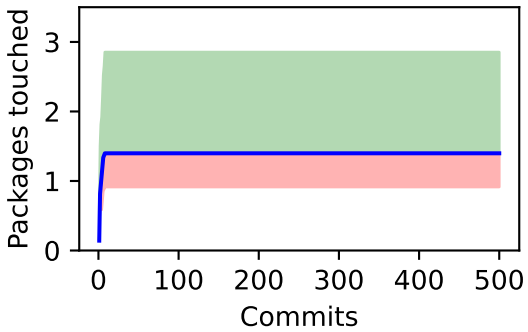


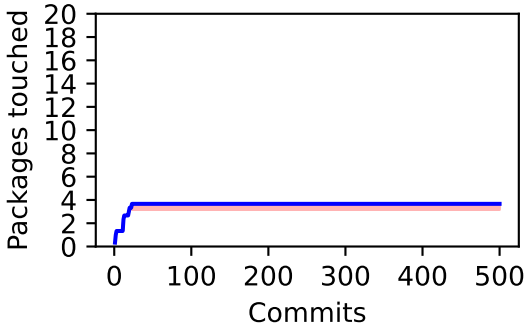
Figure 69: Repository 21 A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for all developers (n=300).



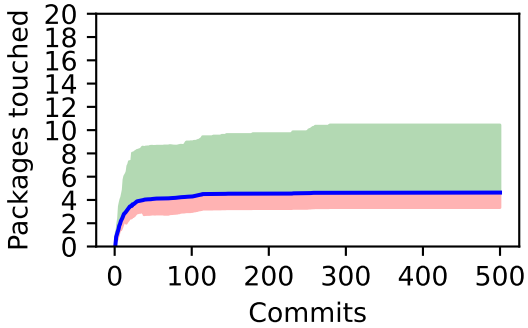
(a) Transient founder developers (n=0)



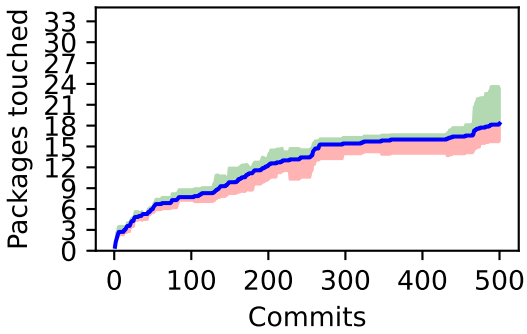
(b) Transient later joiner developers (n=113)



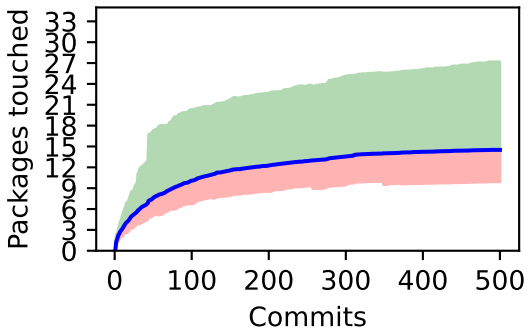
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 70: Repository 21 A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

3.2 Time series components touched - 21 For packages touched for each period

A time series of packages touched on average.

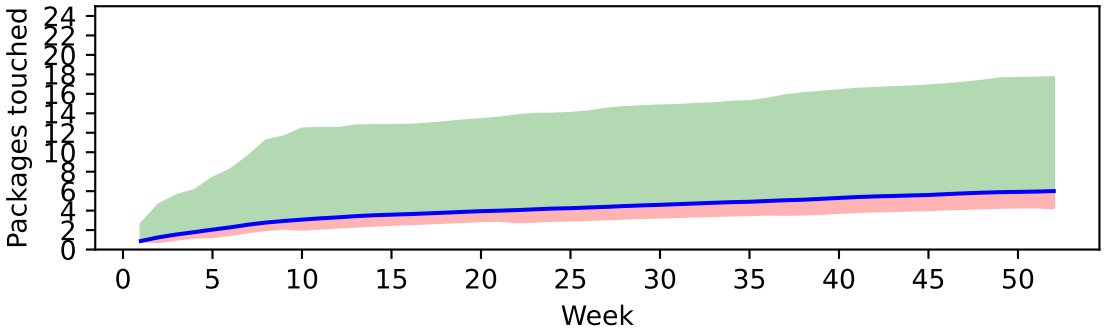
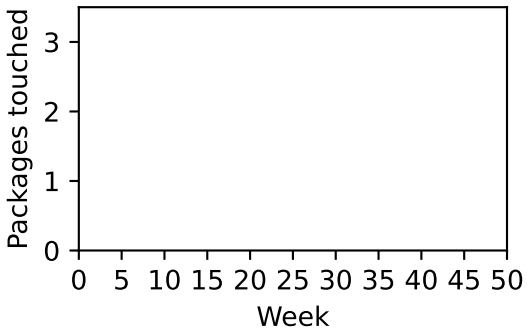
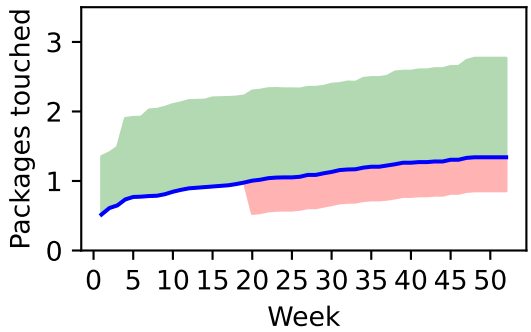


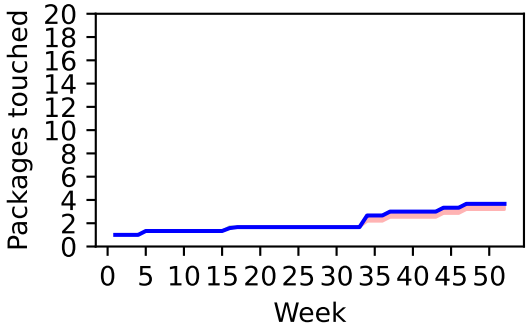
Figure 71: Repository 21 A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for all developers (n=300).



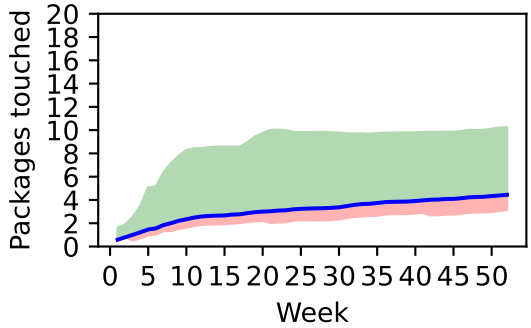
(a) Transient founder developers (n=0)



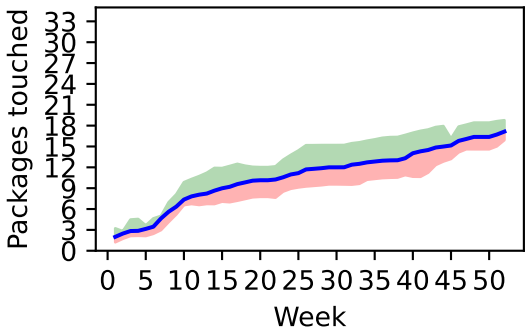
(b) Transient later joiner developers (n=113)



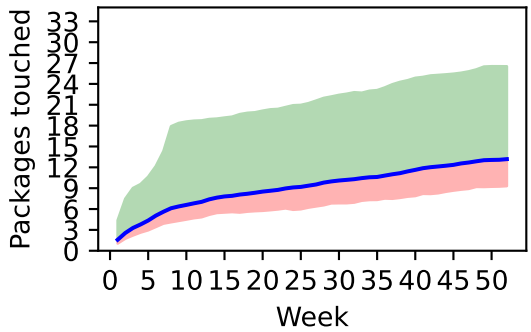
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 72: Repository 21 A time series of the average (mean) total packages touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

3.3 Time series components touched - 21 For classes touched for commit

A time series of classes touched on average.

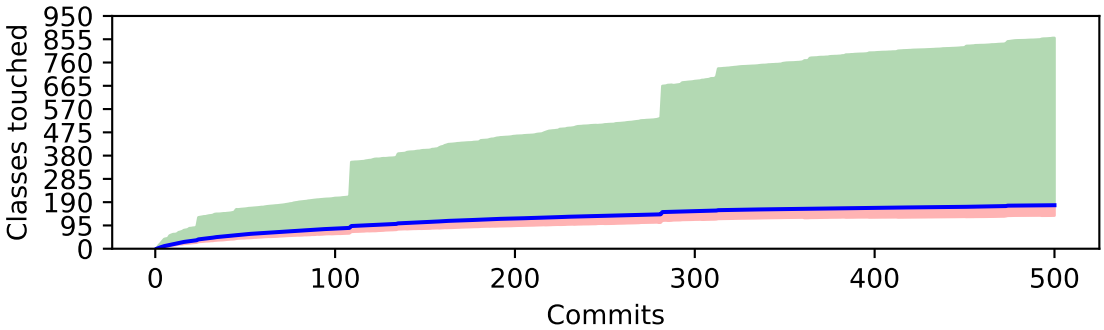
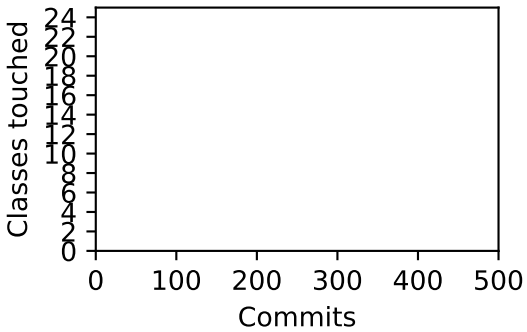
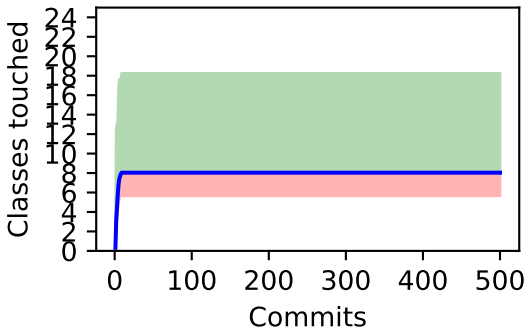


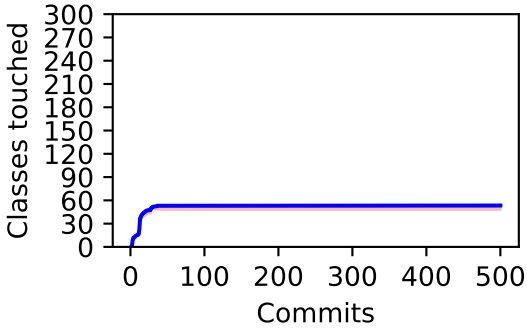
Figure 73: Repository 21 A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for all developers (n=300).



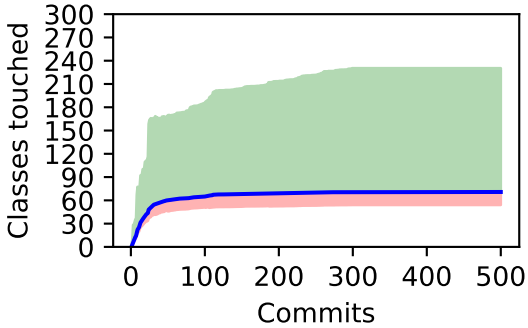
(a) Transient founder developers (n=0)



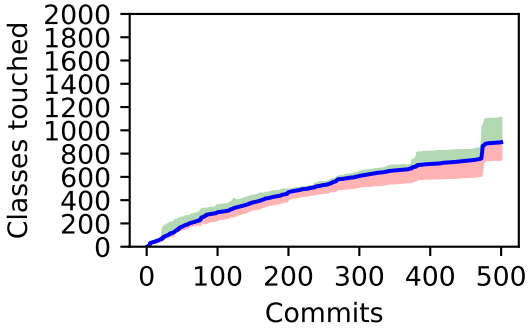
(b) Transient later joiner developers (n=113)



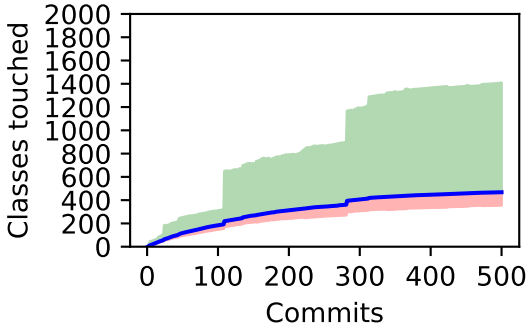
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 74: Repository 21 A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

3.4 Time series components touched - 21 For classes touched for each period

A time series of classes touched on average.

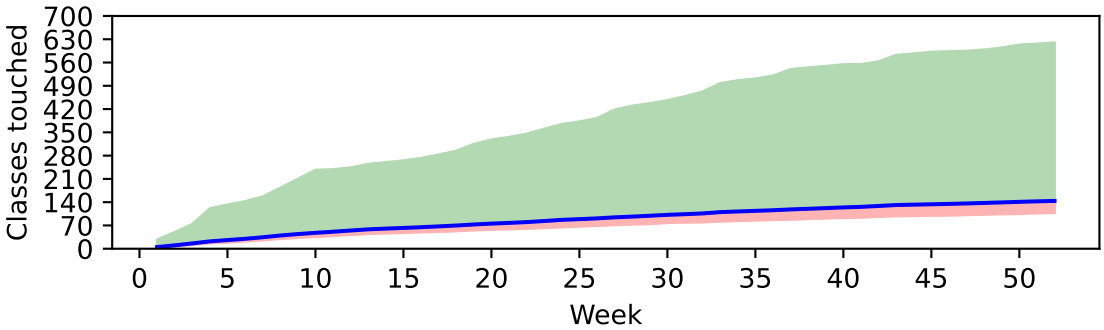
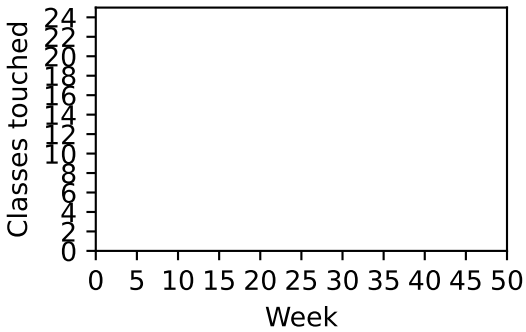
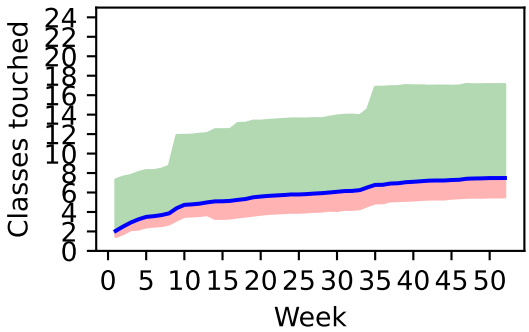


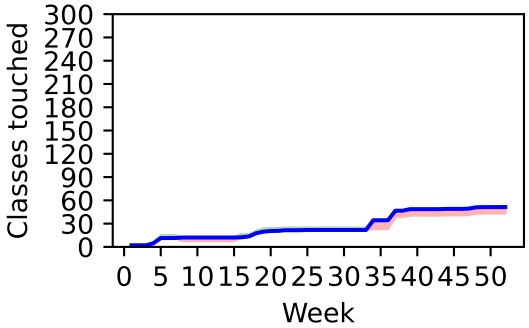
Figure 75: Repository 21 A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for all developers (n=300).



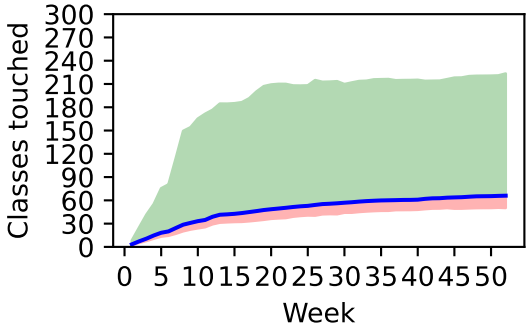
(a) Transient founder developers (n=0)



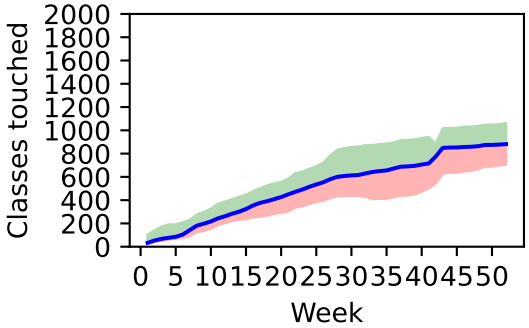
(b) Transient later joiner developers (n=113)



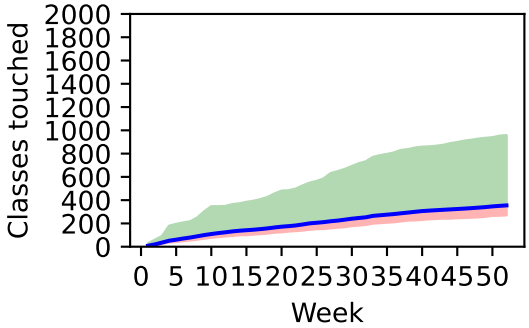
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 76: Repository 21 A time series of the average (mean) total classes touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

3.5 Time series components touched - 21 For methods touched for commit

A time series of methods touched on average.

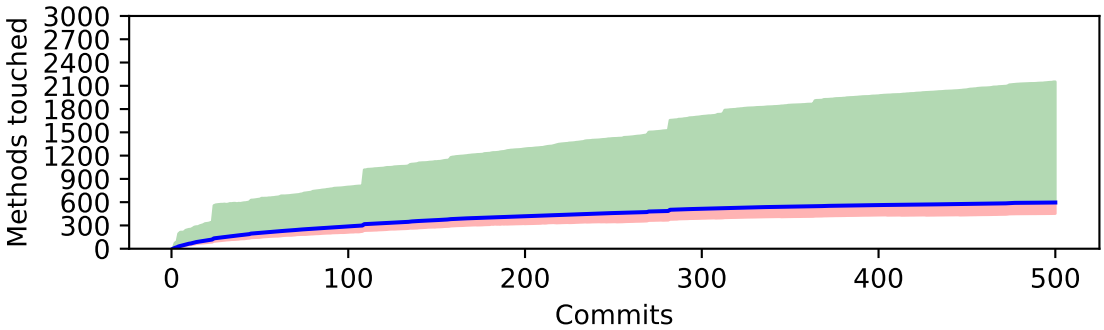
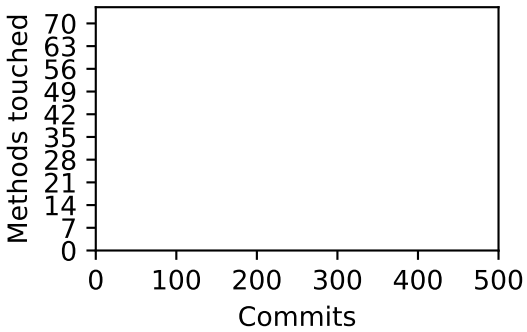
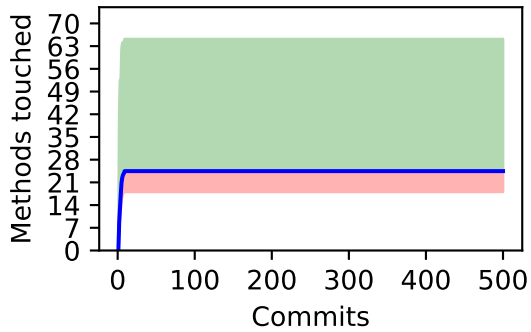


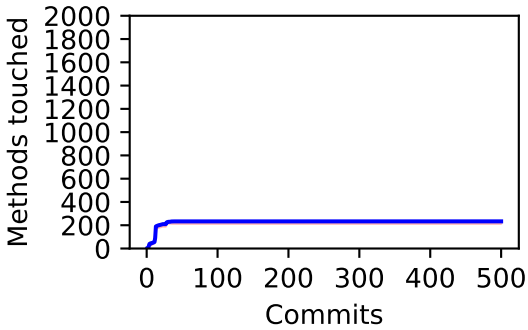
Figure 77: Repository 21 A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for all developers (n=300).



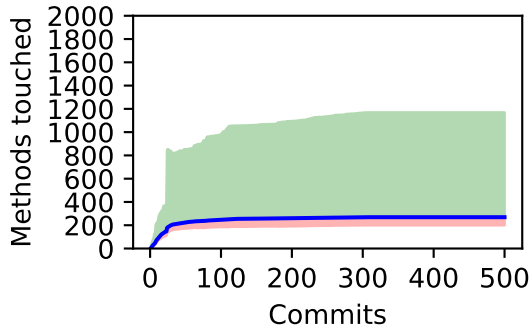
(a) Transient founder developers (n=0)



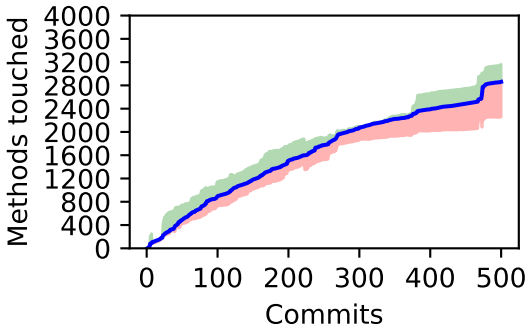
(b) Transient later joiner developers (n=113)



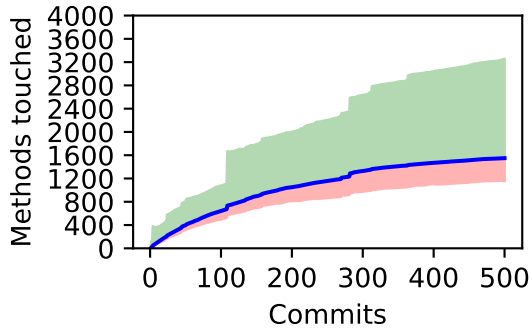
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 78: Repository 21 A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

3.6 Time series components touched - 21 For methods touched for each period

A time series of methods touched on average.

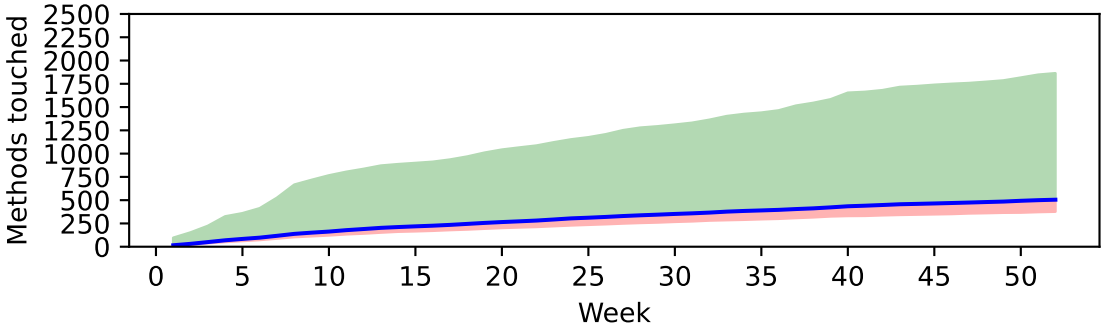
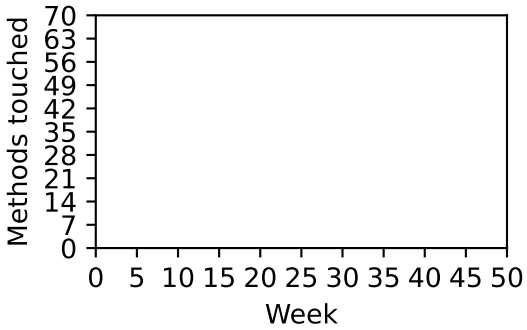
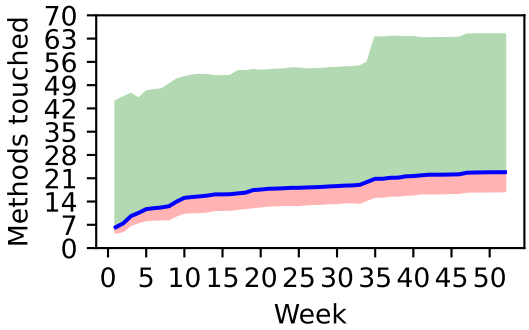


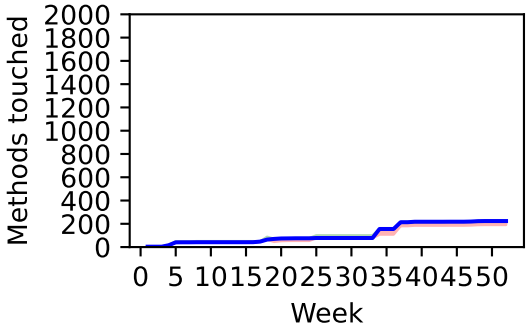
Figure 79: Repository 21 A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for all developers (n=300).



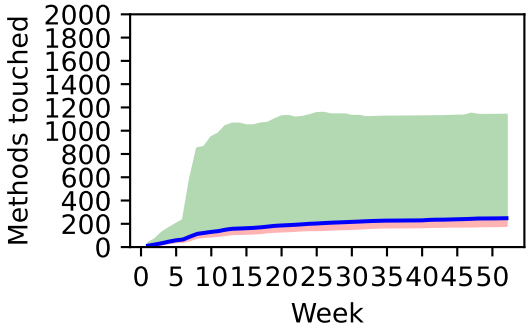
(a) Transient founder developers (n=0)



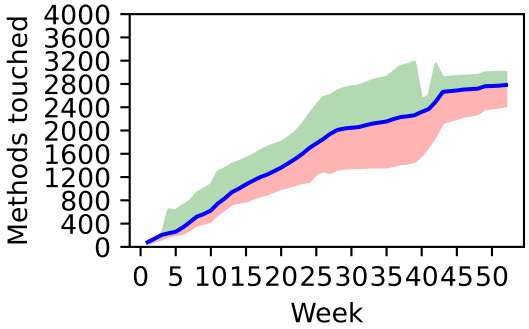
(b) Transient later joiner developers (n=113)



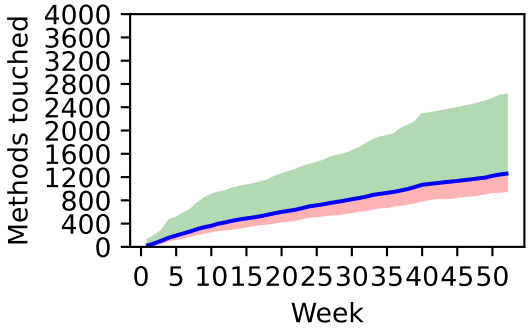
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 80: Repository 21 A time series of the average (mean) total methods touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

3.7 Time series components touched - 21 For packages touched for commit

A time series of packages touched on average.

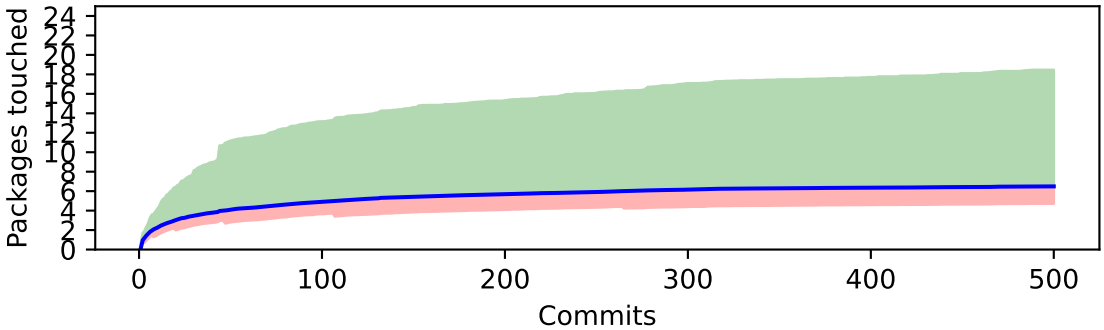
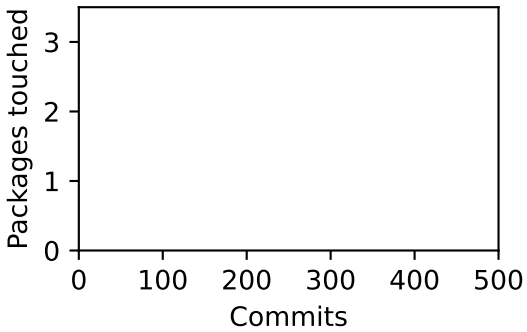
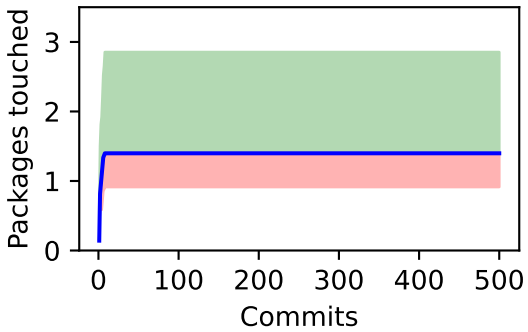


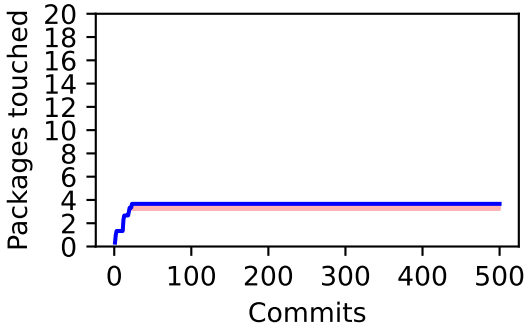
Figure 81: Repository 21 A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for all developers (n=300).



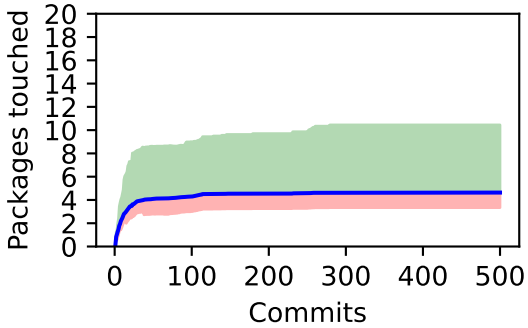
(a) Transient founder developers (n=0)



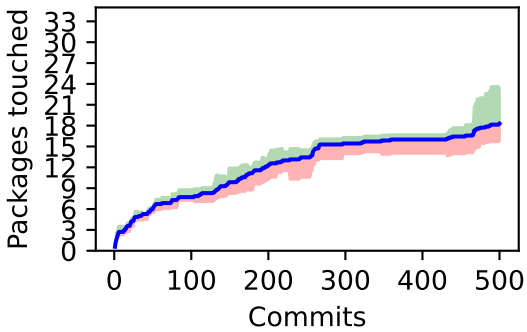
(b) Transient later joiner developers (n=113)



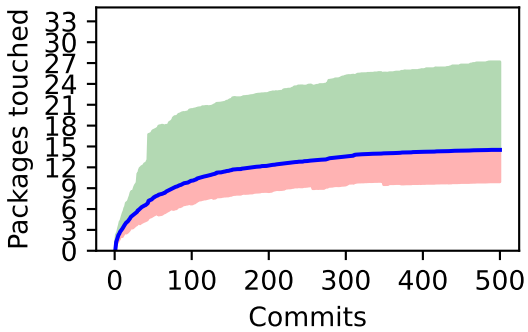
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 82: Repository 21 A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

3.8 Time series components touched - 21 For packages touched for each period

A time series of packages touched on average.

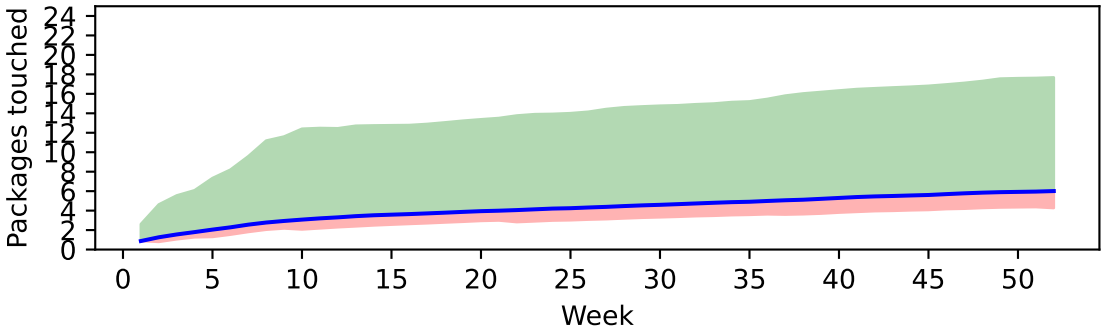
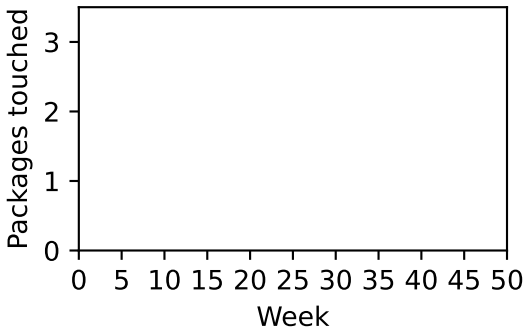
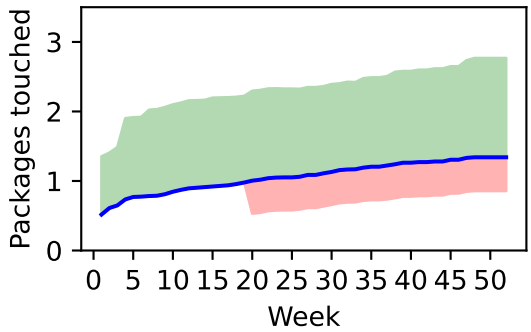


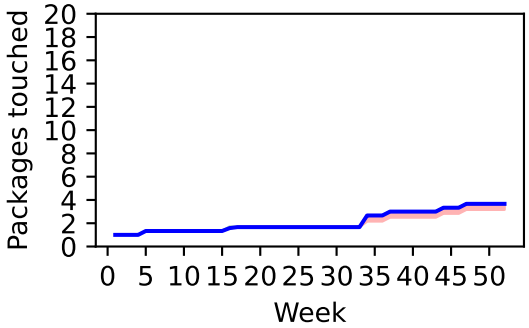
Figure 83: Repository 21 A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for all developers (n=300).



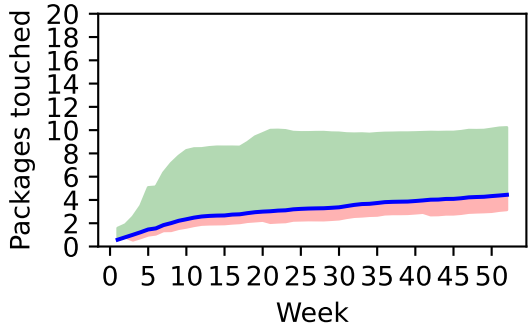
(a) Transient founder developers (n=0)



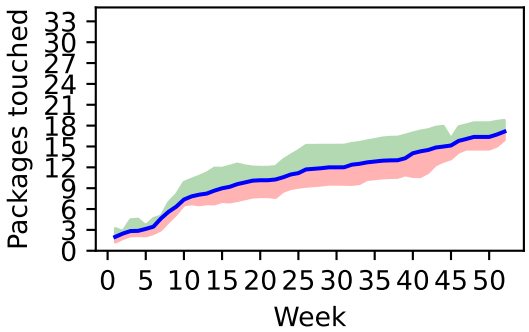
(b) Transient later joiner developers (n=113)



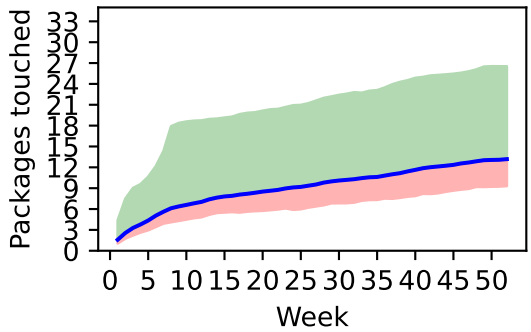
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 84: Repository 21 A time series of the average (mean) total packages touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

3.9 Time series components touched - 21 For classes touched for commit

A time series of classes touched on average.

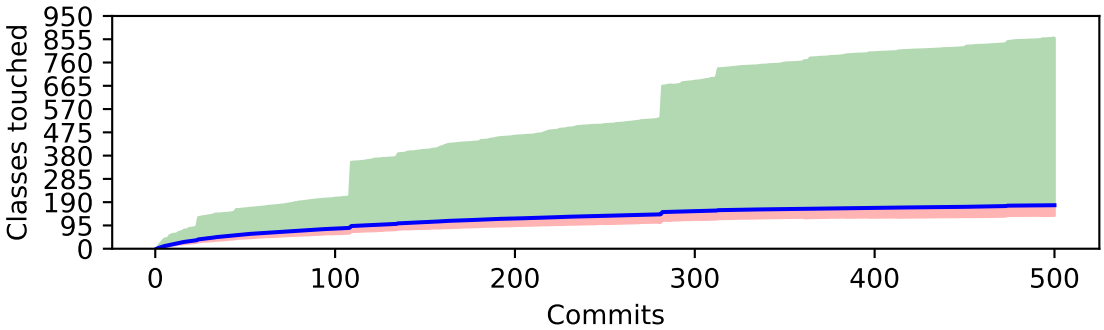
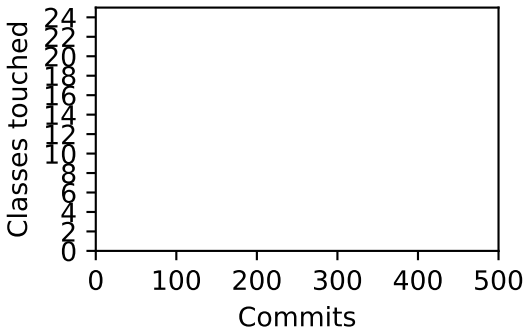
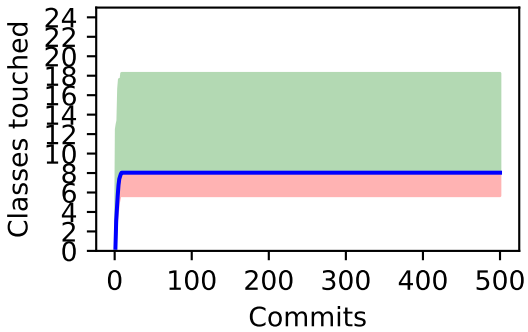


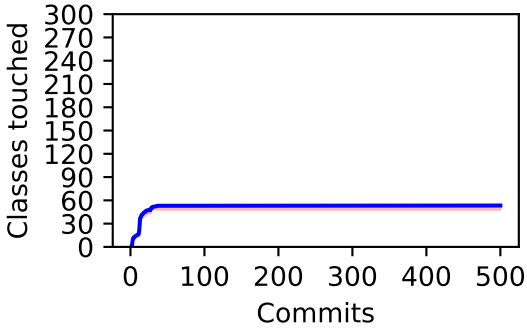
Figure 85: Repository 21 A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for all developers (n=300).



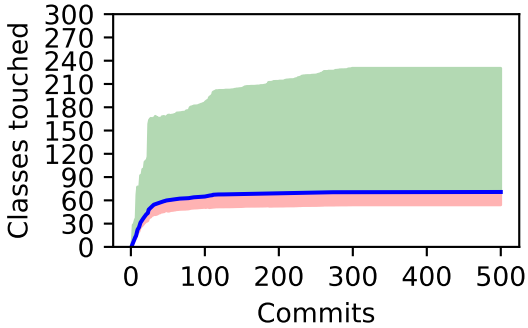
(a) Transient founder developers (n=0)



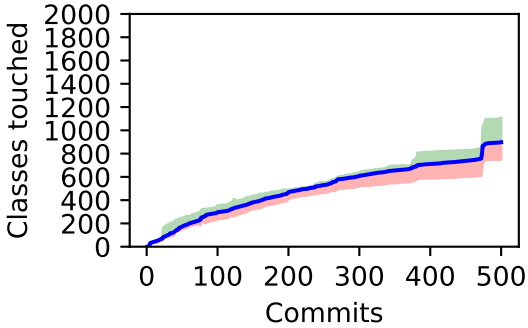
(b) Transient later joiner developers (n=113)



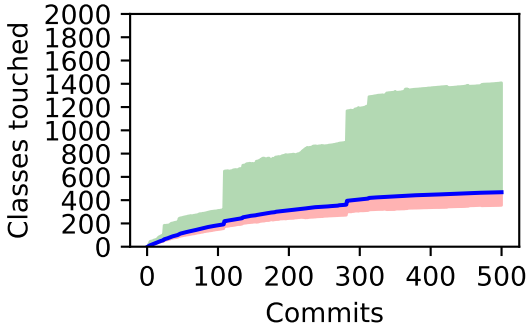
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 86: Repository 21 A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

3.10 Time series components touched - 21 For classes touched for each period

A time series of classes touched on average.

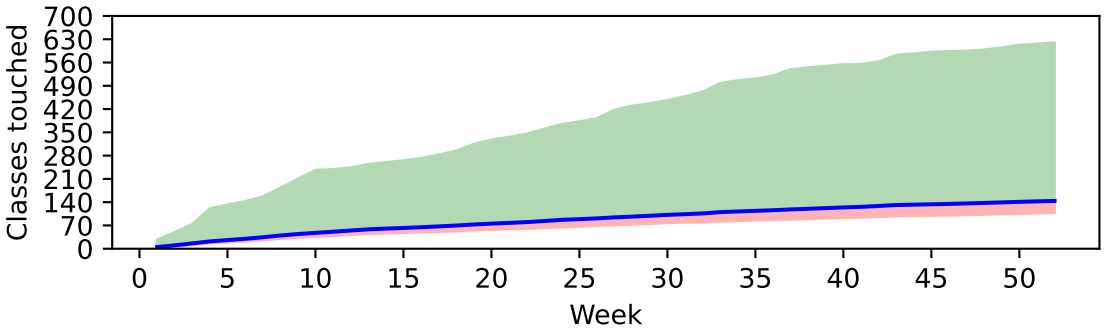
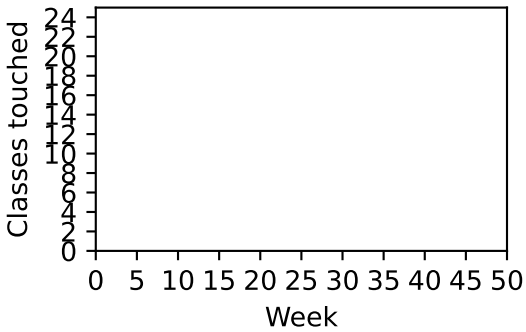
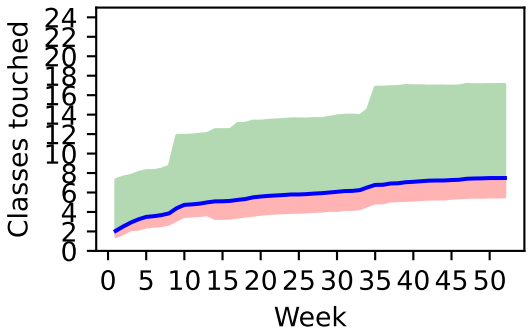


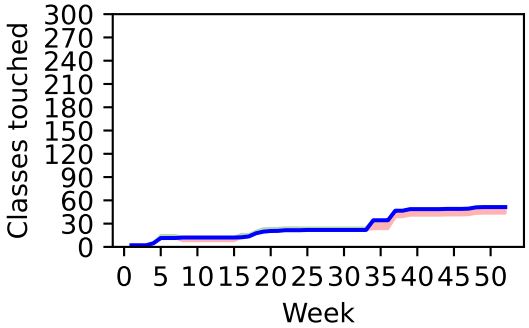
Figure 87: Repository 21 A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for all developers (n=300).



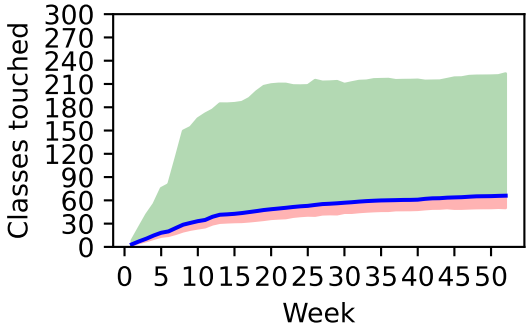
(a) Transient founder developers (n=0)



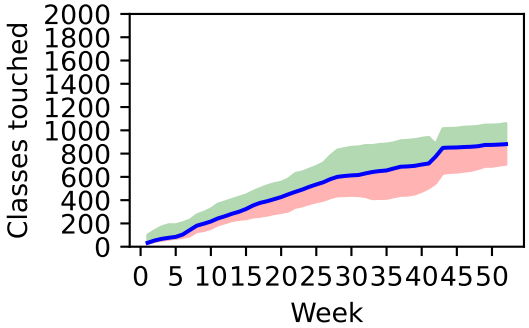
(b) Transient later joiner developers (n=113)



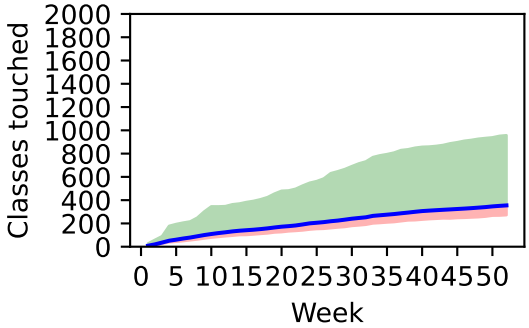
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 88: Repository 21 A time series of the average (mean) total classes touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

3.11 Time series components touched - 21 For methods touched for commit

A time series of methods touched on average.

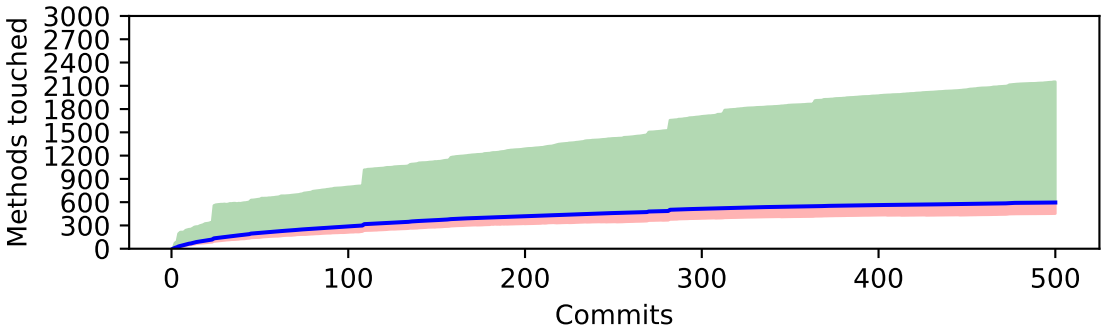
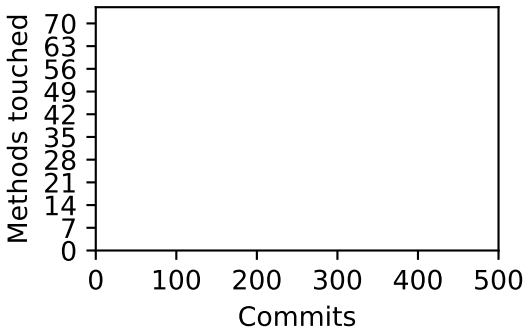
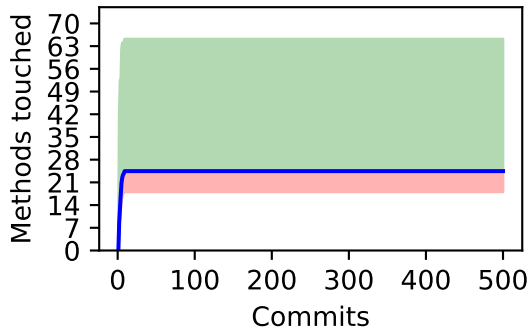


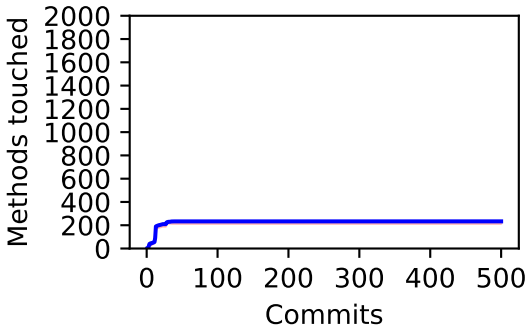
Figure 89: Repository 21 A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for all developers (n=300).



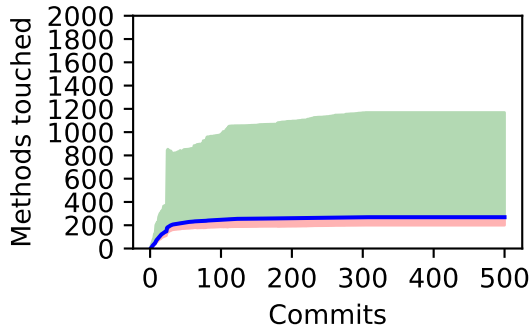
(a) Transient founder developers (n=0)



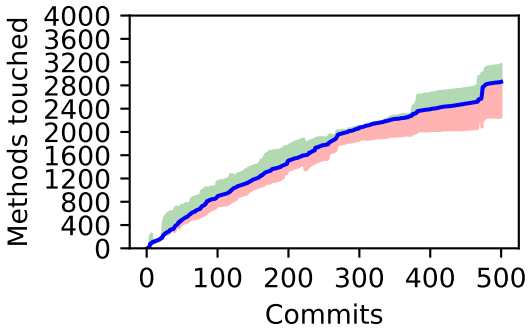
(b) Transient later joiner developers (n=113)



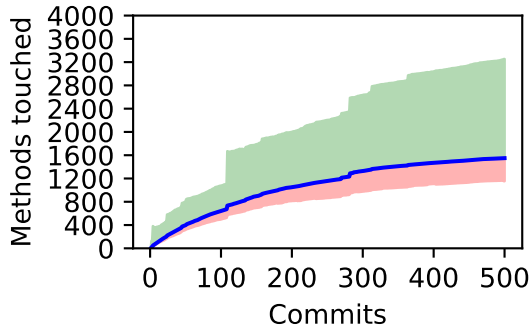
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 90: Repository 21 A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

3.12 Time series components touched - 21 For methods touched for each period

A time series of methods touched on average.

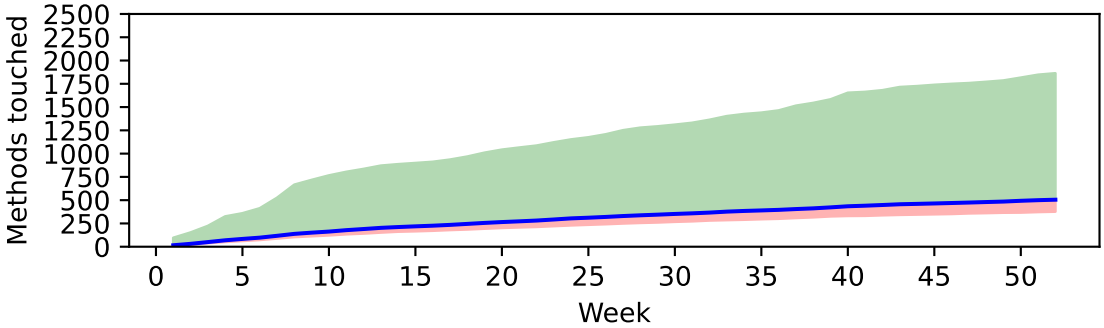
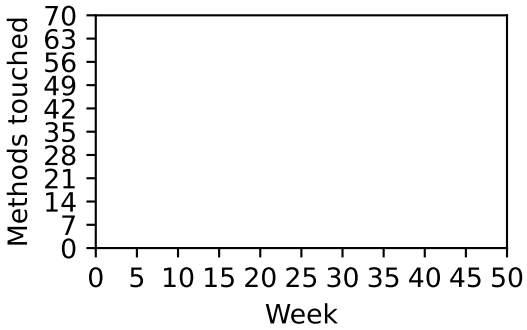
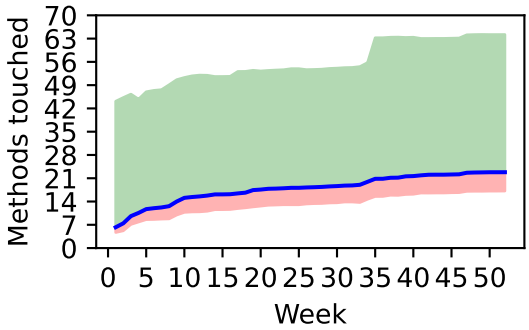


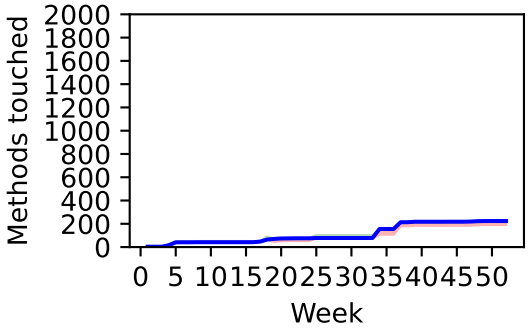
Figure 91: Repository 21 A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for all developers (n=300).



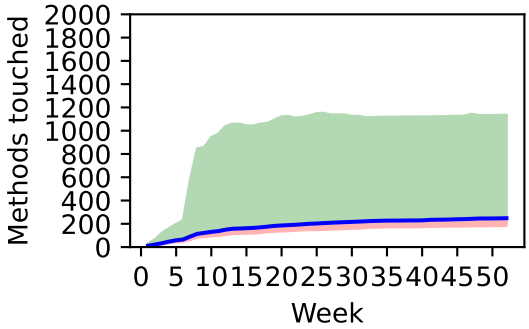
(a) Transient founder developers (n=0)



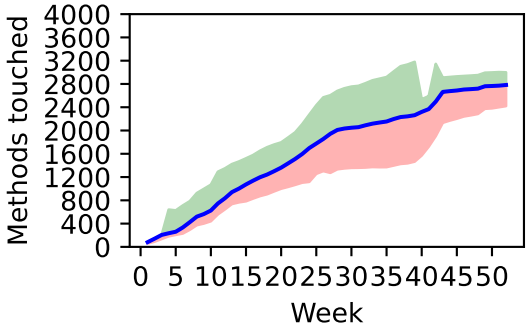
(b) Transient later joiner developers (n=113)



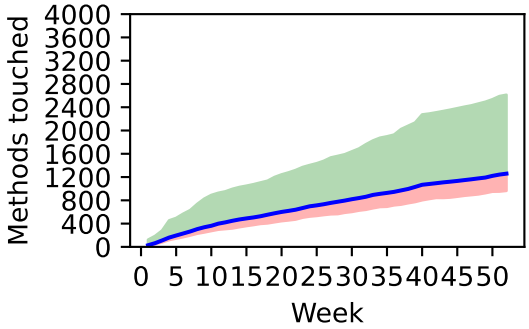
(c) Moderate founder developers (n=3)



(d) Moderate later joiner developers (n=93)



(e) Sustained founder developers (n=7)



(f) Sustained later joiner developers (n=84)

Figure 92: Repository 21 A time series of the average (mean) total methods touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

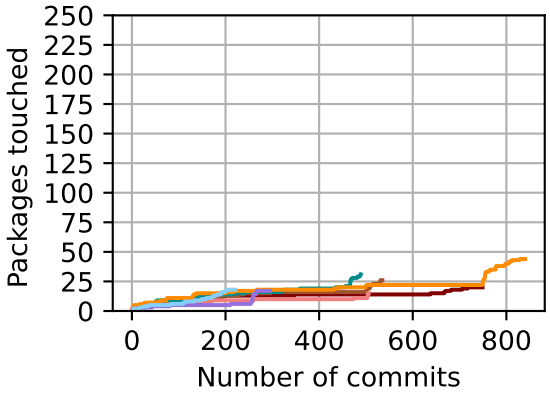
3.13 Time series sustained components touched by developer - 21 For developer commits and components touched.

Table 7: Table of first 7 sustained founder developers in the Scatter Developer graphs.

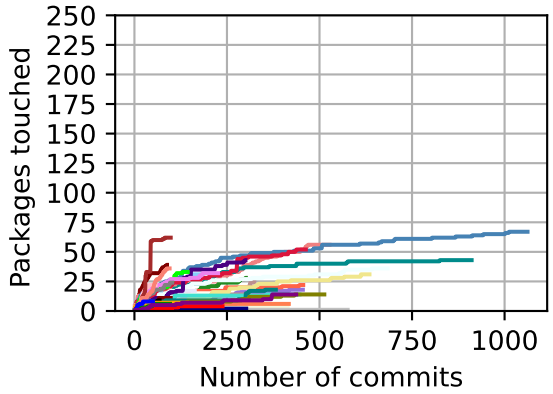
Developer ID	Colour
6245	Maroon
6246	Sienna
6247	LightCoral
6249	DarkCyan
6250	DarkOrange
6251	MediumPurple
6257	LightSkyBlue

Table 8: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

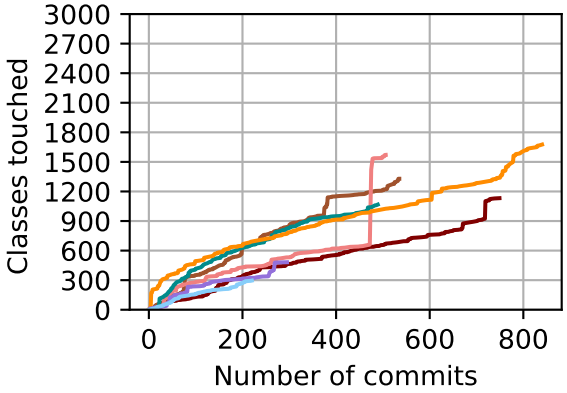
Developer ID	Colour
199	Maroon
6252	Sienna
6253	LightCoral
6259	DarkCyan
6260	DarkOrange
6262	MediumPurple
6263	LightSkyBlue
6264	HotPink
6267	ForestGreen
6271	FireBrick
6272	DodgerBlue
6276	Tomato
6278	SeaGreen
6280	SlateGray
6282	Ivory
6284	Orchid
6288	Chocolate
6295	SteelBlue
6297	RosyBrown
6298	Azure
6302	Tan
6306	Plum
6309	Crimson
6312	Khaki
6313	Lavender
6314	Indigo
6315	Turquoise
6326	Beige
6329	Silver
6333	Gold
6338	Coral
6342	Lime
6343	Maroon
6349	Navy
6352	Teal
6355	Olive
6363	Violet
6368	Gray
6369	Pink
6370	Brown
6373	Black
6374	Yellow
6375	Magenta
6379	Cyan
6391	Salmon
6410	Orange
6411	Red
6417	Green
6418	Blue
6420	Purple



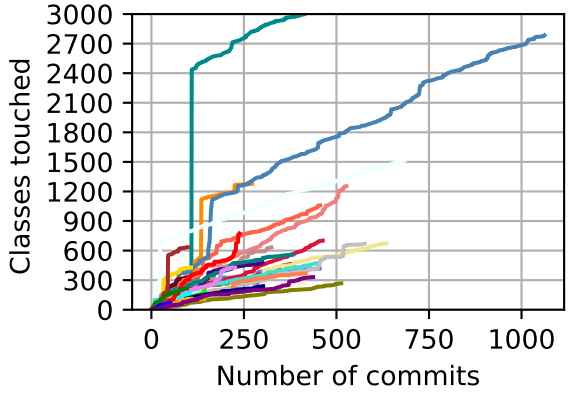
(a) Packages for sustained founder



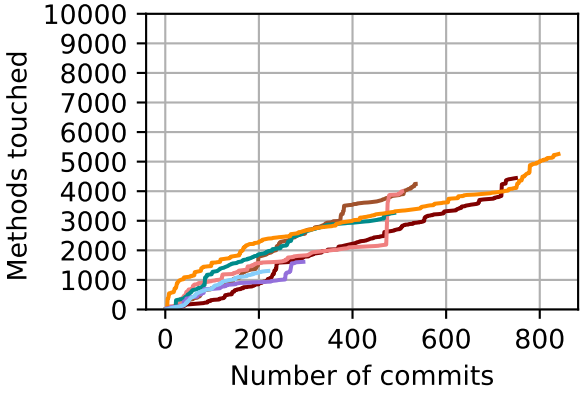
(b) Packages for sustained later joiner



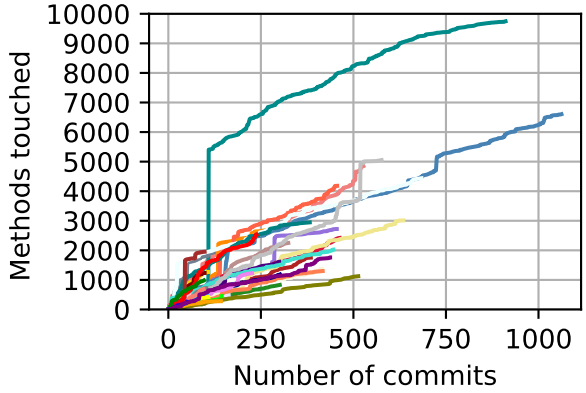
(c) Classes for sustained founder



(d) Classes for sustained later joiner



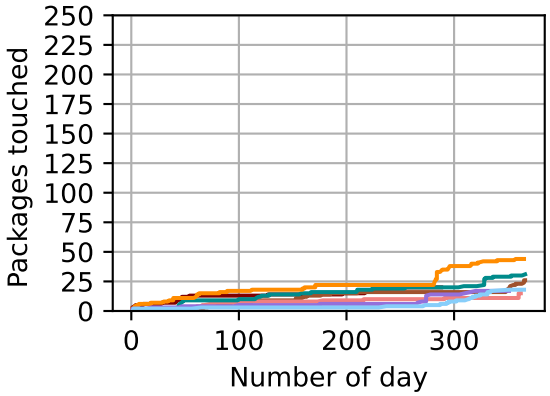
(e) Methods for sustained founder



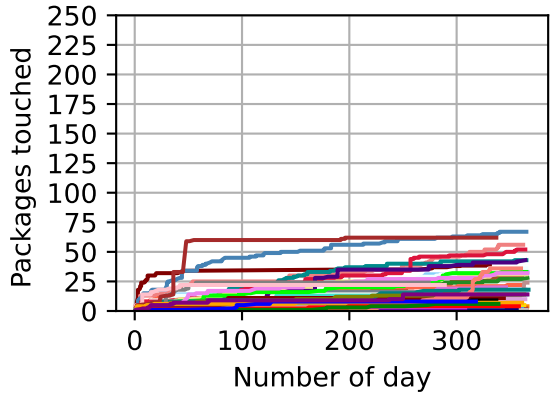
(f) Methods for sustained later joiner

Figure 93: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 7. Sustained later joiner developers with colours are shown in Table 8.

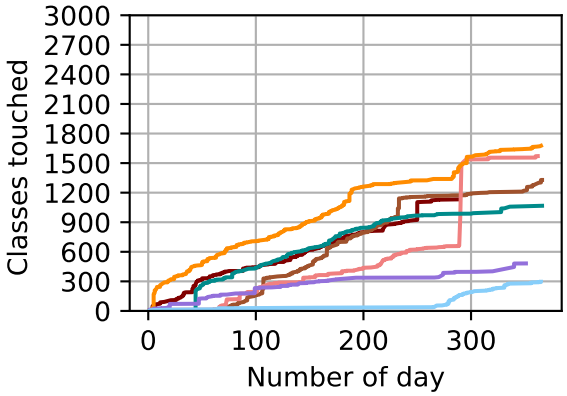
3.14 Time series sustained components touched by developer - 21 For developer day and components touched.



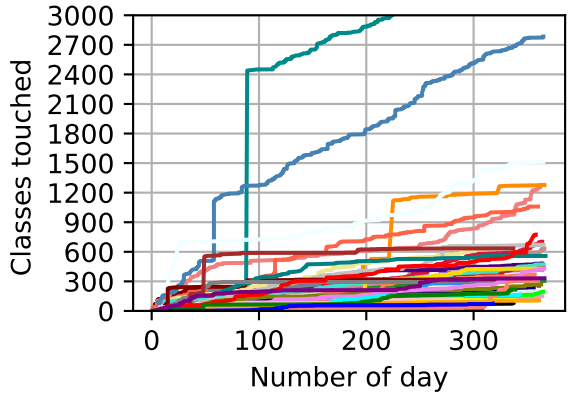
(a) Packages for sustained founder



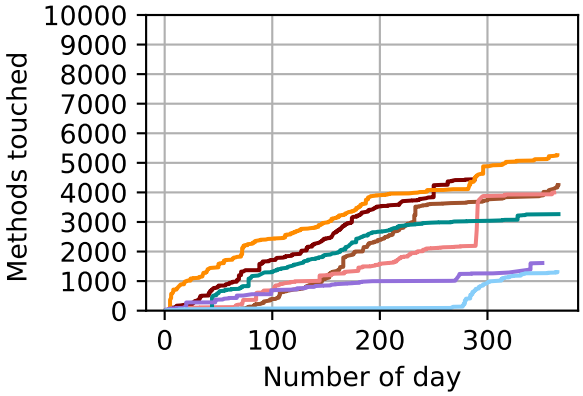
(b) Packages for sustained later joiner



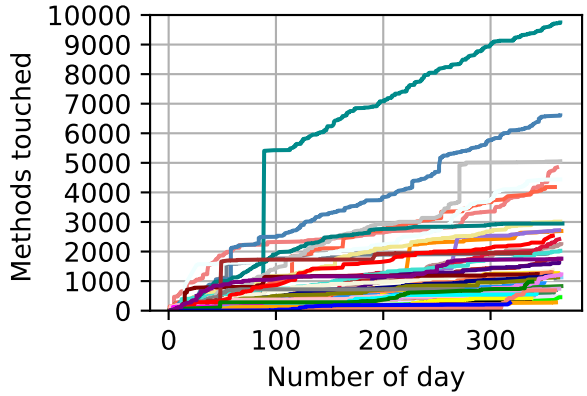
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 94: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 7. Sustained later joiner developers with colours are shown in Table 8.